

Application Note

The Essential Guide for Developing With C2000™ Real-Time Microcontrollers



Matthew Pate

ABSTRACT

Performance, efficiency, flexibility and protection – these are the attributes paramount to power electronics technologies, such as motor control, digital power, renewable energies, lighting, and electrical vehicles. Backed by over 20 years of working with customers developing real-time control applications, the C2000™ real-time Microcontroller (MCU) platform enables developers to cost-efficiently meet all of the above criteria while also differentiating their designs. This application report is intended to provide a deeper look into the components providing differentiation to Real-Time Control Systems and give the next steps for evaluation.

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应用笔记

C2000™实时微控制器开发必备指南

实时微控制器



Matthew Pate

摘要

性能、效率、灵活性和保护——这些是电力电子技术（如电机控制、数字电源、可再生能源、照明和电动汽车）至关重要的属性。凭借20多年与客户合作开发实时控制应用的经验，C2000™ 实时微控制器（MCU）平台使开发者能够以成本效益的方式满足上述所有标准，同时还能使他们的设计脱颖而出。本应用报告旨在深入了解为实时控制系统提供差异化的组件，并给出评估的下一步步骤。

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1 C2000 and Real-Time Control

Starting in 1997, Texas Instruments integrated flash memory, an Analog-to-Digital Converter (ADC), a Digital Signal Processor (DSP) , and Pulse Width Modulation (PWM) units on a single device. The first C2000 real-time control MCU was born.

Over more than two decades, this device family grew and now millions of C2000's can be found in numerous applications in industrial and automotive applications like [Motor Control](#), [Solar Inverters](#), [Digital Power](#), [Electrical Vehicles](#) and more (see [Figure 1-1](#)). There is one tie that binds all the above applications; their real-time nature and the need for a real-time controller.

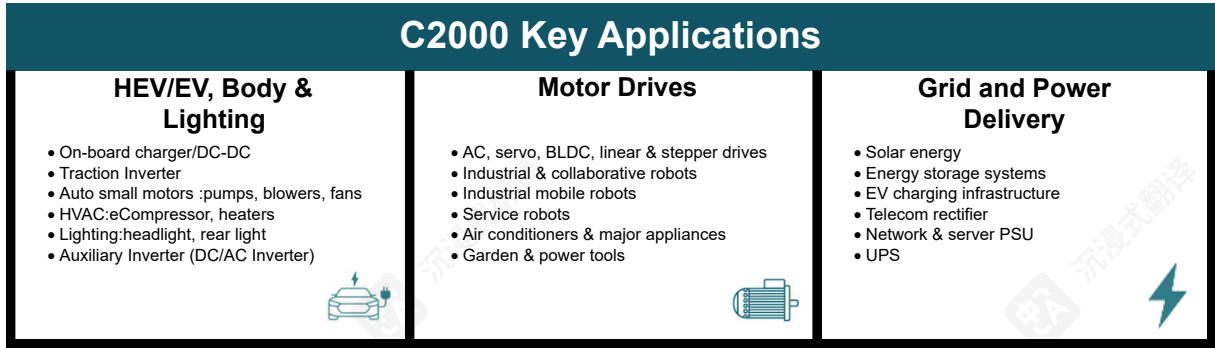


Figure 1-1. Common C2000 Real-Time Control Applications

A real-time control system is typically composed of four main elements: see [Figure 1-2](#):

- **Sensing:** or feedback acquisition. The application needs to measure several key parameters (voltage, current, motor speed, temperature) in an accurate manner and at a very precise moment in time.
- **Processing:** Use the sensing information to apply control algorithms to the incoming data and calculate the next output command.
- **Control:** The command is applied to the system, typically via a PWM unit driving the power electronics system, for example, the motor turns faster, the current to the solar installed system is reduced, the car is accelerating.
- **Interface:** The ability of the device to communicate to other external components. While not necessarily involved in the control of the system, communications to other system components also has to co-exist with the main control loop.

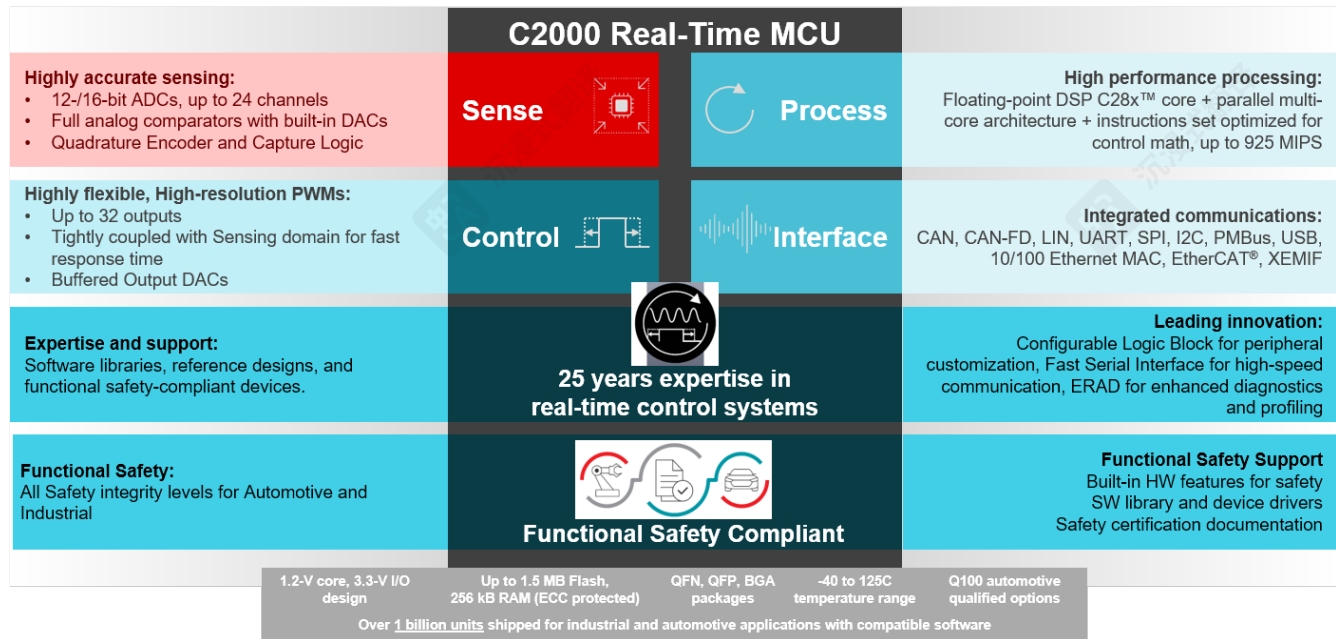


Figure 1-2. C2000 Real-Time MCU Components

1 C2000 与实时控制

自1997年起，德州仪器将闪存、模数转换器（ADC）、数字信号处理器（DSP）和脉冲宽度调制（PWM）单元集成到单个设备上。首款C2000实时控制微控制器诞生。

在超过二十年的时间里，这个设备系列不断发展，现在数百万台 C2000 可以在各种工业和汽车应用中找到，例如电机控制、太阳能逆变器、数字电源、电动汽车等（见图 1-1）。所有这些应用都有一个共同点；它们的实时性和对实时控制器的需求。

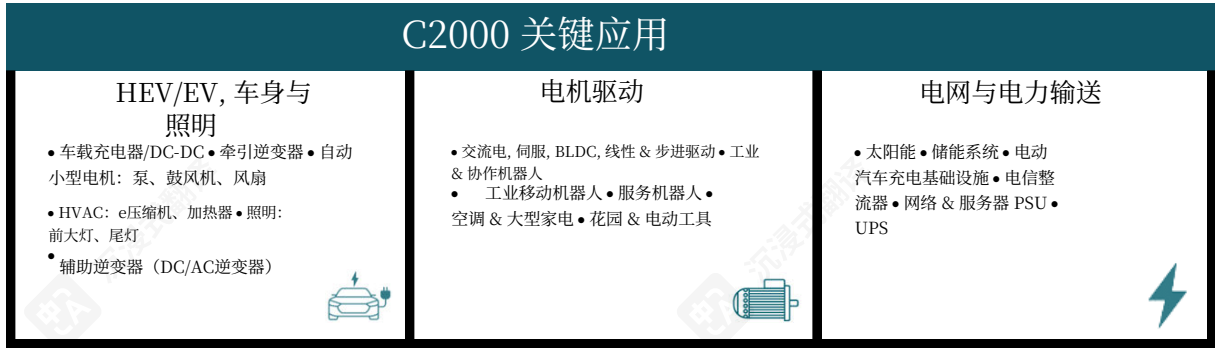


图 1-1. 常见的 C2000 实时控制应用

一个实时控制系统通常由四个主要元素组成：参见图1-2：

- 传感：或反馈获取。应用需要以准确的方式在非常精确的时刻测量几个关键参数（电压、电流、电机速度、温度）。
- 处理：使用传感信息将控制算法应用于传入的数据并计算下一个输出命令。
- 控制：命令应用于系统，通常通过 PWM 单元驱动电力电子系统，例如电机转速加快，太阳能安装系统的电流减少，汽车加速。
- 接口：设备与其他外部组件通信的能力。虽然不一定参与系统的控制，但与其他系统组件的通信也必须与控制循环共存。

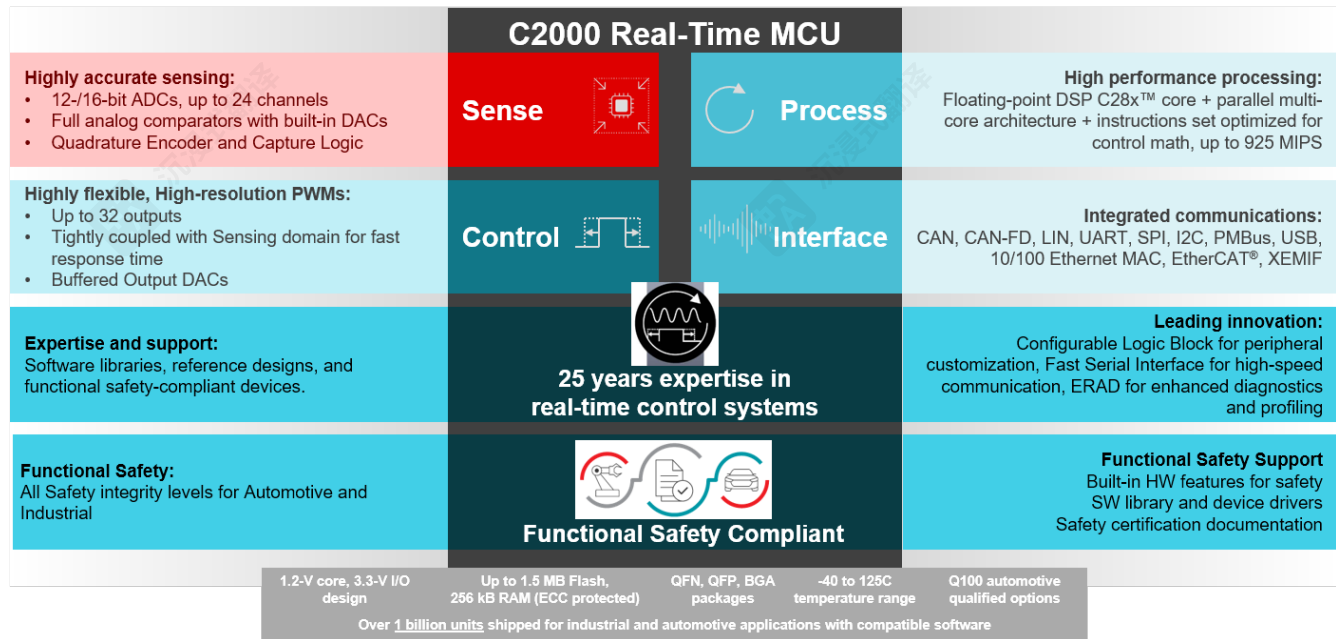


图1-2. C2000实时微控制器组件

The key to real-time control is to minimize the time between Sensing, Processing and Control: this time is defined as the Real-Time Signal Chain. Figure 1-3 illustrates how the entire process is critical to understand the overall system performance of a real-time controller, vs simply looking at the time it takes the main processing unit to complete the control algorithm (step 4).

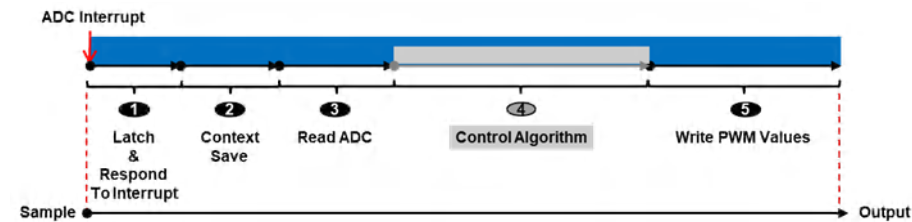


Figure 1-3. Real-Time Signal Chain Components

Many benchmarks, only focus on the time it takes to complete the step 4, typically expressed in Millions Instructions Per Second (MIPS) while the real-time challenge for designers is to minimize the time between Sample to Output: the real-time MCU architecture choice is critical.

The C2000 real-time MCU is a scalable, ultra-low latency, real-time controller platform designed for efficiency in power electronics, such as high power density, high switching frequencies, [GaN and SiC technologies](#) and was designed with the best Real-Time Signal Chain performance in mind and can deliver 2-times more real-time signal chain performance than an Arm®-based architecture.

The following sections showcase these advantages in terms of CPU cycle counts for easy comparison:

- [Section 3.1](#)
- [Section 3.7](#)

For more detailed information on the advantages C2000 brings to the real-time signal chain, including SW benchmarks, see [Signal Chain Benchmarking - A Demonstration of Optimized Real-time Performance of C2000™ MCU](#).

The next sections will zoom in each key elements that enables C2000 MCUs to deliver the best real-time signal performance in the market. Backed up with 25 years of expertise, which has translated in the largest reference design offering for power conversion in the industry, coming with robust [production ready software](#) and [open source hardware documentation](#), designers can now innovate to and build energy system of the future. The C2000 real-time MCU continues to expand with a platform of software compatible device from the low-end to the high-end. Check the [home page on TI.com](#) and register for the [TI newsletter](#) to stay up to date on new innovations from C2000 real-time MCUs.

实时控制的关键在于最小化传感、处理和控制之间的时间：这段时间定义为实时信号链。图1-3说明了整个过程对于理解实时控制器整体系统性能至关重要，而不仅仅是看主处理单元完成控制算法（步骤4）所需的时间。

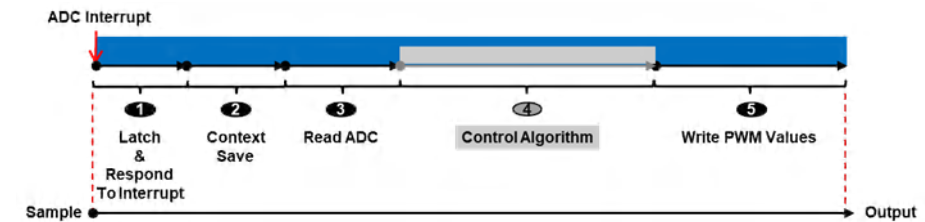


图1-3。实时信号链组件

许多基准测试，仅关注完成步骤4所需的时间，通常以每秒百万条指令（MIPS）表示，而设计人员的实时挑战是尽量缩短采样到输出的时间：实时MCU架构选择至关重要。

C2000实时MCU是一款可扩展、超低延迟的实时控制器平台，专为电力电子领域的效率而设计，具有高功率密度、高开关频率、氮化镓（GaN）和碳化硅（SiC）技术等特点，并旨在提供最佳的实时信号链性能，其性能比基于Arm®的架构高出2倍。

以下部分将展示这些优势，以便于比较CPU周期数：

- [3.1节](#)
- [3.7节](#)

有关C2000为实时信号链带来的优势的详细信息，包括SW基准测试，请参阅Signal Chain Benchmarking - C2000™微控制器优化实时性能的演示。

接下来的章节将聚焦于每个关键要素，这些要素使 C2000 微控制器能够在市场上提供最佳的实时信号性能。凭借 25 年的专业经验，这已经转化为业内最大的电源转换参考设计，并附带强大的生产就绪软件和开源硬件文档，设计师现在可以创新并构建未来的能源系统。C2000 实时微控制器继续扩展，其软件兼容设备平台从低端到高端。请访问 TI.com 的主页并注册 TI 新闻简报，以获取 C2000 实时微控制器最新创新的信息。

1.1 Getting Started Resources

Hardware

Software/Tools

Training

- **Select Part**
 - [Device Selection Guide](#)
 - [Peripheral Reference Guide](#)
- **Pick a C2000 EVM**
 - Launchpads: Low cost evaluation board
 - controlCARDs: Full-featured development board
 - Buy an EVM to get started
- [Reference Designs](#)

- [SW Overview](#)
- **Tool-Chain**
 - Code Composer Studio
 - [Standalone Download](#)
 - [Cloud Based](#)
 - [SysConfig and PinMux Support](#)
- **C2000ware SDK** - Low-level drivers and highly optimized libraries
 - [Digital Power SDK](#)
 - [Motor Control SDKw/ InstaSpin](#)
- **Model-Based Evaluation**
 - [MathWorks Embedded Coder](#)

- [Getting Started](#)
- [C2000 Academy](#)
- [Motor Control Workshop Series](#)
- [Digital Power Training Series](#)
- [EV Training Modules](#)
- [Safety Overview](#)

1.2 Processing

As seen in [Figure 1-4](#), the C2000 real-time MCU uses the C28x DSP(Digital Signal Processor) core as the main processing unit. Capable of both 32-bit float or fixed point operations with dedicated instructions tailored to real-time control applications. Additional components of this sub-system are outlined below:

- [Section 3.4](#): A state machine based 32-bit floating point co-processor capable of independent code execution from the main C28x DSP core
- C28x Extended Instructions:
 - Floating Point Unit (FPU): Supporting 32-bit floating point operations and on select devices supports [Section 3.3](#)
 - [Section 3.1](#): Provides intrinsic instructions to support common trigonometric math functions commonly found in transfroms and torque loop calculations.
 - Viterbi and CRC Unit(VCU): Reducing cycle count for both Viterbi and Cyclical Redundancy Check(CRC) operations found in complex math equations.

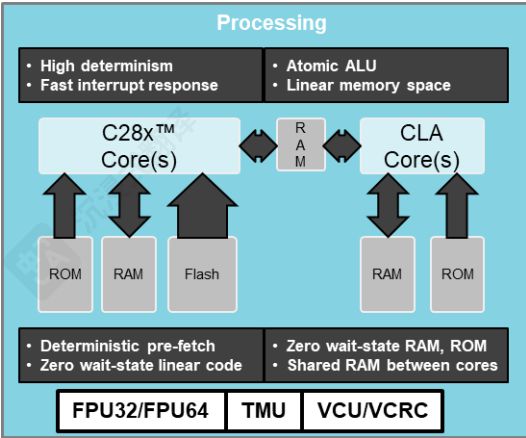


Figure 1-4. C2000 Processing

1.1 获取入门资源

Hardware

Software/Tools

Training

- **选择部件**
 - [设备选择指南](#)
 - [外设参考指南](#)
- **选择一款 C2000 EVM**
 - Launchpads: 低成本评估板
 - 控制CARDs: 全功能开发板
 - 购买EVM开始使用
- [参考设计](#)

- [SW概述](#)
- **工具链**
 - Code Composer Studio
 - [独立下载](#)
 - [基于云](#)
 - [SysConfig and PinMux Support](#)
- C2000ware SDK - 低级驱动和高度优化的库
 - [数字电源SDK](#)
 - [Motor Control SDKw/ InstaSpin](#)
- **Model-Based Evaluation**
 - [MathWorks Embedded Coder](#)

- [入门指南](#)
- [C2000学院](#)
- [电机控制工作坊系列](#)
- [数字电源培训系列](#)
- [电动汽车培训模块](#)
- [安全概述](#)

1.2 处理

如图 1-4 所示，C2000 实时微控制器使用 C28x DSP（数字信号处理器）内核作为主要处理单元。它能够执行 32 位浮点或定点运算，并提供专为实时控制应用设计的专用指令。该子系统的其他组件如下所述：

- 第3.4节：一个基于状态机的32位浮点协处理器，能够独立于主C28x DSP核心执行代码
- C28x扩展指令：
 - 浮点单元（FPU）：支持32位浮点运算，并在部分设备上支持第3.3节
 - 第3.1节：提供内建指令以支持常见于变换和扭矩环计算中的三角数学函数
 - 维特比和CRC单元(VCU)：减少复杂数学方程式中维特比和循环冗余校验(CRC)操作所需的周期数。

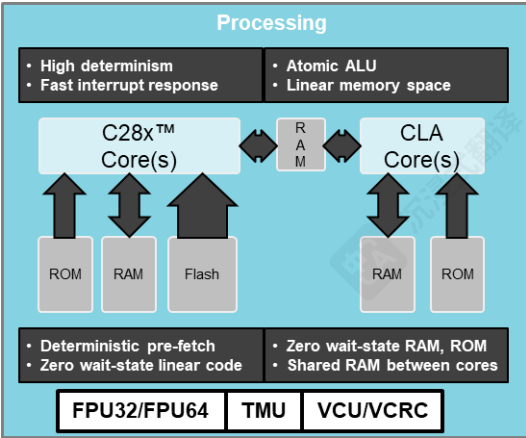


图1-4。C2000处理

1.3 Control

The control sub-system includes modules that will stimulate the system under control. Typically, this is done with [Pulse Width Modulation \(PWM\) outputs](#). This could also be the output of the on-chip Digital-to-Analog Converter (DAC) or just a General-Purpose Input/Output (GPIO) pin.

- PWM – Principle actuation module on the C2000 real-time MCU. Responsible for driving the external Field Effect Transistors (FETs) that exist in most power electronics systems. Supports both standard and high resolution modes for duty-cycle, period, and dead-band control of the waveform
 - All PWMs on a given C2000 have global load capability, ensuring synchronous updates across multiple switch controls
 - [High resolution \(~150ps\) edge placement](#) helps reduce limit cycling in control systems. Available for duty cycle, phase, and deadband placement.
 - Tightly coupled with the analog comparators [to provide over voltage/current protection](#) [CPU independent duty control for systems like Peak Current Mode Control](#) as well as [CPU independent duty control for systems like Peak Current Mode Control](#).
- Buffered DAC – 12-bit DAC capable of driving a defined external load. Typically used to create a bias voltage in the analog domain.
- Configurable Logic Block (CLB) – Group of look up tables and state machine logic that operates on internal signal nodes in the hardware domain. [Can be an endpoint or intermediary step to realize increased system performance](#).

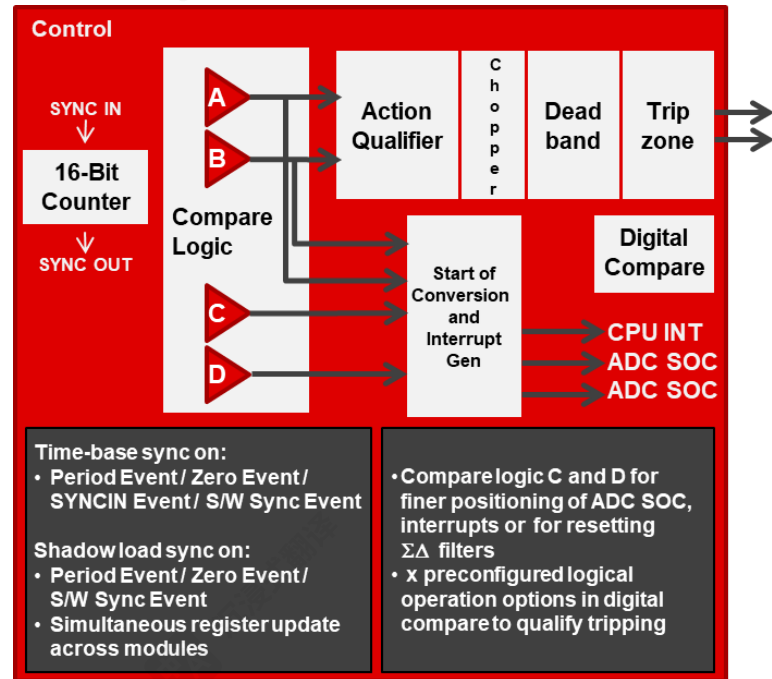


Figure 1-5. PWM Block Diagram

1.3 控制

控制子系统包括将刺激被控系统的模块。通常，这是通过脉冲宽度调制（PWM）输出完成的。这也可能是片上数模转换器（DAC）的输出，或者只是一个通用输入/输出（GPIO）引脚。

- PWM – C2000 实时 MCU 上的原理驱动模块。负责驱动大多数电力电子系统中存在的外部场效应晶体管 (FET)。支持标准和高分辨率模式，用于波形占空比、周期和死区控制
 - 给定 C2000 上的所有 PWM 都具有全局负载能力，确保跨多个开关控制同步更新
 - 高分辨率 (~150ps) 边缘定位有助于减少控制系统中的极限循环。适用于占空比、相位和死区定位
 - 与模拟比较器紧密耦合，以提供过压/电流保护 CPU 独立占空比控制，适用于峰值电流模式控制等系统，以及 CPU 独立占空比控制，适用于峰值电流模式控制等系统
- 缓冲 DAC – 12 位 DAC，能够驱动定义的外部负载。通常用于在模拟域中创建偏置电压
- 可配置逻辑块 (CLB) – 一组查找表和状态机逻辑，在硬件域中对内部信号节点进行操作。可以是实现系统性能提升的端点或中间步骤。

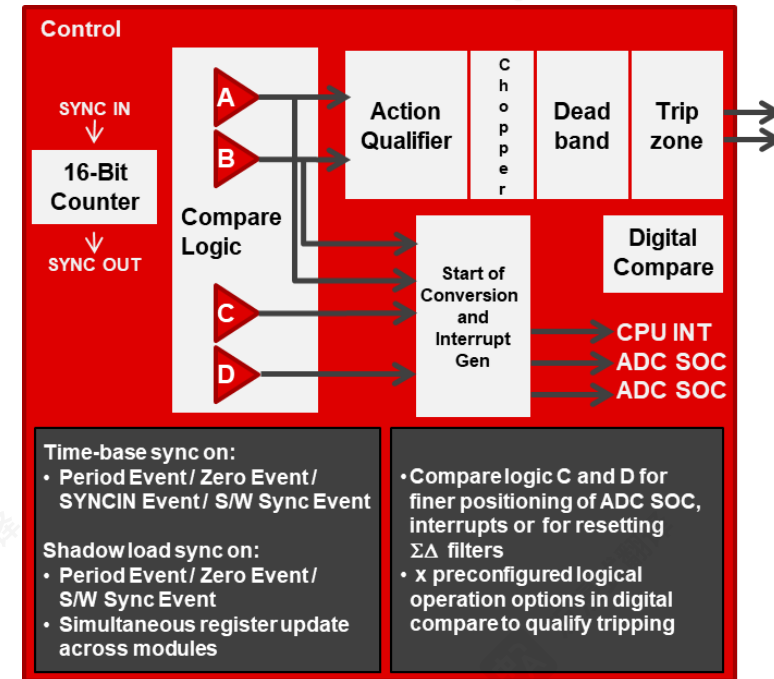


图 1-5. PWM 框图

1.4 Sensing

The [sensing sub-system](#) includes modules that translate the state of the external system under control (analog domain) into data usable by the C2000 real-time controller (digital domain). Often this is the work of the Analog-to-Digital Converters (ADCs) on the MCU, but could also be handled by comparators or demodulators for external ADCs. Other unit converters are included in this domain, such as quadrature encoders and time pulse measurement devices.

- ADC – Multiple [12 or 16-bit ADCs](#) that are used primarily to convert the voltage or current (through a shunt) of the controlled system into the digital domain. Both an internal reference or external references are supported, with sample rates of 3.5MSPS (12-bit) and 1MSPS (16-bit) to quickly translate the system conditions to information that the control system can act upon.
- Comparator (COMP) – Multiple on-chip comparators provide system protection as well as cycle by cycle PWM control by comparing a system voltage to an internal reference point(generated by the internal 12-bit DACs). [Direct connection to the ePWM modules exist to change the output state as quickly as possible without need for CPU intervention.](#)
- Sigma Delta Demodulator – On-chip logic used to decode the serial bit stream output from external sigma delta ADCs. [Often used to cross the isolation boundary of high power systems.](#) Includes hardware threshold detection.
- Quadrature Encoder Pulse Measurement (eQEP) – Counts pulses from a variety of encoders to determine motor shaft position. [Can be coupled with the CLB module for customer encoder solutions in HW.](#)
- Time Capture (eCAP) – Measures the time duration between external pulse events, useful for evaluating Hall Sensors. Both standard and high resolution modes are available.

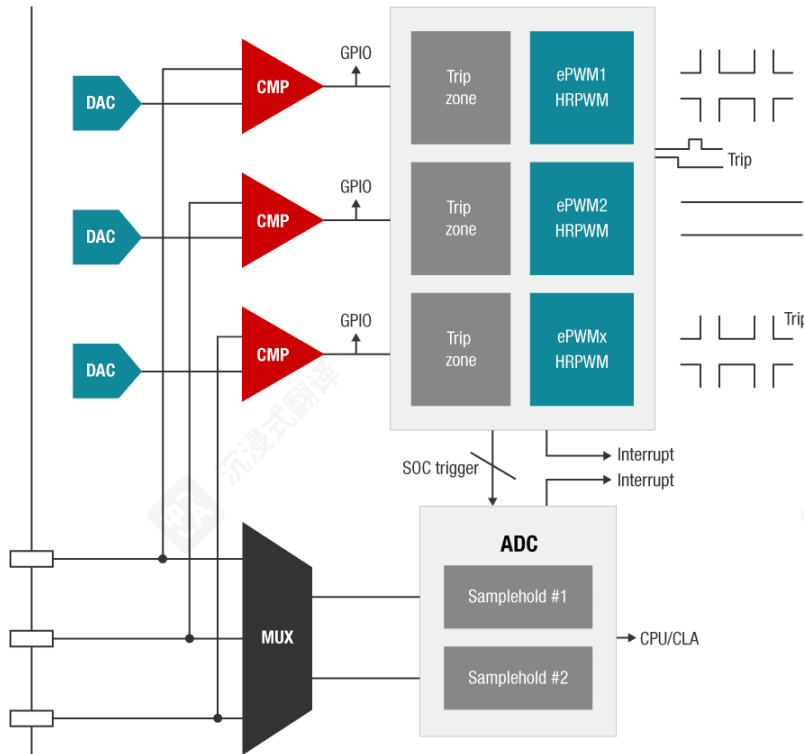


Figure 1-6. C2000 Analog Integration

1.4 感测

传感子系统包括将外部被控系统（模拟域）的状态转换为C2000实时控制器（数字域）可用数据的模块。这通常由 MCU上的模数转换器（ADC）完成，但也可能由比较器或外部ADC的解调器处理。该领域还包括其他单位转换器，如正交编码器和时间脉冲测量设备。

- ADC – 多个12位或16位ADC，主要用于将受控系统的电压或电流（通过分流器）转换为数字域。支持内部参考或外部参考，采样率分别为3.5MSPS（12位）和1MSPS（16位），以快速将系统状态转换为控制s
- 比较器 (COMP) – 多个片上比较器提供系统保护，并可通过将系统电压与内部参考点（由内部12位 DACs生成）进行比较，实现逐周期PWM控制。可直接连接到ePWM模块，以快速改变输出状态，无需CPU干预。
y茎可以作用 u pon.
- Σ - Δ 解调器 – 片上逻辑，用于解码外部 Σ - Δ ADC的串行位流输出。常用于跨越高功率系统的隔离边界。包括硬件阈值检测。
- 正交编码器脉冲测量 (eQEP) – 从各种编码器计数脉冲以确定电机轴位置。可与 CLB 模块配合使用，用于硬件中的客户编码器解决方案。
- 时间捕获 (eCAP) – 测量外部脉冲事件之间的时间间隔，可用于评估霍尔传感器。提供标准和高速模式。

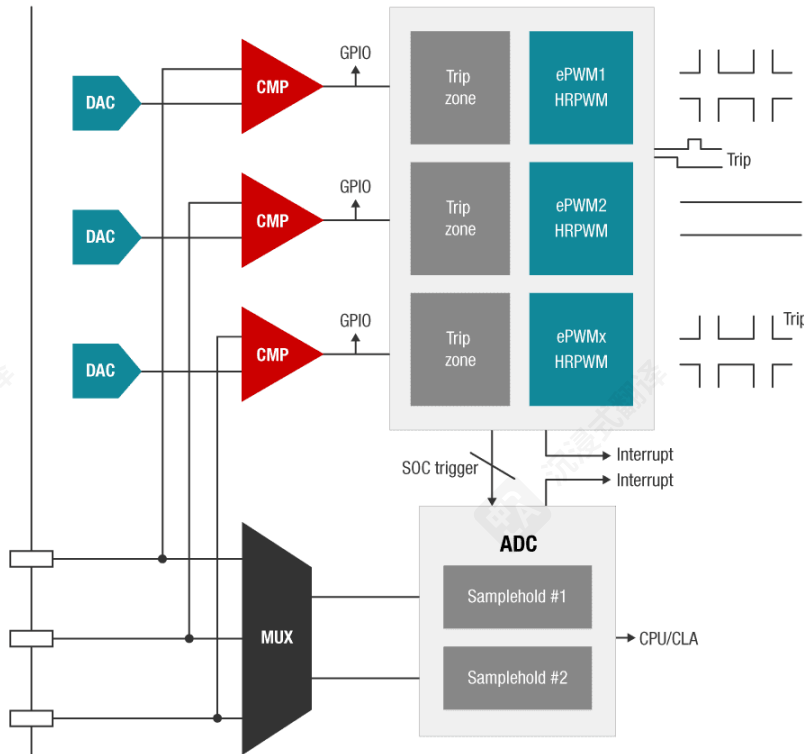


图 1-6. C2000 模拟集成

1.5 Interface

An additional block that, while not typically associated with the real-time control loop of the system, is almost always needed from a system integration point of view. Good integration of the interface block with the CPU is essential to avoid high overhead that could impact the control loop. From serial data streams to multi-channel inputs, as well as industry standard options to proprietary formats, the interface sub-system supports a wide array of communications options.

- Controller Area Network (CAN) – The CAN module supports the Bosch™ CAN protocol standard.
- External Memory InterFace (EMIF) – Parallel data bus typically used to support connections to SDRAM as well as wide bus peripherals.
- EtherCAT Slave Controller (EtherCAT) – This module allows for the C2000 MCU to act as a slave node in an EtherCAT network.
- Ethernet – 10/100 Mbps Ethernet controller and physical interface for external communications across this bus.
- Fast Serial Interface (FSI) – 2 or 3 line simplex serial data transmit or receive. Designed to meet both the high speed (100Mbps) as well as the variable latency introduced when crossing an isolation boundary.
- Host Interface Controller (HIC) - Grants ability for other devices to control/interact with the C2000 peripherals
- Inter-Integrated Circuit (I2C) – Interface/controller for an I2C bus
- Serial Peripheral Interface (SPI) – Interface/controller for a standard Serial Peripheral Interface bus.
- Universal Asynchronous Receiver/Transmitter (UART) – Interface/controller for Universal Asynchronous Receiver Transmitter bus
- Universal Serial Bus (USB) – USB 2.0 MAC and PHY used to interface to standard USB network.

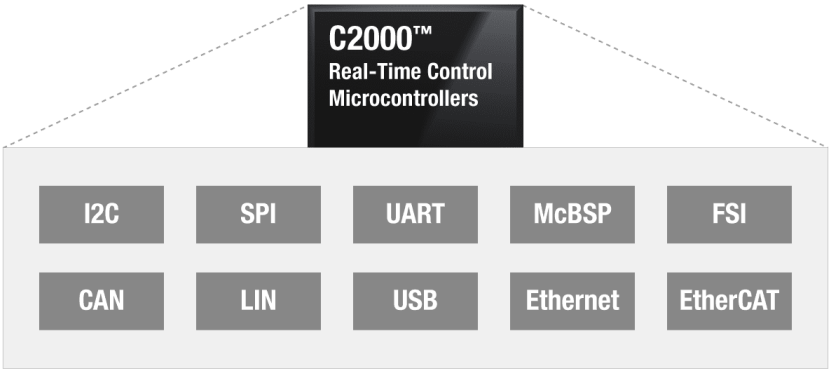


Figure 1-7. C2000 MCU Supported Interfaces

Note

Peripheral counts, as well as features, may vary from device to device. For a complete listing of the number of a peripherals on a device, see the data sheets referenced at the end of each section. For the feature sets supported on a given device, see the [C2000 Real-Time Control Peripherals Reference Guide](#).

1.5 接口

一个附加模块，虽然通常不与系统的实时控制循环相关联，但从系统集成的角度来看几乎总是需要的。接口模块与CPU的良好集成对于避免可能影响控制循环的高开销至关重要。从串行数据流到多通道输入，以及行业标准选项到专有格式，接口子系统支持广泛的通信选项。

- 控制器局域网 (CAN) – CAN 模块支持博世™ CAN 协议标准。
- 外部内存接口 (EMIF) – 并行数据总线通常用于支持连接到 SDRAM 以及宽总线外设。
- EtherCAT 从控制器 (EtherCAT) – 该模块允许 C2000 MCU 在 EtherCAT 网络中充当从节点。
- 以太网 – 10/100 Mbps 以太网控制器和物理接口，用于通过此总线进行外部通信。
- 快速串行接口 (FSI) – 2 或 3 线单工串行数据发送或接收。设计用于满足高速 (100Mbps) 以及跨越隔离边界时引入的变量延迟。
- 主机接口控制器 (HIC) - 授予其他设备控制/交互 C2000 外设的能力
- 内部集成电路 (I2C) – I2C 总线的接口/控制器
- 串行外围接口 (SPI) – 标准串行外围接口总线的接口/控制器。
- 通用异步收发器 (UART) – 通用异步收发器总线的接口/控制器
- 通用串行总线 (USB) – USB 2.0 MAC 和 PHY，用于与标准 USB 网络接口。

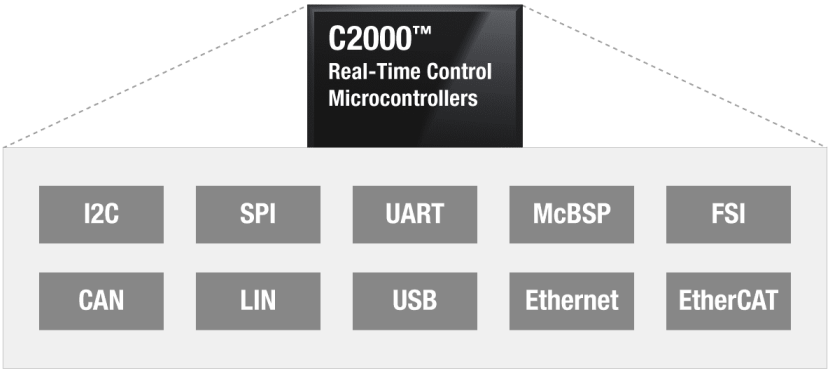


图 1-7。C2000 MCU 支持的接口

Note

外设数量以及特性可能因设备而异。要获取设备上外设数量的完整列表，请参阅每个章节末尾引用的数据表。要了解特定设备支持的特性集，请参阅 C2000 实时控制外设参考指南。

1.6 Functional Safety

According to the International Electrotechnical Commission (IEC), safety is defined as freedom from unacceptable risk of physical injury or damage to thehealth of people, either directly or indirectly, as a resultof damage to property or to the environment. The IEC defines functional safety as the part of overall safetythat depends on a system or equipment operating correctly in response to its inputs.

With over 300 safety mechanisms defined and independently assessed by TUV SUD for its effectiveness, C2000 MCUs provide the required diagnostic coverage to meet a random hardware capability of SIL 2/ASIL B at a component level. Functional safety manuals provide detailed information on the safety mechanisms, techniques for achieving non-interference between elements and avoiding dependent failures, to aid customers in the development of compliant systems up to SIL 3/ASIL D.

Examples of functional safety enablers across the C2000 device can be seen in [Functional Safety Enablers](#).

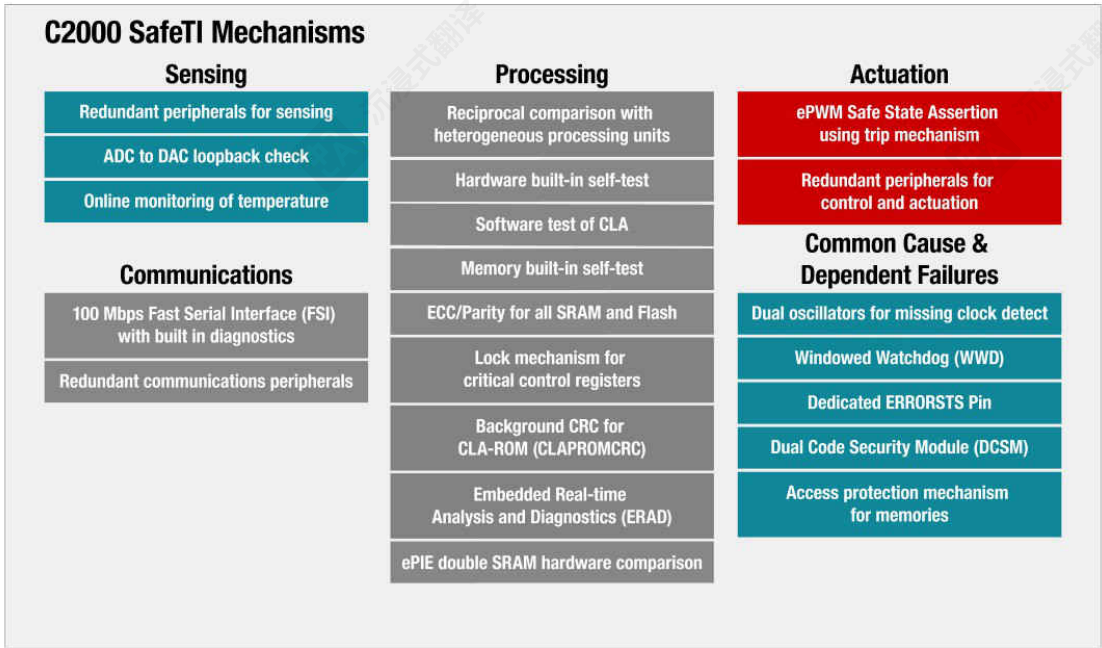


Figure 1-8. Functional Safety Enablers

For more detailed information please see the following technical documents:

- [Industrial Functional Safety forC2000™ Real-Time Microcontrollers](#)
- [Automotive Functional Safety forC2000™ Real-Time Microcontrollers](#)

1.6 功能安全

根据国际电工委员会（IEC）的定义，安全是指免受因财产或环境损坏而直接或间接导致人体身体伤害或健康损害的不可接受风险。IEC 将功能安全定义为整体安全的一部分，该部分取决于系统或设备在响应其输入时正确运行。

C2000 MCU已定义超过300种安全机制，并由TUV SUD独立评估其有效性，为组件级别的随机硬件能力提供所需的诊断覆盖率，以满足SIL 2/ASIL B要求。功能安全手册提供了关于安全机制、实现元素间非干扰和避免相关故障的技术细节，以帮助客户开发符合SIL 3/ASIL D要求的系统。

C2000设备中功能安全使能器的示例可以在《功能安全使能器》中找到。

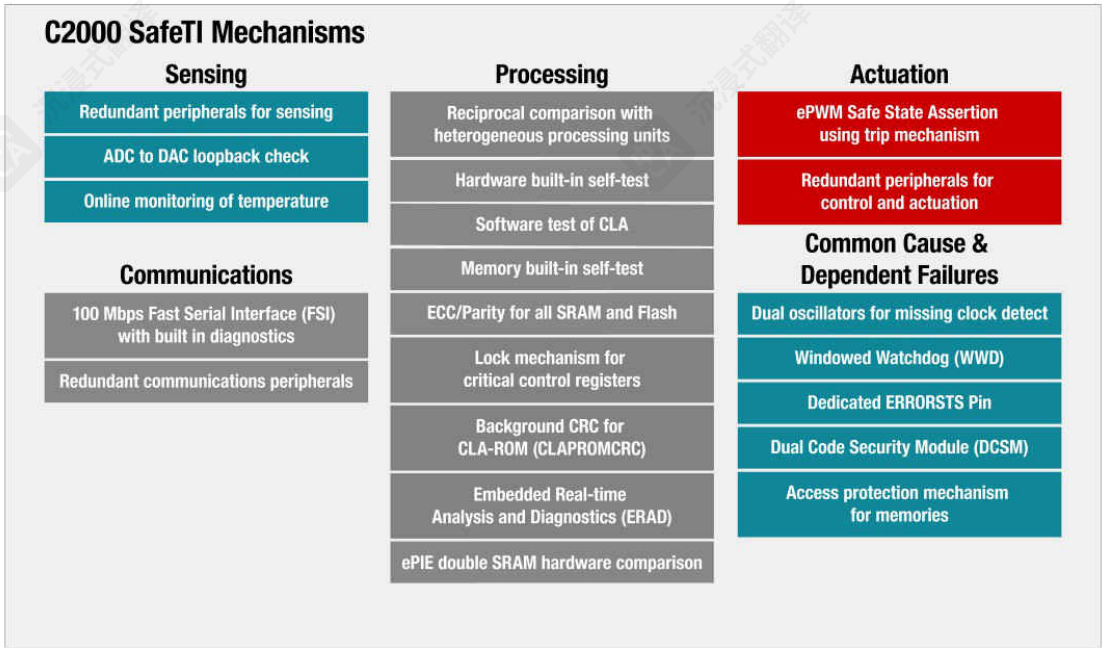


图1-8. 功能安全使能器

更多详细信息请参阅以下技术文档：

- [《C2000™实时微控制器工业功能安全》](#)
- [C2000™ 实时微控制器的汽车功能安全](#)

2 Sensing Key Technologies

2.1 Accurate Digital Domain Representation of Analog Signals

2.1.1 Value Proposition

Many MCUs have integrated ADCs as part of their sensing subsystem. The ability of the ADC to accurately convert the analog domain to the digital space is one of the most crucial aspects of the MCU in order to realize a proper control system. The data sheet specifications for a C2000 MCU ADC are such that performance in the system can be properly evaluated prior to system implementation.

2.1.2 In Depth

The first step when selecting an MCU for a real-time control system is relatively straightforward process; comparing the components of the MCU to the system needs. There are questions of memory size, CPU speed, communications standards used, analog content, number of I/Os, and so forth. When looking at the fit for an analog module like the ADC, it can appear straightforward to base the decision on sampling rate, number of inputs, and bit level. In practice, however, there is much more to this decision.

Too often ADC selection is based solely on the top level specifications, only to realize during development there are limitations to the system performance due to the ADC itself:

- Will the system be using the analog inputs for frequency analysis? Then, AC specifications like SNR and THD become important to consider when picking an MCU with an on-chip ADC.
- Is overall accuracy a key care about? Looking at the DC specifications like INL, Gain, and Offset are key parameters to consider.

A quick summary of ADC specifications and their relevance to the system:

- AC Specifications: Parameters related to how accurately the converter can resolve the fundamental frequency tone of a signal from other noise sources. Includes SNR, SINAD, THD, and SFDR all expressed in dB. Also includes ENOB, which is the SINAD translated into number of bits. Typically SINAD and ENOB based on SINAD are considered when choosing an ADC, the importance will vary depending on the end application.
- DC Specifications: Parameters related to the accuracy of the converter as it applies to representing an analog input in the digital domain. Includes Gain, Offset, DNL, and INL. The weighted summation of the Gain, Offset, and INL are often referred to as "Total Unadjusted Error" ([Equation 1](#)). This equation is typically used to determine the real-world impact of these parameters on the accuracy of a conversion.

$$\sum \sqrt{(\text{Err}_{\text{gain}})^2 + (\text{Err}_{\text{offset}})^2 + (\text{Err}_{\text{INL}})^2} \quad (1)$$

where

- Err_{gain} is the maximum gain error of the ADC in LSBs
- $\text{Err}_{\text{offset}}$ is the maximum offset error ADC in LSBs
- Err_{INL} is the maximum INL error of the ADC in LSBs

An example of how the C2000 ADC is specified and the parameters can be seen in [Table 2-1](#), a dynamic link to this same table in the data sheet is located [here](#).

One final aspect of all the parameters that C2000 devices list in the data sheet is what is implied by the inclusion of the parameter itself. For parameters that have a MIN/MAX, these are assured specs over the full operational range and lifetime of the device. The typical (TYP) column is also significant for all parameters, as it represents the mean performance of a parameter across its operational range.

2 感知关键技术

2.1 模拟信号精确的数字域表示

2.1.1 价值主张

许多 MCU 都集成了 ADC 作为其传感子系统的一部分。ADC 将模拟域准确转换为数字空间的能力是 MCU 实现正确控制系统最关键的因素之一。C2000 MCU ADC 的数据表规格可以在系统实施之前正确评估系统中的性能。

2.1.2 深入探讨

为实时控制系统选择MCU的第一步是一个相对简单的过程；比较MCU的组件与系统的需求。有内存大小、CPU速度、使用的通信标准、模拟内容、I/O数量等等问题。在考虑像ADC这样的模拟模块的适配性时，基于采样率、输入数量和位级来做出决定可能看起来很简单。然而，实际上，这个决定还有更多内容。

ADC的选择往往仅基于顶层规格，直到开发过程中才发现由于ADC本身导致系统性能存在限制：

- 系统会使用模拟输入进行频率分析吗？那么，在选择带片上ADC的MCU时，SNR和THD等AC规格就变得重要需要考虑。
- 整体精度是一个关键考虑因素吗？查看INL、增益和偏移等DC规格是关键参数需要考虑。

ADC规格的快速总结及其与系统的相关性：

- AC规格：与转换器能够从其他噪声源中准确解析信号的基本频率成分的能力相关的参数。包括以dB表示的SNR、SINAD、THD和SFDR。还包括ENOB，它是将SINAD转换为位数。在选择ADC时，通常基于SINAD的SINAD和ENOB会被考虑，其重要性会根据最终应用而变化。
- DC规格：与转换器在将模拟输入表示为数字域中的精度相关的参数。包括增益、偏移、DNL和INL。增益、偏移和INL的加权总和通常被称为“未调整总误差”（公式1）。这个公式通常用于确定这些参数对转换精度实际影响的计算。

$$\sum \sqrt{(\text{Err}_{\text{gain}})^2 + (\text{Err}_{\text{offset}})^2 + (\text{Err}_{\text{INL}})^2} \quad (1)$$

哪里

- Err_{gain} 是ADC的最大增益误差，单位为LSBs
- $\text{Err}_{\text{offset}}$ 是ADC的最大偏移误差，单位为LSBs
- Err_{INL} 是ADC的最大INL误差，单位为LSBs

C2000 ADC的规格说明示例以及参数可见于表2-1，数据表中对此表的动态链接位于此处。

所有C2000设备在数据表中列出的参数的最后一个方面是参数本身包含所隐含的内容。对于具有MIN/MAX的参数，这些是设备在整个工作范围和寿命期间的保证规格。典型（TYP）列对所有参数也都很重要，因为它代表了参数在其工作范围内的平均性能。

Table 2-1. TMS320F28379D 16-Bit ADC Specifications

Parameter	Test Conditions	Min	Typ	Max	Unit
ADC conversion cycles		29.6		31	ADCCLKs
Power-up time (after setting ADCPWDNZ to first conversion)				500	μs
Gain error		−64	±9	64	LSBs
Offset error		−16	±9	16	LSBs
Channel-to-channel gain error			±6		LSBs
Channel-to-channel offset error			±3		LSBs
ADC-to-ADC gain error	Identical V _{REFHI} and V _{REFLO} for all ADCs		±6		LSBs
ADC-to-ADC offset error	Identical V _{REFHI} and V _{REFLO} for all ADCs		±3		LSBs
DNL		> −1	±0.5	1	LSBs
INL		−3	±1.5	3	LSBs
SNR	V _{REFHI} = 2.5 V, f _{in} = 10 kHz		87.6		dB
THD	V _{REFHI} = 2.5 V, f _{in} = 10 kHz		−93.5		dB
SFDR	V _{REFHI} = 2.5 V, f _{in} = 10 kHz		95.4		dB
SINAD	V _{REFHI} = 2.5 V, f _{in} = 10 kHz		86.6		dB
ENOB	V _{REFHI} = 2.5 V, f _{in} = 10 kHz, single ADC		14.1		bits
	V _{REFHI} = 2.5 V, f _{in} = 10 kHz, synchronous ADCs		14.1		
	V _{REFHI} = 2.5 V, f _{in} = 10 kHz, asynchronous ADCs		Not supported		
PSRR	V _{DDA} = 3.3-V DC + 200 mV DC up to Sine at 1 kHz		77		dB
PSRR	V _{DDA} = 3.3-V DC + 200 mV Sine at 800 kHz		74		dB
CMRR	DC to 1 MHz		60		dB
V _{REFHI} input current			190		μA
ADC-to-ADC isolation	V _{REFHI} = 2.5 V, synchronous ADCs	−2		2	LSBs
	V _{REFHI} = 2.5 V, asynchronous ADCs		Not supported		

2.1.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

2.1.4 Hardware Platforms and Software Examples

All controlCARDs for their specific C2000 MCU have been verified to reproduce the DS specifications for the on-chip ADC

- [TMDSCNCD28388D](#)
- [TMDSCNCD28379D](#)
- [TMDSCNCD280049C](#)
- [TMDSCNCD280025](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)

表2-1。TMS320F28379D 16位ADC规格

参数	测试条件	Min	Typ	Max	Unit
ADC转换周期		29.6		31	ADCCLKs
上电时间（设置后） ADCPWDNZ 至首次转换)				500	μs
增益误差		−64	±9	64	LSBs
偏移误差		−16	±9	16	LSBs
通道间增益误差			±6		LSBs
通道间偏移误差			±3		LSBs
ADC 至 ADC 增益误差	所有 ADC 的 V _{REFHI} 和 V _{REFLO} 相同		±6		LSBs
ADC 之间的偏移误差	相同的 V _{REFHI} 和 V _{REFLO} 适用于所有 ADCs		±3		LSBs
DNL		> −1	±0.5	1	LSBs
INL		−3	±1.5	3	LSBs
SNR	V _{REFHI} = 2.5 V, f _{in} = 10 kHz		87.6		dB
THD	V _{REFHI} = 2.5 V, f _{in} = 10 kHz		−93.5		dB
SFDR	V _{REFHI} = 2.5 V, f _{in} = 10 kHz		95.4		dB
SINAD	V _{REFHI} = 2.5 V, f _{in} = 10 kHz		86.6		dB
ENOB	V _{REFHI} = 2.5 V, f _{in} = 10 kHz, single ADC		14.1		bits
	V _{REFHI} = 2.5 V, f _{in} = 10 kHz, 同步 ADC		14.1		
	V _{REFHI} = 2.5 V, f _{in} = 10 kHz, 异步 ADC		Not 支持		
PSRR	V _{DDA} = 3.3-V DC + 200 mV DC 至 1 kHz 正弦波		77		dB
PSRR	V _{DDA} = 3.3-V DC + 200 mV 800 kHz 的正弦波		74		dB
CMRR	DC至1 MHz		60		dB
V _{REFHI} 输入电流			190		μA
ADC至ADC隔离	V _{REFHI} = 2.5 V, 同步ADC	−2		2	LSBs
	V _{REFHI} = 2.5 V, 异步 ADCs		Not 支持		

2.1.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

2.1.4 硬件平台和软件示例

所有针对其特定C2000微控制器的controlCARDs均已验证能够重现片上ADC的DS规格

- [TMDSCNCD28388D](#)
- [TMDSCNCD28379D](#)
- [TMDSCNCD280049C](#)
- [TMDSCNCD280025](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)

2.1.5 Documentation

- Texas Instruments: [TMS320F2838x Real-Time Microcontrollers With Connectivity Manager Data Sheet](#) (see the *C28x Analog Peripherals* section)
- Texas Instruments: [TMS320F2837xD Dual-Core Microcontrollers Data Sheet](#) (see the *Analog Peripherals* section)
- Texas Instruments: [TMS320F28004x Microcontrollers Data Sheet](#) (see the *Analog Peripherals* section)
- Texas Instruments: [TMS320F28003x Microcontrollers Data Sheet](#) (see the *Analog Peripherals* section)
- Texas Instruments: [TMS320F28002x Microcontrollers Data Sheet](#) (see the *Analog Peripherals* section)
- Texas Instruments: [TMS320F280013x Microcontrollers Data Sheet](#) (see the *Analog Peripherals* section)
- Texas Instruments: [TMS320F280015x Microcontrollers Data Sheet](#) (see the *Analog Peripherals* section)

2.2 Optimizing Acquisition Time vs Circuit Complexity for Analog Inputs

2.2.1 Value Proposition

Control systems have the need to interface with a variety of feedback and monitoring sources. Signal sources in these systems differ in their ability to drive a capacitive input circuit like those typically found in the sample-and-hold (S+H) input circuit of an analog-to-digital converter (ADC). The ADCs on C2000 devices allow the acquisition time of the S+H to be individually configured for each input channel over a wide range. This allows the system to simultaneously interface with a mix of high-performance and low-cost signal sources.

2.2.2 In Depth

The inputs of an ADC are typically modeled as a switched capacitor circuit where the hold capacitor inside the ADC, C_h , needs to be charged from an unknown voltage to a value close to the input voltage during the acquisition time. An example, taken from the [TMS320F2837xD](#) device, is shown in [Figure 2-1](#).

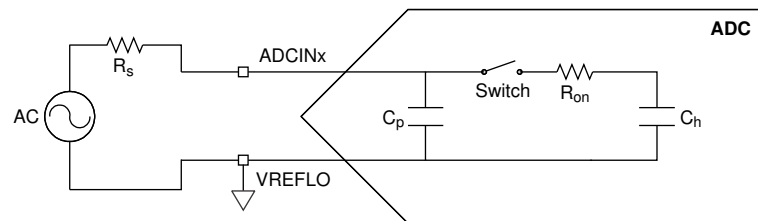


Figure 2-1. Single-Ended Input Model

The required acquisition time for charging C_h is determined by the external impedance of passive components, bandwidth of any buffers or sensors, the internal ADC input parasitics, and the resolution of the ADC.

The system designer can make a variety of trade-offs with respect to external circuit cost and complexity vs settling speed, for example:

- **Adding/upgrading the op-amp buffer driving the ADC inputs:** Lowering acquisition time through better charge transfer to the sample and hold capacitor inside the ADC
- **Increasing the amount of resistance and/or capacitance seen by the ADC input:** Helps reduce noise by adding additional low-pass filtering at the expense of a longer acquisition time
- **Tolerating less accuracy:** Alternatively, using a smaller acquisition window to decrease the sampling time, at the expense of accuracy/resolution.

With all the above possible trade-offs, it is difficult to select a single acquisition time that is appropriate for all analog inputs in the system. [C2000 ADCs](#) allow a separate acquisition window to be selected for each channel, giving the system designer a great deal of flexibility to make whatever speed vs signal conditioning circuit cost vs accuracy trade-offs they would like.

2.1.5 文档

- 德州仪器: TMS320F2838x 带连接管理器的实时微控制器数据表 (参见C28x 模拟外设部分)
- 德州仪器: TMS320F2837xD 双核微控制器数据表 (参见模拟外设部分)
- 德州仪器: TMS320F28004x 微控制器数据表 (参见模拟外设部分)
- 德州仪器: TMS320F28003x 微控制器数据表 (参见模拟外设部分)
- 德州仪器: TMS320F28002x 微控制器数据表 (参见模拟外设部分)
- 德州仪器: TMS320F280013x 微控制器数据表 (参见模拟外设部分)
- 德州仪器: TMS320F280015x 微控制器数据表 (参见模拟外设部分)

2.2 优化模拟输入的采集时间与电路复杂度

2.2.1 价值主张

控制系统需要与各种反馈和监控源接口。这些系统中的信号源在驱动电容输入电路 (如通常在模数转换器 (ADC) 的采样保持 (S+H) 输入电路中发现的电路) 的能力上有所不同。C2000设备上的ADC允许S+H的采集时间针对每个输入通道进行单独配置。这允许系统同时与高性能和低成本信号源混合接口。

2.2.2 深入了解

ADC的输入通常建模为开关电容电路, 其中ADC内部的保持电容 C_h 需要在采集时间内从未知电压充电到接近输入电压的值。一个例子取自TMS320F2837xD设备, 如图2-1所示。

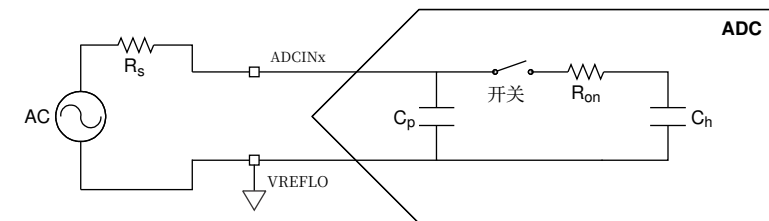


图2-1. 单端输入模型

充电 C_h 所需的采集时间由无源元件的外部阻抗、任何缓冲器或传感器的带宽、内部 ADC 输入寄生参数以及 ADC 的分辨率决定。

系统设计者可以在外部电路成本和复杂性相对于建立速度方面做出各种权衡, 例如:

- 为 ADC 输入添加/升级运算放大器缓冲器: 通过更好的电荷转移至 ADC 内部的采样和保持电容来降低采集时间
- 增加 ADC 输入看到的电阻和/或电容的量: 通过在牺牲更长采集时间的情况下添加额外的低通滤波来帮助减少噪声
- 容忍较低的精度: 或者, 使用更小的采集窗口来减少采样时间, 以牺牲精度/分辨率为代价。

在所有上述可能的权衡中, 为系统中的所有模拟输入选择一个合适的单一采集时间非常困难。C2000 ADC 允许为每个通道选择一个单独的采集窗口, 这为系统设计者提供了很大的灵活性, 使他们可以根据需要做出速度与信号调理电路成本与精度之间的权衡。

The acquisition window (controlled by the ACQPS field of the ADC SOC configuration register) can also be configured over a wide range of values and with a small step size as shown in [Table 2-2](#).

Table 2-2. Range of Acquisition Time Configuration (per Channel)

C2000 MCU	Device SYSCLK	Minimum S+H Time	Maximum S+H Time	S+H Time Configuration Resolution
TMS320F28004x and TMS320F28002x	100 MHz	80 ns	5.1 μ s	10.00 ns
TMS320F2807x , TMS320F28003x , TMS320F280013x , TMS320F280015x	120 MHz	75 ns	4.3 μ s	8.33 ns
TMS320F2837xD and TMS320F2837xS	200 MHz	75 ns	2.6 μ s	5.00 ns

There are a variety of ways to model the ADC input, [Texas Instruments offers free tools to help design the ADC input driver circuit](#) as well as [instructional videos on proper front end component selection](#).

2.2.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

2.2.4 Hardware Platforms and Software Examples

- [TMDSCNCDF28388D](#)
- [TMDSCNCDF28379D](#)
- [TMDSCNCDF280049C](#)
- [TMDSCNCD280039C](#)
- [TMDSCNCD280025](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)
- [F2838xD ADC SW Example](#)
- [F2837xD ADC SW Example](#)
- [F28004x ADC SW Example](#)
- [F28002x ADC SW Example](#)
- [F280013x ADC SW Example](#)
- [F280015x ADC SW Example](#)

2.2.5 Documentation

- [Charge-Sharing Driving Circuits for C2000 ADCs](#)
- [ADC Input Circuit Evaluation for C2000 MCUs](#)
- [SAR ADC Input Driver Design](#)
- [TI Precision Labs - ADCs: Introduction to SAR ADC Front-End Component Selection](#)
- [TMS320F2838xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Choosing an Acquisition Window Duration* section)
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Choosing an Acquisition Window Duration* section)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Choosing an Acquisition Window Duration* section)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Choosing an Acquisition Window Duration* section)

采集窗口（由 ADC SOC 配置寄存器的 ACQPS 字段控制）也可以在很宽的范围内配置，并且具有小的步长，如表 2-2 所示。

表2-2。采集时间配置范围（每个通道）

C2000 MCU	设备 SYSCLK	最小S+H时间	最大S+H Time	S+H 时间配置分辨率
TMS320F28004x 和 TMS320F28002x	100 MHz	80 ns	5.1 μ s	10.00 ns
TMS320F2807x , TMS320F28003x , TMS320F280013x , TMS320F280015x	120 MHz	75 ns	4.3 μ s	8.33 ns
TMS320F2837xD 和 TMS320F2837xS	200 MHz	75 ns	2.6 μ s	5.00 ns

有多种方法可以建模ADC输入，德州仪器提供免费工具来帮助设计ADC输入驱动电路，以及关于正确选择前端组件的教学视频。

2.2.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

2.2.4 硬件平台和软件示例

- [TMDSCNCDF28388D](#)
- [TMDSCNCDF28379D](#)
- [TMDSCNCDF280049C](#)
- [TMDSCNCD280039C](#)
- [TMDSCNCD280025](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)
- [F2838xD ADC SW 示例](#)
- [F2837xD ADC SW 示例](#)
- [F28004x ADC SW 示例](#)
- [F28002x ADC SW 示例](#)
- [F280013x ADC SW 示例](#)
- [F280015x ADC SW 示例](#)

2.2.5 文档

- [C2000 ADC 的电荷分配驱动电路](#)
- [C2000 MCU 的 ADC 输入电路评估](#)
- [SAR ADC 输入驱动器设计](#)
- [TI 精密实验室 - ADCs：SAR ADC 前端组件选择入门](#)
- [TMS320F2838xD 实时微控制器技术参考手册](#)（更多信息，请参阅选择采集窗口持续时间部分）
- [TMS320F2837xD 实时微控制器技术参考手册](#)（更多信息，请参阅选择采集窗口持续时间部分）
- [TMS320F28004x 实时微控制器技术参考手册](#)（更多信息，请参阅选择采集窗口持续时间部分）
- [TMS320F28003x 实时微控制器技术参考手册](#)（更多信息，请参阅选择采集窗口持续时间部分）

- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Choosing an Acquisition Window Duration* section)
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Choosing an Acquisition Window Duration* section)
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Choosing an Acquisition Window Duration* section)

2.3 Hardware Based Monitoring of Dual-Thresholds Using a Single Pin Reference

2.3.1 Value Proposition

C2000 MCUs help to mitigate the cost and complexity of using multiple comparators to monitor a feedback signal by providing single pin access to two embedded voltage comparators per each Comparator Subsystem (CMPSS) module.

2.3.2 In Depth

Control systems commonly use voltage comparators to monitor feedback signals for threshold crossing events. These crossing events can represent a variety of states that range from nominal to critical conditions. A single feedback signal may sometimes be monitored by multiple comparators in order to trigger a custom response for each state.

Consider a simple hysteresis controller ([Figure 2-2](#)) that uses two threshold levels to define the actuation on-off behavior:

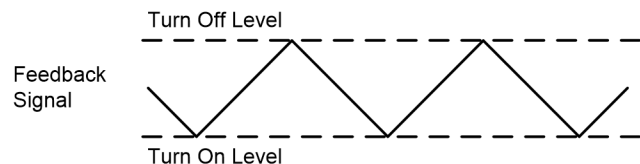


Figure 2-2. Threshold Levels in a Hysteresis Controller

1. A low-level "floor" threshold defines when the actuation should turn on, and
2. A high-level "ceiling" threshold defines when the actuation should turn off

A comparator-based monitoring and triggering scheme for this hysteresis controller can be implemented using a single CMPSS pin as shown in [Figure 2-3](#). Similarly, a second CMPSS pin can be used to detect both over-voltage and under-voltage fault conditions for system protection.

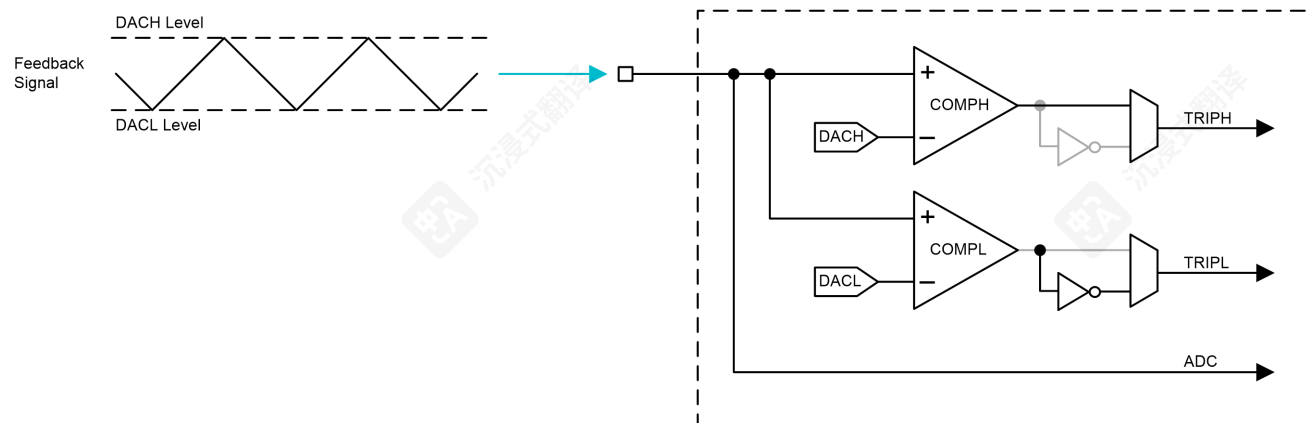


Figure 2-3. CMPSS Block Diagram

Each CMPSS comparator is provided with its own voltage reference DAC, output conditioning logic, and unique trip signals for independent operation. Additionally, each CMPSS pin is also assigned to an ADC channel that can sample the pin voltage in parallel with comparator monitoring. These ADC samples can be used to influence sophisticated system behaviors, and to serve as a redundant form of voltage monitoring.

- TMS320F28002x 实时微控制器技术参考手册（更多信息，请参阅选择采集窗口持续时间部分）
- TMS320F280013x 实时微控制器技术参考手册（更多信息，请参阅选择采集窗口持续时间部分）
- TMS320F280015x 实时微控制器技术参考手册（更多信息，请参阅选择采集窗口持续时间部分）

2.3 基于硬件的双阈值单引脚参考监控

2.3.1 价值主张

C2000 MCUs 帮助降低使用多个比较器来监控反馈信号的成本和复杂性，通过为每个比较器子系统（CMPSS）模块提供每个模块两个嵌入式电压比较器的单引脚访问。

2.3.2 深入解析

控制系统通常使用电压比较器来监控反馈信号以检测阈值跨越事件。这些跨越事件可以表示从正常到临界条件等各种状态。单个反馈信号有时可能由多个比较器监控，以便为每种状态触发自定义响应。

考虑一个简单的迟滞控制器（图2-2），它使用两个阈值来定义执行器的开-关行为：

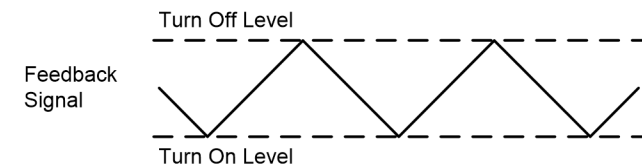


图2-2。迟滞控制器中的阈值

1. 一个低电平"地板"阈值定义了执行器何时应该开启，并且
2. 一个高电平"天花板"阈值定义了执行器何时应该关闭

基于比较器的监控和触发方案可用于此迟滞控制器，如图2-3所示。类似地，第二个CMPSS引脚可用于检测过压和欠压故障条件以进行系统保护。

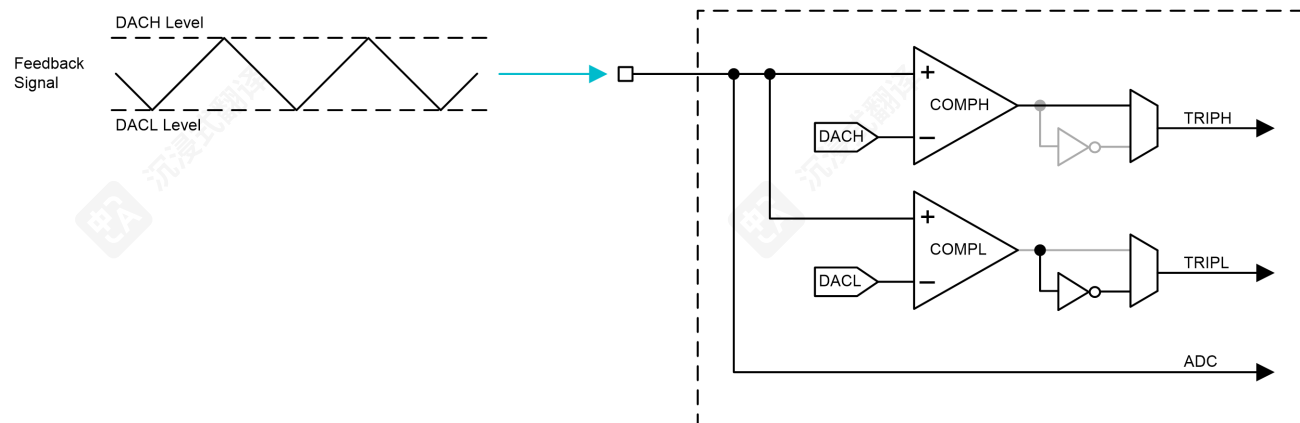


图2-3. CMPSS框图

每个CMPSS比较器都配备了独立的电压参考DAC、输出调理逻辑和独特的跳闸信号，以实现独立运行。此外，每个CMPSS引脚也被分配到一个ADC通道，该通道可以与比较器监控并行采样引脚电压。这些ADC采样值可用于影响复杂的系统行为，并作为电压监控的冗余形式。

The multi-function capability of the CMPSS pin demonstrates a significant advantage in resource optimization over other embedded solutions with dedicated pin functionality; the optimization advantage is even greater when compared to discrete solutions that require localized resources like power supplies and reference voltages. System cost and complexity can be reduced by utilizing the full resources available from each C2000 MCU pin.

2.3.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

2.3.4 Hardware Platforms and Software Examples

- [TIDM-DC-DC-BUCK](#)
- [TIDM-DC-DC-BUCK Example SW](#)
- [TIDM-02002](#)
- [TIDM-02002 Example SW](#)
- [TIDM-1022](#)
- [TIDM-1022 Software Example](#)
- [TMDXIDDK379D](#)
- [TMDXIDDK379D Software Examples](#)

2.3.5 Documentation

- [TMS320F2838x Real-Time Microcontrollers With Connectivity Manager Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F2837xD Dual-Core Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F28004x Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F28003x Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F28002x Real-Time Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F280013x Real-Time Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F280015x Real-Time Microcontrollers Data Sheet](#)(for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F2838xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#) (see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#) (see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#)(see the *Comparator Subsystem (CMPSS)* chapter)

CMPSS引脚的多功能能力在资源优化方面显著优于其他具有专用引脚功能的嵌入式解决方案；与需要本地化资源（如电源和参考电压）的离散解决方案相比，优化优势更为明显。通过利用每个C2000 MCU引脚提供的全部资源，可以降低系统成本和复杂性。

2.3.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

2.3.4 硬件平台和软件示例

- [TIDM-DC-DC-BUCK](#)
- [TIDM-DC-DC-BUCK 示例软件](#)
- [TIDM-02002](#)
- [TIDM-02002 示例软件](#)
- [TIDM-1022](#)
- [TIDM-1022 软件示例](#)
- [TMDXIDDK379D](#)
- [TMDXIDDK379D 软件示例](#)

2.3.5 文档

- [TMS320F2838x 实时微控制器带连接管理器数据表](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F2837xD 双核微控制器数据表](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F28004x 微控制器数据表](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F28003x 微控制器数据表](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F28002x 实时微控制器数据表](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F280013x 实时微控制器数据手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F280015x 实时微控制器数据手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F2838xD 实时微控制器技术参考手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F2837xD 实时微控制器技术参考手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F28004x 实时微控制器技术参考手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F28003x 实时微控制器技术参考手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F28002x 实时微控制器技术参考手册](#)（请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F280013x 实时微控制器技术参考手册](#)（请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F280015x 实时微控制器技术参考手册](#)(请参阅比较器子系统 (CMPSS) 章节)

2.4 Resolving Tolerance and Aging Effects During ADC Sampling

2.4.1 Value Proposition

The ADC result is often manipulated mathematically before its used in the control law of a given system. This is typically done with some additional operations by the CPU, adding increased latency to the system as well as loading the CPU for such operations. The C2000 MCU as the ability to correct for this in hardware with no CPU overhead and no impact to ADC sample rate.

2.4.2 In Depth

Prior to using the ADC result in control calculations, it is often necessary to remove any known offset introduced by external factors such as component tolerances or layout differences (Table 2-3). While aspects of the above issues can be partially addressed through PCB layout or choosing higher tolerance/stable resistors there are always deviations from the ideal.

Table 2-3. Typical Resistor Tolerance Over Time and System Impact

Life Cycle Stage	Total Tolerance	Associated 12-Bit Error
Purchase	±0.05%	±2 LSBs
Post Assembly	±0.5%	±20 LSBs
Post Storage/Moisture	±0.75%	±30 LSBs
Temp Coeff and EOL	±1.00%	± 40 LSBs

C2000 MCUs implement an integrated hardware block to correct up to a 10-bit signed value co-incident to the ADC conversion process, saving valuable cycles in the system. The cycle value to the system is effectively doubled, as the ADC sample rate is maintained and no CPU cycles are used to perform the correction. Saturation is built in as well.

For the implementation of the offset correction in addition to the other modules included in the ADC Post Processing Block, see Figure 2-4.

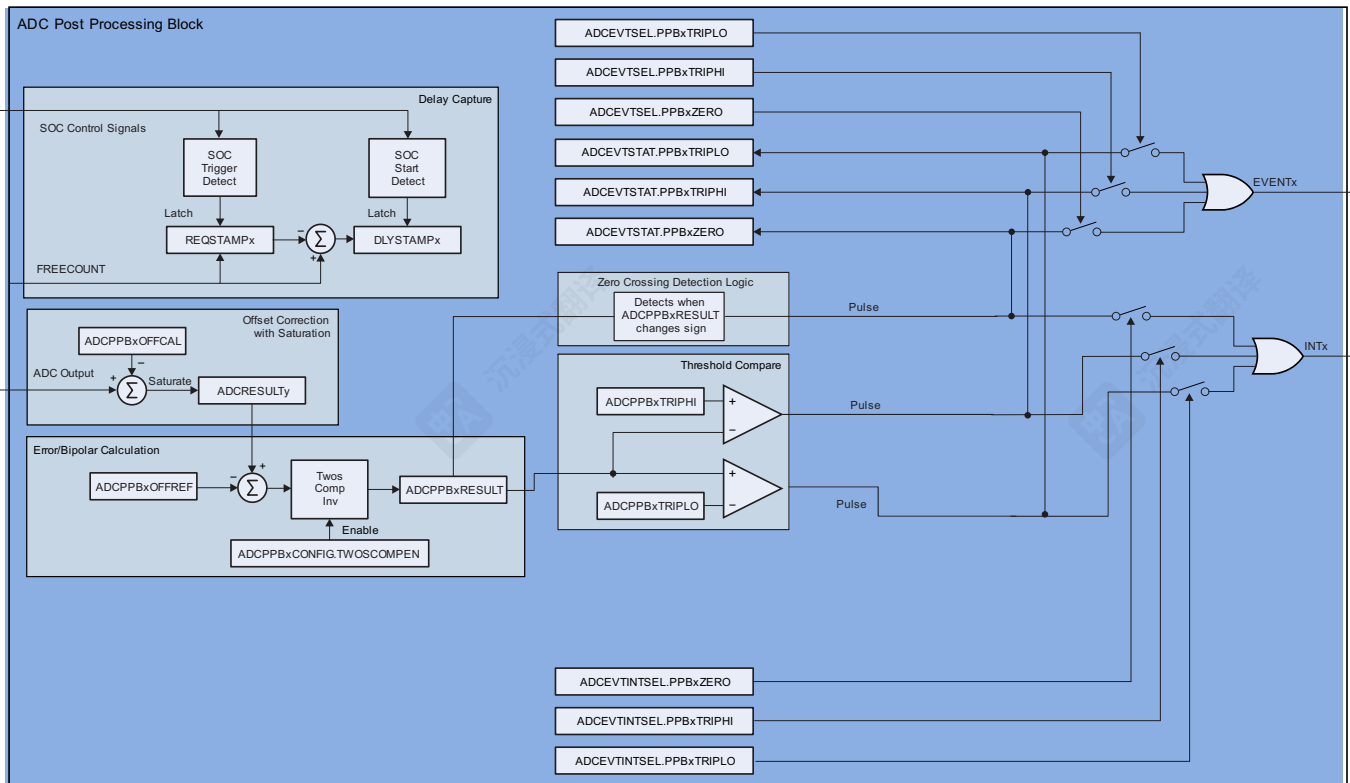


Figure 2-4. ADC Post Processing Block on TMS320F2837xD

2.4 在ADC采样期间解决容差和老化效应

2.4.1 价值主张

ADC结果在使用于给定系统的控制律之前，通常会被进行数学上的操作。这通常由CPU通过一些额外的操作来完成，从而增加了系统的延迟，并给CPU带来负载。C2000微控制器能够在硬件中纠正这一问题，无需CPU开销，且不会影响ADC采样率。

2.4.2 深入探讨

在使用ADC结果进行控制计算之前，通常需要消除由外部因素（如组件容差或布局差异）引入的已知偏移（见表2-3）。虽然上述问题的某些方面可以通过PCB布局或选择更高容差/更稳定的电阻来部分解决，但始终存在与理想值的偏差。

表2-3. 典型电阻随时间推移的容差和系统影响

生命周期阶段	总容差	关联12位错误
采购	±0.05%	±2 LSBs
组装后	±0.5%	±20 LSBs
组装后储存/湿度	±0.75%	±30 LSBs
温度系数和EOL	±1.00%	± 40 LSBs

C2000 MCU实现了一个集成硬件模块，用于校正与ADC转换过程同时发生的10位有符号值，从而节省系统中的宝贵周期。由于ADC采样率保持不变且未使用CPU周期执行校正，系统周期值实际上翻倍。饱和度也已内置。

除 ADC 后处理模块中包含的其他模块外，关于偏移校正的实现，请参见图 2-4。

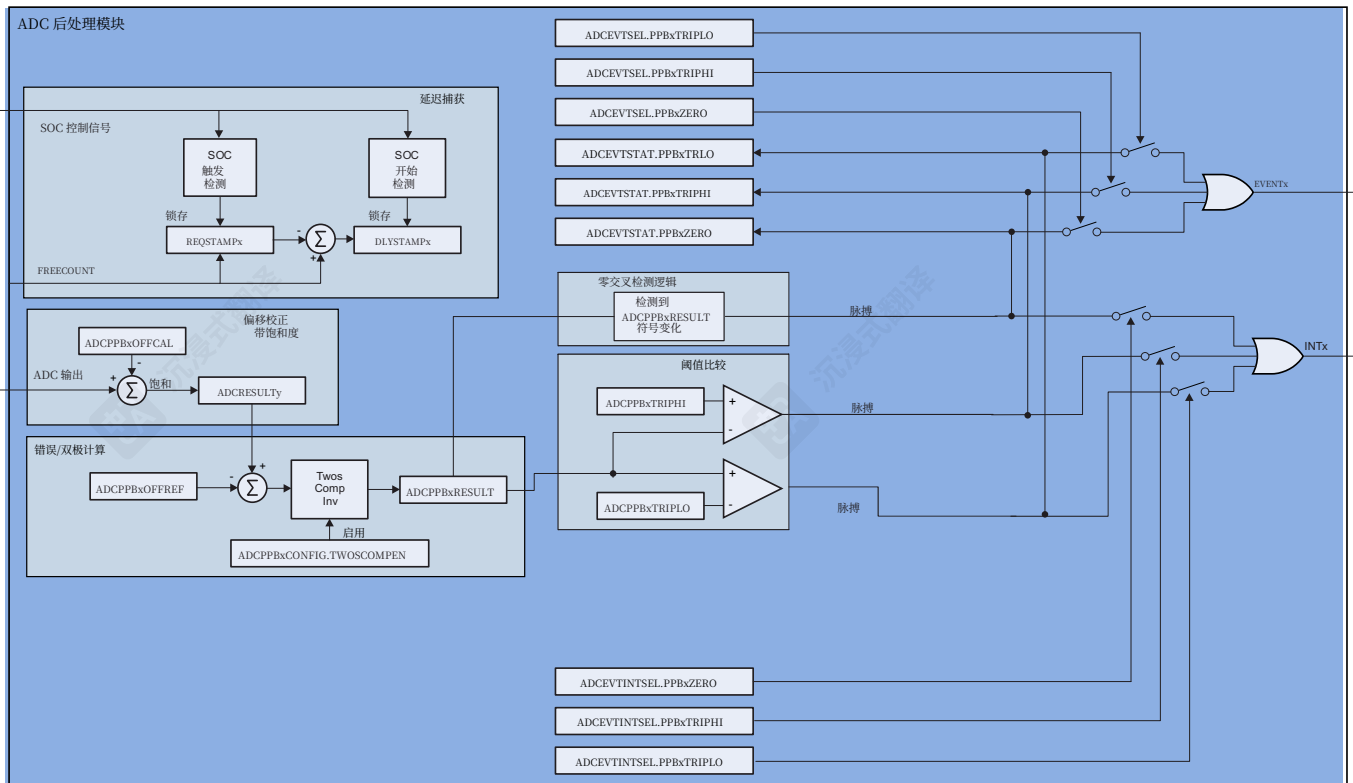


图2-4。TMS320F2837xD上的ADC后处理模块

2.4.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

2.4.4 Hardware Platforms and Software Examples

- [TMDSCNCDF28388D](#)
- [TMDSCNCDF28379D](#)
- [TMDSCNCDF280049C](#)
- [TMDSCNCD280039C](#)
- [TMDSCNCD280025](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)
- [F2838xD ADC PPB SW Example](#)
- [F2837xD ADC PPB SW Example](#)
- [F28004x ADC PPB SW Example](#)
- [F28003x ADC PPB SW Example](#)
- [F28002x ADC PPB SW Example](#)
- [F280013x ADC PPB SW Example](#)
- [F280015x ADC PPB SW Example](#)

2.4.5 Documentation

- [TMS320F2838xD Real-Time Microcontrollers Technical Reference Manual](#) - see the *Post-Processing Blocks* section in the *Analog-to-Digital Converter (ADC)* chapter
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#) - see the *Post-Processing Blocks* section in the *Analog-to-Digital Converter (ADC)* chapter
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#) - see the *Post-Processing Blocks* section in the *Analog-to-Digital Converter (ADC)* chapter
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#) - see the *Post-Processing Blocks* section in the *Analog-to-Digital Converter (ADC)* chapter
- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#) - see the *Post-Processing Blocks* section in the *Analog-to-Digital Converter (ADC)* chapter
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#) - see the *Post-Processing Blocks* section in the *Analog-to-Digital Converter (ADC)* chapter
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#) -see the *Post-Processing Blocks* section in the *Analog-to-Digital Converter (ADC)* chapter

2.5 Realizing Rotary Sensing Solutions Using C2000 Configurable Logic Block

2.5.1 Value Proposition

As mentioned in the article: On-Chip Hardware Customization ([Section 4.3](#)), the C2000 Configurable Logic Block (CLB) technology enables systems designers to implement new logic within the C2000 device and eliminate external circuitry. A specific example for a real time control system is an integrated interface to digital rotary encoders; using the CLB, developers have the freedom to integrate an industry-standard encoder communications protocol or an application-specific customized protocol inside the C2000 MCU without additional external circuitry.

2.4.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

2.4.4 硬件平台和软件示例

- [TMDSCNCDF28388D](#)
- [TMDSCNCDF28379D](#)
- [TMDSCNCDF280049C](#)
- [TMDSCNCD280039C](#)
- [TMDSCNCD280025](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)
- [F2838xD ADC PPB SW 示例](#)
- [F2837xD ADC PPB SW 示例](#)
- [F28004x ADC PPB SW 示例](#)
- [F28003x ADC PPB SW 示例](#)
- [F28002x ADC PPB SW 示例](#)
- [F280013x ADC PPB SW 示例](#)
- [F280015x ADC PPB SW 示例](#)

2.4.5 文档

- TMS320F2838xD 实时微控制器技术参考手册 - 请参阅模拟数字转换器 (ADC) 章节中的后处理模块部分
- TMS320F2837xD 实时微控制器技术参考手册 - 请参阅模拟数字转换器 (ADC) 章节中的后处理模块部分
- TMS320F28004x 实时微控制器技术参考手册 - 请参阅后处理模块模拟数字转换器 (ADC) 章节中的部分
- TMS320F28003x 实时微控制器技术参考手册 - 请参阅后处理模块在模数转换器 (ADC) 章节中的部分
- TMS320F28002x 实时微控制器技术参考手册 - 请参阅后处理模块在模数转换器 (ADC) 章节中的部分
- TMS320F280013x 实时微控制器技术参考手册 - 请参阅模数转换器 (ADC) 章节中的后处理模块部分
- TMS320F280015x 实时微控制器技术参考手册 - 请参阅模数转换器 (ADC) 章节中的后处理模块部分

2.5 使用 C2000 可配置逻辑块实现旋转传感解决方案

2.5.1 价值主张

如文章中所述：片上硬件定制（第 4.3 节），C2000 可配置逻辑块 (CLB) 技术使系统设计人员能够在 C2000 器件内部实现新逻辑，并消除外部电路。一个实时控制系统中的具体示例是集成接口到数字旋转编码器；使用 CLB，开发者可以自由地在 C2000 MCU 内部集成行业标准编码器通信协议或特定于应用的定制协议，而无需额外的外部电路。

2.5.2 In Depth

A digital rotary encoder is a device that converts the position of a shaft to a digital signal. There are two main types of encoders:

- Absolute Encoders (Figure 2-5): The output of an absolute encoder indicates the current angular position in a message sent back to the master as defined by a particular protocol.
- Incremental Encoders: The output of an incremental encoder provides information in a train of modulated pulses that are typically further processed by the system master into information such as speed, distance and position.

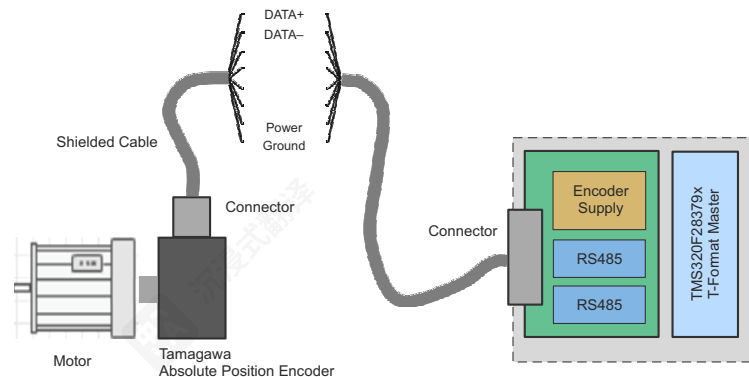


Figure 2-5. Industrial Servo Drive With T-Format Absolute Position Encoder Interface

C2000 CLB technology enables an integrated solution to interface to the most popular digital rotary position encoders, eliminating the necessity for external field-programmable gate arrays (FPGAs) or application-specific integrated circuits (ASICs). The [PositionManager BoosterPack](#) plug-in module is a flexible low voltage platform intended for evaluating various encoder interfaces and designed to work with multiple C2000 LaunchPad development kits such as the [LAUNCHXL-F28379D](#), [LAUNCHXL-F280049C](#), or the [LAUNCHXL-F280025C](#).

Below are examples of encoders that are available for evaluation today as part of the [Motor Control SDK](#) software package. Future updates are planned to add examples for both the BiSS-C and EnDAT22 protocols.

- T-Format Absolute Encoder Interface:

The Tamagawa T-Format protocol is a popular digital, bidirectional interface for absolute encoders. The easy-to-use library and example software delivered with [TIDM-1011](#) demonstrates Tamagawa's T-Format standard. In this example, the T-Format absolute encoder interface is integrated into the C2000 using on-chip resources such as the CLB, SPI, and GPIO as shown in [Figure 2-5](#). This TI Design includes the following features:

- A T-Format encoder interface library and CRC library that implements protocol commands
- A demonstration project (with full source code access) to exercise the T-Format commands
- Fully supported by the C2000 [CLB Tool](#) integrated inside [Code Composer Studio](#)

- T-Format Evaluation in High-Bandwidth Current Loop Applications:

With the recent C2000 MCUs such as [TMS320F2837x](#) and [TMS320F28004x](#), it is possible to implement Fast Current Loop (FCL) algorithms that provide a high current loop bandwidth with the same external hardware as used in classical Field Oriented Control (FOC) methods. TI has developed the FCL algorithm on these MCUs and implemented it on the [DesignDRIVE IDDK](#) platform. The T-Format encoder interface has been integrated into an evaluation implementation of FCL algorithms on C2000 devices.

[Quick Response Control of PMSM Using Fast Current Loop](#) studies the frequency response analysis of current loops in real time and also verifies the interface logic for the T-format encoder interface implemented in [TIDM-1011](#). The position loop in this example can be closed using a QEP encoder or a T-format encoder and FCL can be implemented in both cases.

2.5.2 深入了解

数字旋转编码器是一种将轴的位置转换为数字信号的设备。编码器主要有两种类型：

- 绝对编码器（图2-5）：绝对编码器的输出表示当前角位置，并通过特定协议发送回主设备。
- 增量编码器：增量编码器的输出提供一系列调制脉冲的信息，系统主设备通常进一步处理这些信息，以获取速度、距离和位置等数据。

图2-5。工业伺服驱动与T格式绝对位置编码器接口

C2000 CLB技术能够提供集成解决方案，以接口到最流行的数字旋转位置编码器，消除了对外部现场可编程门阵列（FPGA）或专用集成电路（ASIC）的需求。PositionManager BoosterPack插件模块是一个灵活的低电压平台，用于评估各种编码器接口，并设计用于与多个C2000 LaunchPad开发套件（如LAUNCHXL-F28379D、LAUNCHXL-F280049C或LAUNCHXL-F280025C）配合使用。

以下是作为Motor Control SDK软件包一部分可供今天评估的编码器示例。计划在未来更新中添加BiSS-C和EnDAT22协议的示例。

- T格式绝对编码器接口：

Tamagawa T格式协议是一种流行的数字双向绝对编码器接口。随TIDM-1011提供的易于使用的库和示例软件展示了Tamagawa的T格式标准。在此示例中，T格式绝对编码器接口通过CLB、SPI和GPIO等片上资源集成到C2000中，如图2-5所示。此TI设计包括以下功能：

- 一个实现协议命令的T-Format编码器接口库和CRC库
- 一个演示项目（可访问完整源代码）用于练习T-Format命令
- 完全由集成在Code Composer Studio内的C2000 CLB工具支持

- T-Format在高带宽电流环应用中的评估：

随着最新的C2000 MCU如TMS320F2837x和TMS320F28004x，可以实现提供与经典磁场定向控制（FOC）方法相同的外部硬件的高电流环（FCL）算法的快速电流环（FCL）算法。TI在这些MCU上开发了FCL算法，并在DesignDRIVE IDDK平台上实现了它。T-Format编码器接口已集成到C2000设备的FCL算法评估实现中。

快速响应控制永磁同步电机（PMSM）使用快速电流环研究实时电流环的频率响应分析，并验证了在TIDM-1011中实现的T格式编码器接口的接口逻辑。此示例中的位置环可以使用QEP编码器或T格式编码器关闭，并且FCL在这两种情况下都可以实现。

Pulse Train Output (PTO) QepDiv and PulseGen:

Incremental rotary encoders output a pulse train to indicate that the shaft being monitored has moved. The system master typically processes this pulse train to determine information such as speed, distance and position. In the QepDiv implementation, position information is sent from the encoder to the Enhanced Quadrature Encoder Pulse (eQEP) module on a C2000 MCU. In the PulseGen case a custom pulse stream is generated to meet the system needs. The PTO-QepDiv example demonstrates how the CLB can be used to generate a divided pulse stream from these eQEP inputs as shown in Figure 2-6. The divided pulse stream can then be sent to another device in the system. Both the QepDiv and PulseGen examples are documented in the [C2000 Position Manager PTO API Reference Guide](#).

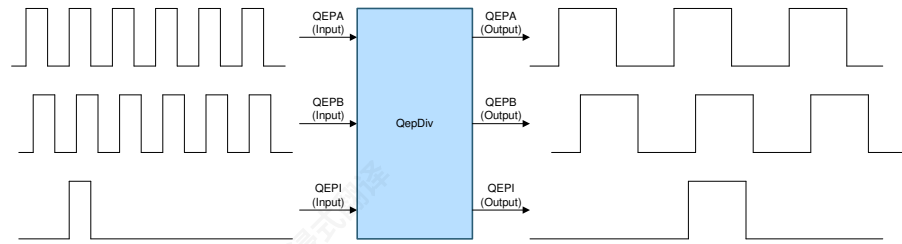


Figure 2-6. QepDiv Input and Output Diagram

2.5.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

2.5.4 Hardware Platforms and Software Examples

- [Tamagawa T-Format Absolute-Encoder Master Interface Reference Design for C2000™ MCUs](#)
- [C2000 Position Manager PTO API Reference Guide](#)
- [MotorControl software development kit \(SDK\) for C2000 MCUs](#)
- [Position Manager BoosterPack \(BOOSTXL-POSMGR\)](#)
- [C2000™ DesignDRIVE Development Kit for Industrial Motor Control \(TMDXIDDK379D\)](#)

2.5.5 Documentation

- [Training: How the C2000 Configurable Logic Block \(CLB\) tool integrates custom logic in my design](#)
- [CLB Tool User's Guide](#)
- [Designing With the C2000 Configurable Logic Block \(CLB\)](#)
- [How to Migrate Custom Logic From an FPGA/CPLD to C2000 Microcontrollers](#)
- [Fast Current Loop Library](#)

2.6 Smart Sensing Across An Isolation Boundary

2.6.1 Value Proposition

High voltage control systems often make use of a sigma-delta ADC in order to easily pass the analog information of the high voltage domain to the lower voltage domain where the MCU exists. Before the data stream can be used, it must be processed by a filter and demodulator. This logic exists on the C2000 MCU and is called the Sigma Delta Filter Module (SDFM) module. The high and low comparators inside the SDFM on a C2000 MCU can actuate the PWMs without CPU intervention saving valuable time to better control the system.

2.6.2 In Depth

A sigma-delta type ADC is by design an over-sampling architecture. The ADC itself is a single bit design that over-samples the signal of interest to produce a higher order output. The modulation of this type of converter results in a binary output and, hence, a serial data stream. This serial data stream is then sent to the filter/demodulator on the C2000 MCU for re-construction into a higher bit order digital representation of the sampled signal (Figure 2-7).

脉冲序列输出 (PTO) QepDiv 和脉冲发生器:

增量式旋转编码器输出脉冲序列以指示所监控的轴已移动。系统主控通常处理此脉冲序列以确定速度、距离和位置等信息。在QepDiv实现中, 位置信息从编码器发送到C2000 MCU上的增强型四象限编码器脉冲 (eQEP) 模块。在PulseGen情况下, 会生成自定义脉冲流以满足系统需求。PTO-QepDiv示例展示了如何使用CLB从这些eQEP输入生成分频脉冲流, 如图2-6所示。然后可以将分频脉冲流发送到系统中的另一个设备。QepDiv和PulseGen示例都在C2000位置管理器PTO API参考指南中有文档记录。

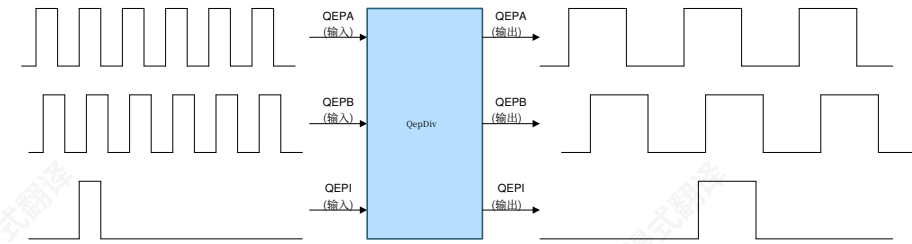


图 2-6. QepDiv 输入和输出图

2.5.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

2.5.4 硬件平台和软件示例

- [Tamagawa T-Format 绝对编码器 主接口参考设计 for C2000™ MCUs](#)
- [C2000 位置管理器 PTO API 参考指南](#)
- [MotorControl 软件开发套件 \(SDK\) for C2000 MCUs](#)
- [位置管理器 BoosterPack \(BOOSTXL-POSMGR\)](#)
- [C2000™ DesignDRIVE 开发套件 for 工业电机控制 \(TMDXIDDK379D\)](#)

2.5.5 文档

- [培训: C2000 可配置逻辑块 \(CLB\) 工具如何在我的设计中集成自定义逻辑](#)
- [CLB 工具用户指南](#)
- [使用 C2000 可配置逻辑块 \(CLB\) 进行设计](#)
- [如何将自定义逻辑从 FPGA/CPLD 迁移到 C2000 微控制器](#)
- [快速电流环库](#)

2.6 跨隔离边界智能传感

2.6.1 价值主张

高压控制系统通常使用sigma-delta ADC, 以便轻松地将高压域的模拟信息传递到存在MCU的低压域。在使用数据流之前, 必须通过滤波器和解调器进行处理。这种逻辑存在于C2000 MCU上, 并称为Sigma Delta滤波器模块 (SDFM) 模块。C2000 MCU上的SDFM内部的高低压比较器可以在不干预CPU的情况下驱动PWM, 从而节省宝贵的时间, 以更好地控制系统。

2.6.2 深入了解

一个sigma-delta类型的ADC在设计上是一种过采样架构。该ADC本身是一种单比特设计, 通过过采样目标信号来产生更高阶的输出。这种类型转换器的调制结果是一个二进制输出, 因此是一个串行数据流。这个串行数据流随后被发送到C2000微控制器的滤波器/解调器, 以重建为采样信号更高位数的数字表示 (图2-7)。

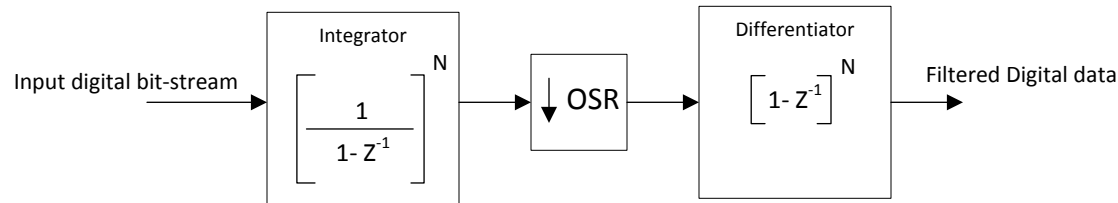


Figure 2-7. Filter and Demodulator Inside the C2000 SDFM

At this point an interrupt is generated to the MCU, informing the other domains there is new data to process and act upon.

The SDFM module on the C2000 MCU is unique in that not only can the converted data be read post filtering, but the PWMs can be switched based on this data automatically. Each SDFM module contains four channels. Inside each channel exists two filters: a primary filter that produces the SDFM data and a secondary filter containing both high and low limit comparators (Figure 2-8). This allows the system to control the PWM signals without waiting for CPU intervention, resulting in both lower latency as well as lower overall CPU utilization.

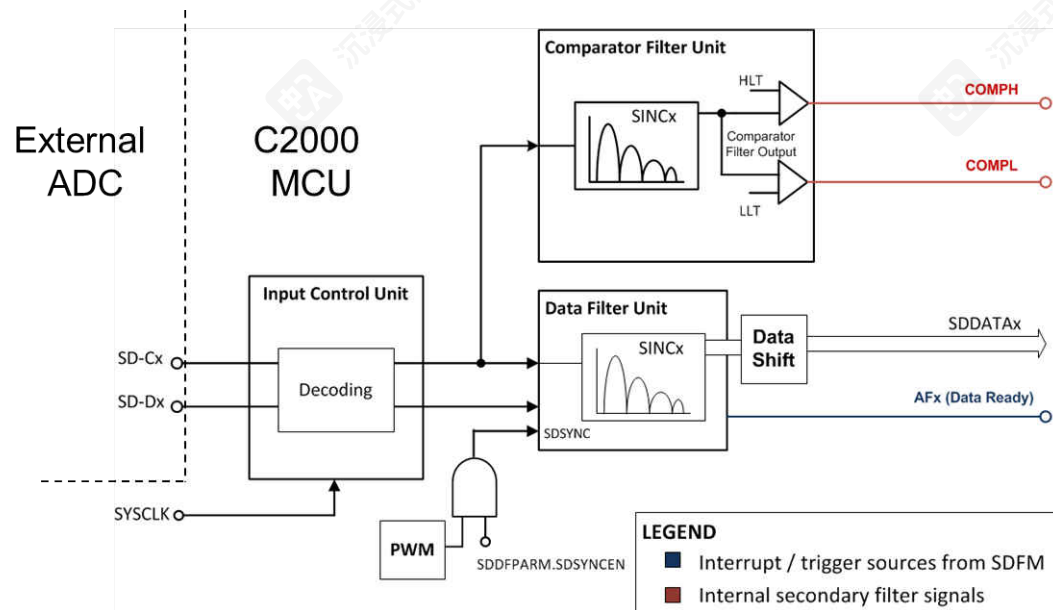


Figure 2-8. SDFM Module With Both Primary and Secondary Filter Blocks on the TMS320F2837xD MCU

2.6.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)

2.6.4 Hardware Platforms and Software Examples

- [TMDSCNCD28388D controlCARD evaluation module](#)
- [LAUNCHXL-F28379D LaunchPad](#)
- [LAUNCHXL-F280049C LaunchPad](#)
- [TMDSCNCD280039C controlCARD evaluation module](#)
- [SDFM_filter_sync_cpuread SW example for F2838xD/S](#)
- [SDFM_pwm_sync_cpuread SW example for the F2838xD/S](#)

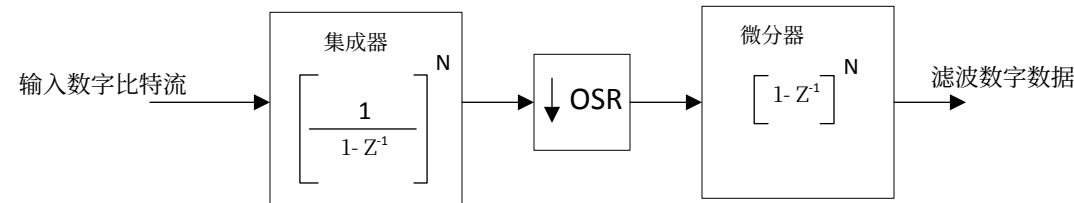


图2-7。C2000 SDFM内部的滤波器和解调器

此时会向微控制器生成一个中断，通知其他域有新数据需要处理和执行。

C2000微控制器的SDFM模块独特之处在于，不仅可以在滤波后读取转换数据，还可以根据这些数据自动切换PWM。每个SDFM模块包含四个通道。每个通道内部有两个滤波器：一个产生SDFM数据的主滤波器，以及一个包含高低限比较器的次级滤波器（图2-8）。这使得系统能够在不等待CPU干预的情况下控制PWM信号，从而实现更低的延迟和更低的总体CPU利用率。

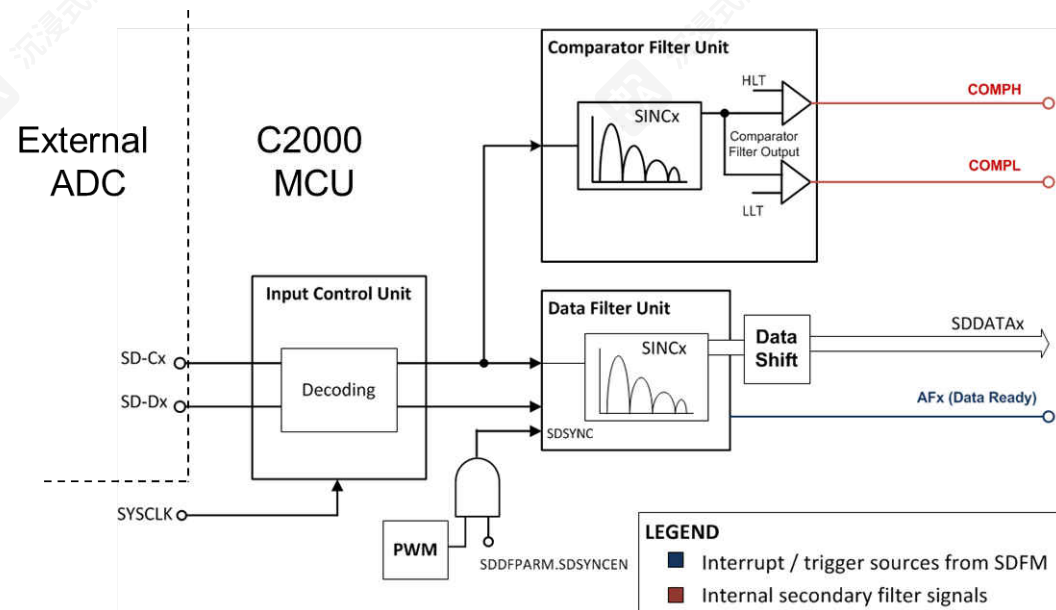


图2-8。TMS320F2837xD微控制器上的SDFM模块（包含主滤波器和次级滤波器块）

2.6.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)

2.6.4 硬件平台和软件示例

- [TMDSCNCD28388D controlCARD 评估模块](#)
- [LAUNCHXL-F28379D 启动板](#)
- [LAUNCHXL-F280049C 启动板](#)
- [TMDSCNCD280039C 控制卡评估模块](#)
- [SDFM_filter_sync_cpuread SW example for F2838xD/S](#)
- [SDFM_pwm_同步_cpuread SW示例，适用于F2838xD/S](#)

2.6.5 Documentation

- [TMS320F2838x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the SDFM chapter)
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the SDFM chapter)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the SDFM chapter)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the SDFM chapter)
- [Real-Time controllers get new connectivity capabilities](#)

2.7 Enabling Intra-Period Updates in High Bandwidth Control Topologies

2.7.1 Value Proposition

One of the most important considerations in designing a real-time control system is what the minimum sample to output delay the system can achieve (Figure 2-9). From the basics of providing stable control of the real-time system to realizing increased system efficiency the on-chip ADCs inside every C2000 real-time microcontroller are designed to minimize their effective time delay to the system in a variety of ways. Looking beyond the basics of a discrete ADCs and comparing to the C2000 on-chip ADCs there are differentiated features as a result of decades of involvement in real-time control.

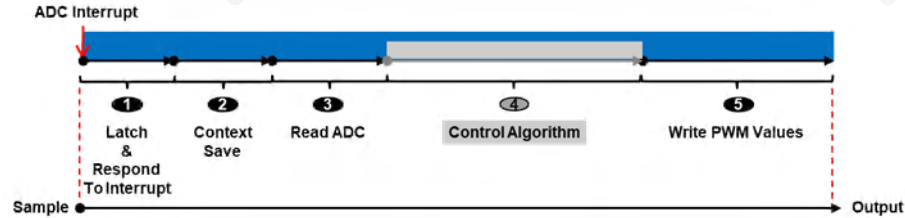


Figure 2-9. Real-Time Signal Chain

2.7.2 In Depth

While choosing a discrete ADC for a system many factors are considered: raw converter speed, DC error terms like gain and offset, and AC error terms like SNR, THD(Shown in:Section 2.1). Beyond those parameters are the aspects of system integration, conversion speed with respect to the control loop, and sampling capability that can make or break a real-time control design.

System integration: One advantage of an integrated ADC is the lack of an external communications bus that adds complexity to the system in both latency of the comms channel but also noise considerations for bus routing. Once the ADC conversion is complete, minimizing the delay into the control scheme of the system is vital. The ADC results are accessible by all CPU cores, CLA cores, as well as DMAs.

Another common delay is introduced due to the latency introduced when the CPU is notified that the ADC conversion is complete. The C2000 ADCs implement a selectable early interrupt that takes advantage of the time the ADC incurs between the start of a sample and the completion of the analog to digital conversion. Since the ADC is integrated the C2000 MCU has the ability to synchronize to either the end of the analog sampling phase or the full conversion complete. This allows the C28x CPU or CLA to continue to do other tasks or calculations in parallel to the ADC conversion and use the ADC conversion as soon as it is available. Not only are wasted cycles minimized, but results in a more stable system control as the converted value used by the CPU is closer in time to the real world signal state.

2.6.5 文档

- TMS320F2838x 实时微控制器技术参考手册（更多信息，请参阅 SDFM 章节）
- TMS320F2837xD 实时微控制器技术参考手册（更多信息，请参阅 SDFM 章节）
- TMS320F28004x 实时微控制器技术参考手册（更多信息，请参阅 SDFM 章节）
- TMS320F28003x 实时微控制器技术参考手册（更多信息，请参阅 SDFM 章节）
- [实时控制器获得新的连接能力](#)

2.7 在高带宽控制拓扑中启用周期内更新

2.7.1 价值主张

在设计实时控制系统时，最重要的考虑因素之一是系统可以实现的最小采样到输出延迟（图2-9）。从提供实时系统稳定控制到实现系统效率提升，C2000实时微控制器内部的片上ADC通过多种方式设计，以最小化其有效时间延迟。超越离散ADC的基础，并与C2000片上ADC进行比较，由于数十年来参与实时控制，存在差异化的特性。

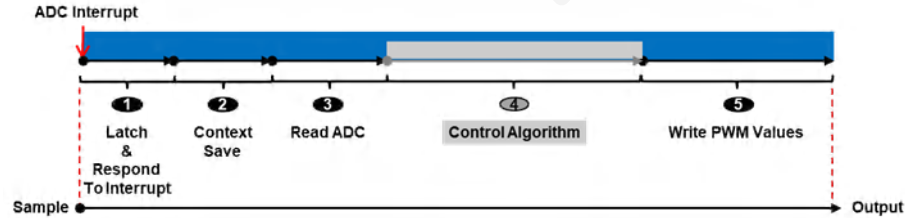


图 2-9。实时信号链

2.7.2 深入探讨

在选择系统的离散 ADC 时，会考虑许多因素：原始转换器速度、直流误差项（如增益和偏移）以及交流误差项（如 SNR、THD（显示在：第 2.1 节））。除此之外，系统集成的方面、控制循环中的转换速度以及采样能力，都可能决定实时控制设计的成败。

系统集成：集成式 ADC 的一个优势是缺少外部通信总线，这增加了系统复杂性，不仅体现在通信通道的延迟上，还体现在总线布线时的噪声考虑上。一旦 ADC 转换完成，将延迟最小化到系统的控制方案中至关重要。ADC 结果可被所有 CPU 核心、CLA 核心，以及 DMA 访问。

另一个常见的延迟是由于CPU在收到ADC转换完成通知时引入的延迟。C2000 ADC实现了一种可选的早期中断，利用ADC在采样开始和模拟到数字转换完成之间产生的时间。由于ADC是集成的，C2000 MCU能够与模拟采样阶段结束或完整转换完成同步。这使得C28x CPU或CLA能够并行执行其他任务或计算，并在ADC转换可用时立即使用它。这不仅最大限度地减少了浪费的周期，还使系统控制更加稳定，因为CPU使用的转换值在时间上更接近真实世界信号状态。

Conversion Speed:One aspect that is common in the evaluation of any ADC, discrete or integrated, is its conversion speed. This is how long it takes the ADC to convert the sampled analog signal into a digital representation. This is especially important in high bandwidth systems as it is beneficial to update the system intracycle vs waiting for the next full cycle of the control loop to happen (Figure 2-10). This is shown the (Fast Current Loop examples). The ADC on the C2000 MCU is capable of generating a conversion in as little as 260ns after the sample completes. This allows enough time for the C28x or CLA core to calculate the new PWM frequencies well ahead of the next control loop period allowing for the intra-cycle update to happen. The ePWM module plays a role here as well, allowing the PWM to be updated by writing to the active registers vs the shadow resigster, which would pend the update until the next full period.

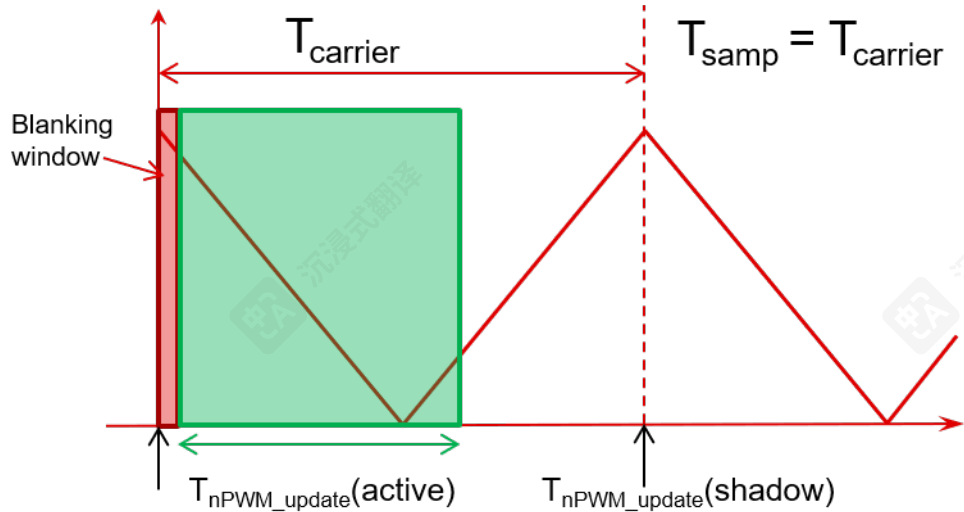


Figure 2-10. Intra-Period Update of PWM

Simultaneous sampling: Each C2000 device referenced in this guide has at least 2 if not more ADCs. Looking at common real-time control topologies this allows simultaneous sampling of the motor currents. For the case of a system that uses a resolver for posistion sensing this would allow both the sine and cosine feedbacks to be sampled at the same point in time. Table 2-4 shows the potential effect on the system if such inputs are not sampled at the same point in time

Table 2-4. Potential Phase Error Caused by Sample Delay

Control Loop Frequency	Phase Error With 500ns Sample Lag	Phase Error With 1µs Sample Lag
10 kHz	1.8 degrees	3.6 degrees
20 kHz	3.6 degrees	7.2 degrees
50 kHz	8.9 degrees	17.8 degrees

2.7.3 Device List

- TMS320F2838xD/S
- TMS320F2837xD/S
- TMS320F2807x
- TMS320F28004x
- TMS320F28003x
- TMS320F28002x
- TMS320F280013x
- TMS320F280015x

2.7.4 Hardware Platforms and Software Examples

- TIDM-02007 Dual-axis motor drive using fast current loop (FCL) and SFRA on a single MCU reference design
- BOOSTXL-3PHGANINV 48-V Three-Phase Inverter With Shunt-Based In-Line Motor Phase Current Sensing Evaluation Module

转换速度：任何ADC（无论是离散的还是集成的）评估中的一个常见方面是其转换速度。这是ADC将采样到的模拟信号转换为数字表示所需的时间。在高带宽系统中，这一点尤其重要，因为更新系统周期内而不是等待控制循环的下一个完整周期发生（图2-10）是有益的。这体现在（快速电流环示例）中。C2000 MCU上的ADC能够在采样完成后在260ns内完成转换。这允许C28x或CLA核心在下一个控制循环周期之前有足够的时间计算新的PWM频率，从而实现周期内更新。ePWM模块也起作用，允许通过写入活动寄存器而不是影子寄存器来更新PWM，这会延迟更新直到下一个完整周期。

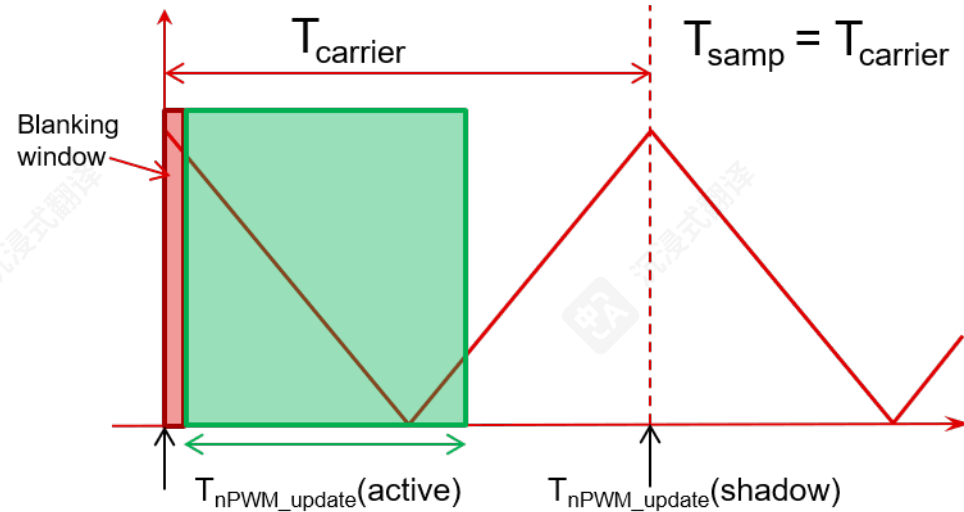


图 2-10. PWM 的周期内更新

同时采样：本指南中提到的每个 C2000 设备至少有 2 个（如果不是更多）ADC。从常见的实时控制拓扑来看，这允许同时采样电机电流。对于使用编码器进行位置传感的系统，这允许正弦和余弦反馈在同一时间点进行采样。表 2-4 显示了如果这些输入不在同一时间点进行采样，对系统可能产生的影响

表 2-4. 样本延迟引起的潜在相移误差

控制环频率	500ns 样本延迟时的相移误差	1µs 样本延迟时的相移误差
10 kHz	1.8 degrees	3.6 degrees
20 kHz	3.6 degrees	7.2 degrees
50 kHz	8.9 度	17.8 度

2.7.3 设备列表

- TMS320F2838xD/S
- TMS320F2837xD/S
- TMS320F2807x
- TMS320F28004x
- TMS320F28003x
- TMS320F28002x
- TMS320F280013x
- TMS320F280015x

2.7.4 硬件平台和软件示例

- TIDM-02007 双轴电机驱动使用快速电流环 (FCL) 和单 MCU 参考设计上的 SFRA
- BOOSTXL-3PHGANINV 48-V 三相逆变器带基于并联的串联电机相电流检测评估模块

- [BOOSTXL-3PHGANINV examples in C2000Ware Motor Control SDK](#)
- [TMDXIDDK379D C2000 DesignDRIVE Development Kit for Industrial Motor Control](#)

2.7.5 Documentation

- [Performance Analysis of Fast Current Loop \(FCL\) in Servo](#)
- [Quick Response Control of PMSM Using Fast Current Loop](#)

2.8 Accurate Monitoring of Real-Time Control System Events Without the Need for Signal Conditioning

2.8.1 Value Proposition

As mentioned in [Section 4.4](#), quick detection of over-shoots and under-shoots is critical to the operation of a real-time control systems. However in reality, the transients of concern often have coupled noise, which can cause false trips and shut downs resulting in reliability issues in system functionality.

One approach typically used to remedy this is external passive RC filters to remove the high frequency noise from the signal of interest. This has drawbacks of BOM cost, PCB space, and slowing down the signal of interest. The CMPSS module on the C2000 MCU has configurable filters integrated in the device to filter out noise events without the need for an external RC filter.

2.8.2 In Depth

The CMPSS filter works on a principle of majority vote. It captures a window of high/low trip events from the comparator and sets the output to high or low depending on the threshold setting. The number of trip events captured by the window and the threshold setting used to determine the output are both configurable. Due to the configurability, the application can trade-off between how much noise to reject and how much the final trip is delayed. [Figure 2-11](#) is a high level illustration of the filter in work. For further details, please refer to the CMPSS chapter in the TRM.

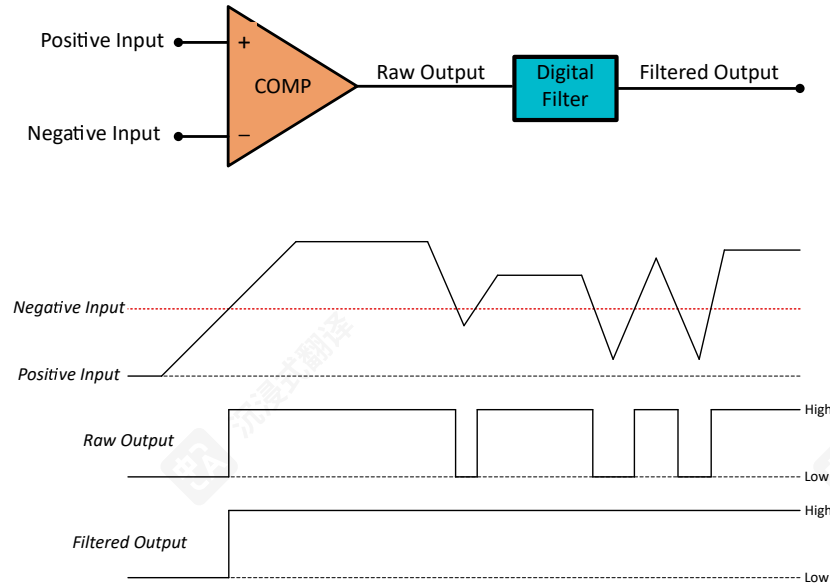


Figure 2-11. CMPSS Raw vs Filtered Output

2.8.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)

- [BOOSTXL-3PHGANINV 示例在 C2000Ware 电机控制 SDK 中](#)
- [TMDXIDDK379D C2000 DesignDRIVE 工业电机控制开发套件](#)

2.7.5 文档

- [伺服快速电流环（FCL）性能分析](#)
- [使用快速电流环对PMSM的快速响应控制](#)

2.8 无需信号调理的实时控制系统事件精确监控

2.8.1 价值主张

如第4.4节所述，快速检测超调和下冲对实时控制系统的运行至关重要。然而在实际中，所关心的瞬态往往存在耦合噪声，这可能导致误动作和停机，从而引发系统功能可靠性问题。

通常用于解决此问题的一种方法是外部无源RC滤波器，以去除目标信号的高频噪声。但这存在BOM成本、PCB空间占用以及减慢目标信号速度的缺点。C2000微控制器的CMPSS模块集成了可配置的滤波器，无需外部RC滤波器即可滤除噪声事件。

2.8.2 深入了解

CMPSS 滤波器基于多数投票原理工作。它从比较器捕获高/低跳变事件窗口，并根据阈值设置将输出设置为高或低。窗口捕获的跳变事件数量和用于确定输出的阈值设置都是可配置的。由于可配置性，应用程序可以在拒绝多少噪声和最终跳变延迟多少之间进行权衡。图 2-11 是滤波器工作的高层次说明。有关详细信息，请参阅 TRM 中的 CMPSS 章节。

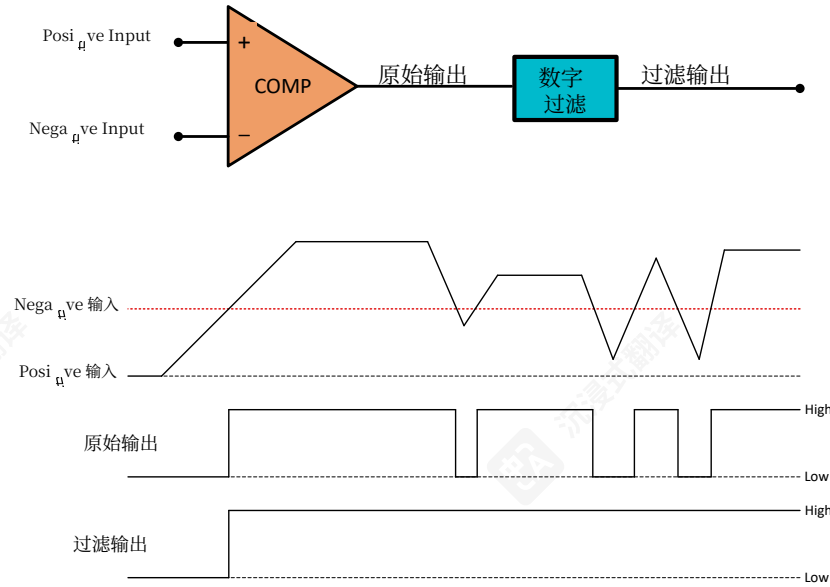


图 2-11. CMPSS 原始输出与滤波输出

2.8.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)

- [TMS320F280015x](#)

2.8.4 Hardware Platforms and Software Examples

- [TMDSCNCDF28388D](#)
- [TMDSCNCDF28379D](#)
- [TMDSCNCDF280049C](#)
- [TMDSCNCD280039C](#)
- [TMDSCNCD280025](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)
- [F2838xD CMPSS Digital Filter SW Example](#)
- [F2837xD CMPSS Digital FilterSW Example](#)
- [F28004x CMPSS Digital Filter SW Example](#)
- [F28003x CMPSS Digital Filter SW Example](#)
- [F28002x CMPSS Digital Filter SW Example](#)
- [F280013x CMPSS Digital Filter SW Example](#)
- [F280015x CMPSS Digital Filter SW Example](#)

2.8.5 Documentation

- [TMS320F2838x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [CMPSS](#) chapter)
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [CMPSS](#) chapter)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [CMPSS](#) chapter)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [CMPSS](#) chapter)
- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [CMPSS](#) chapter)
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [CMPSS](#) chapter)
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [CMPSS](#) chapter)

3 Processing Key Technologies

3.1 Accelerated Trigonometric Math Functions

3.1.1 Value Proposition

Trigonometric functions are used heavily in real-time control systems, both in power applications as well as motor control. Park Transforms ([Figure 3-1](#)), Space Vector Generation, and resolver angle are a few of these examples that rely on trigonometric math. The Trigonometric Math Unit (TMU) on C2000 MCUs enables [an extended instruction set](#) targeted at 32-bit floating-point trigonometry based calculations. This results both faster performance and smaller code size for Trigonometric functions.

$$\text{Park} \begin{bmatrix} i_d \\ i_q \\ i_o \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} X \begin{bmatrix} i_\alpha \\ i_\beta \\ i_o \end{bmatrix}$$

Figure 3-1. Park Transform

3.1.2 In Depth

Many common mathematical techniques in real-time control rely on the use of trigonometric functions: sine, cosine, and arc tangent are all examples. The TMU adds dedicated instructions to the C28x core for these functions as well as their inverse, that supersede the standard C library calls. As shown in [Figure 3-2](#), significant cycle count reductions are possible using the TMU based instructions vs their Fast RTS counterparts.

- [TMS320F280015x](#)

2.8.4 硬件平台和软件示例

- [TMDSCNCDF28388D](#)
- [TMDSCNCDF28379D](#)
- [TMDSCNCDF280049C](#)
- [TMDSCNCD280039C](#)
- [TMDSCNCD280025](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)
- [F2838xD CMPSS 数字滤波器 SW 示例](#) • [F2837xD CMPSS 数字滤波器SW 示例](#) • [F28004x CMPSS 数字滤波器 SW 示例](#) • [F28003x CMPSS 数字滤波器 SW 示例](#) • [F28002x CMPSS 数字滤波器 SW 示例](#) • [F280013x CMPSS Di](#)

- [F280015x CMPSS 数字滤波器 SW 示例](#)

2.8.5 文档

- [TMS320F2838x 实时微控制器技术参考手册](#)（更多信息，请参阅 [CMPSS](#) 章节）
- [TMS320F2837xD 实时微控制器技术参考手册](#)（更多信息，请参阅 [CMPSS](#) 章节）
- [TMS320F28004x 实时微控制器技术参考手册](#)（更多信息，请参阅 [CMPSS](#) 章节）
- [TMS320F28003x 实时微控制器技术参考手册](#)（更多信息，请参阅 [CMPSS](#) 章节）
- [TMS320F28002x 实时微控制器技术参考手册](#)（更多信息，请参阅 [CMPSS](#) 章节）
- [TMS320F280013x 实时微控制器技术参考手册](#)（更多信息，请参阅 [CMPSS](#) 章节）
- [TMS320F280015x 实时微控制器技术参考手册](#)（更多信息，请参阅 [CMPSS](#) 章节）

3 处理关键技术

3.1 加速三角数学函数

3.1.1 价值主张

三角函数在实时控制系统中被广泛使用，无论是在电力应用还是电机控制中。帕克变换（图3-1）、空间矢量生成和解算器角度是其中几个依赖三角数学的例子。C2000微控制器上的三角数学单元（TMU）支持针对32位浮点三角计算扩展指令集。这为三角函数带来了更快的性能和更小的代码尺寸。

$$\text{Park} \begin{bmatrix} i_d \\ i_q \\ i_o \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} X \begin{bmatrix} i_\alpha \\ i_\beta \\ i_o \end{bmatrix}$$

图3-1。帕克变换

3.1.2 深入了解

实时控制中的许多常见数学技术依赖于三角函数的使用：正弦、余弦和反正切都是例子。TMU 为 C28x 核心添加了用于这些函数及其反函数的专用指令，这些指令取代了标准 C 库调用。如图 3-2 所示，与它们的 Fast RTS 对应指令相比，使用基于 TMU 的指令可以显著减少周期数。

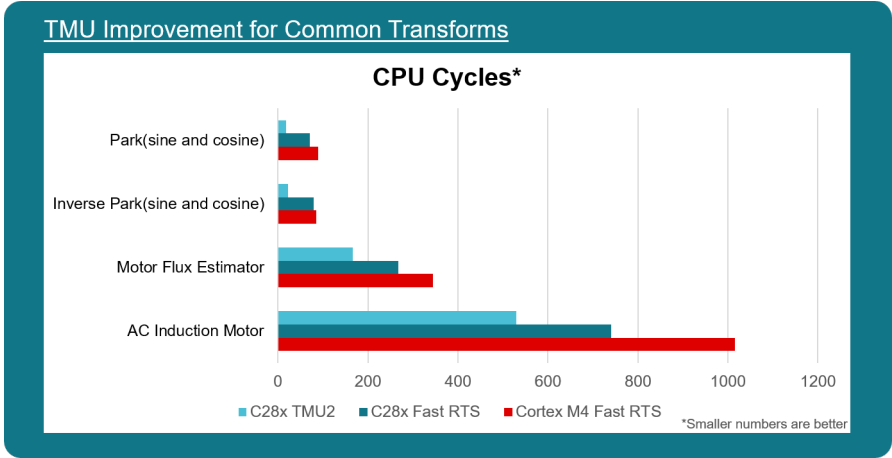


Figure 3-2. TMU Improvement for Common Transforms

There is also single instruction support for both square root as well as floating point division. These are often used in conjunction with the trigonometric functions previously listed. A full list of supported instructions and their cycle counts can be seen in [Table 3-1](#). These instructions are inserted automatically by the C compiler on devices that have a TMU.

Table 3-1. TMU Supported Instructions Summary

Operation	C Equivalent Operation	C28x Pipeline Cycles
Multiply by 2*pi	a = b * 2pi	2 cycles + Sine/Cosine function
Divide by 2*pi	a = b / 2pi	2 cycles + Sine/Cosine function
Divide	a = b / c	5 cycles
Square Root	a = sqrt(b)	5 cycles
Sin Per Unit	a = sin(b*2pi)	4 cycles
Cos Per Unit	a = cos(b*2pi)	4 cycles
Arc Tangent Per Unit	a = atan(b)/2pi	4 cycles
Arc Tangent 2 and Quadrant Operation	Operation to assist in calculating ATANPU2	5 cycles

Note

While the C2000 MCUs listed below all have a TMU module, the C compiler used to generate the target code must have the correct options selected to utilize this hardware. This is controlled in the Processor Options of the C2000 Compiler via the TMU support drop down, as well as using the "relaxed" setting for the floating point mode under C2000 Compiler → Optimizations. Specific TMU based functions can be called explicitly as inline functions in the C source if that is preferable to a global setting.

3.1.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

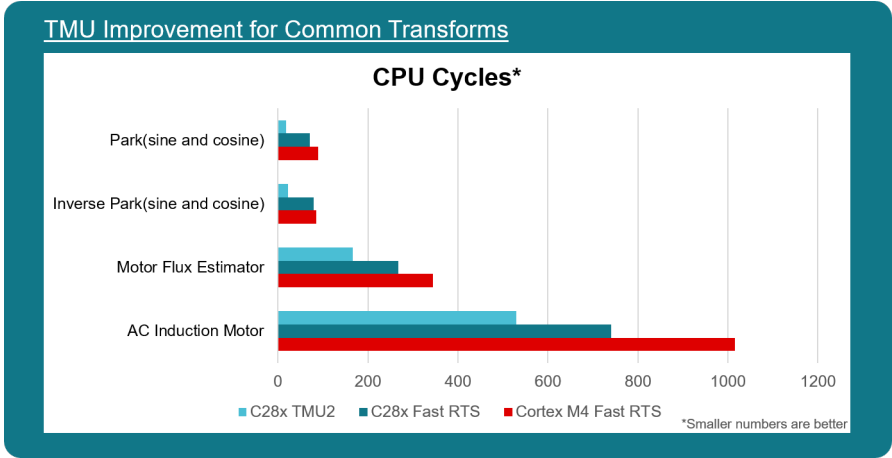


图 3-2. 常用变换的 TMU 改进

还有对平方根和浮点除法的单指令支持。这些通常与之前列出的三角函数一起使用。支持的所有指令及其周期数可以在表 3-1 中看到。这些指令由 C 编译器在具有 TMU 的设备上自动插入。

表 3-1. TMU 支持的指令摘要

操作	C 语言等效操作	C28x 流水线周期
乘以 2*pi	a = b * 2pi	2个周期 + 正弦/余弦函数
除以 2*pi	a = b / 2pi	2个周期 + 正弦/余弦函数
除以	a = b / c	5 cycles
平方根	a = sqrt(b)	5 cycles
正弦每单位	a = sin(b*2pi)	4 cycles
Cos Per Unit	a = cos(b*2pi)	4 cycles
Arc Tangent Per Unit	a = 反正切(b)/2pi	4 个周期
反正切 2 和象限运算	辅助计算的运算 ATANPU2	5 个周期

Note

虽然下面列出的 C2000 微控制器都配备了 TMU 模块，但用于生成目标代码的 C 编译器必须选择正确的选项才能利用这块硬件。这通过 C2000 编译器的处理器选项中的 TMU 支持下拉菜单进行控制，以及在 C2000 编译器 → 优化下的浮点模式中使用 "宽松" 设置。如果希望使用内联函数而不是全局设置，可以在 C 源代码中显式调用基于 TMU 的特定函数。

3.1.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

3.1.4 Hardware Platforms and Software Examples

- TIDM-1007 Interleaved CCM Totem Pole Bridgeless Power Factor Correction (PFC) Reference Design
- TIDM-HV-1PH-DCAC Single-Phase Inverter Reference Design With Voltage Source and Grid Connected Modes
- TMDXIDDK379D C2000 DesignDRIVE Development Kit for Industrial Motor Control
- TMDSHVMTRINSPIN High Voltage Motor Control Kit with InstaSPIN-FOC and InstaSPIN-MOTION enabled for F280049C device lab7 and lab8

3.1.5 Documentation

- [Enhancing the Performance Capabilities of the C2000™ MCU Family](#)
- [Signal Chain Benchmarking - A Demonstration of Optimized Real-time Performance of C2000™ MCU](#)
- [TMS320C28x Extended Instruction Sets Technical Reference Manual](#)

3.2 Fast Onboard Integer Division

3.2.1 Value Proposition

Compared to other standard arithmetic operations, accurate integer division are typically more complex and have much higher cycles counts in a MCU. Additionally in control algorithms, there is need for more linear division operations, such as Euclidean and Modulus that often consume additional CPU overhead. The FID module addresses this by supporting these special division definitions through dedicated HW, which enables control algorithms without the typical additional overhead associated with these methods. Finally, the FID supports various size and signed operands with optimum cycle counts.

3.2.2 In Depth

There are multiple definitions for division and modulo operations according to the programming language and computer science literature, each of these definitions provide different mathematical properties that can be beneficially employed in the application context. Truncated division is the standard division definition widely used in many programming languages like C. In this definition, the remainder will always have the sign of numerator. The transfer function of truncated division is shown in [Figure 3-3](#), as can be observed that the function is non-periodic since there is a “platform” around zero point.

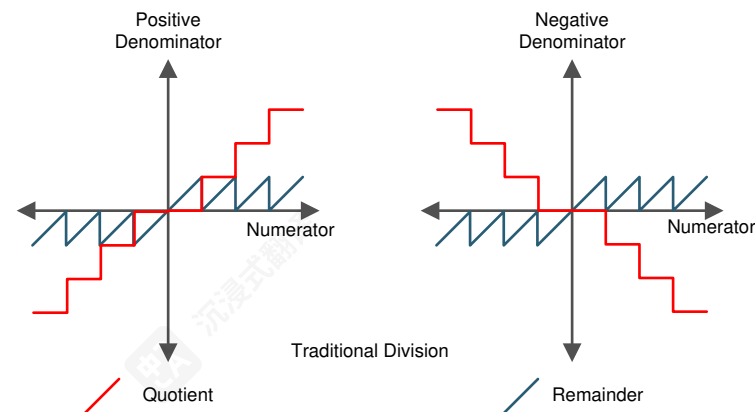


Figure 3-3. Truncated Division Function

Due to the non-linearity around zero point, the function is not preferred in control applications. Thus non-conventional definitions of division and modulo operations are sometimes preferred for better linearity and periodicity. The transfer function of Floored/Modulo ([Figure 3-4](#)) and Euclidean division ([Figure 3-5](#)) functions are shown below. In the modulo division function the remainder will always have the sign of denominator, thus the function is linear around the zero point. In the Euclidean division, the remainder is always positive so the division function is linear around zero point and also the modulo function is periodic.

3.1.4 硬件平台和软件示例

- TIDM-1007 交错式 CCM 无桥式功率因数校正 (PFC) 参考设计
- TIDM-HV-1PH-DCAC 带电压源和并网模式的单相逆变器参考设计
- TMDXIDDK379D C2000 DesignDRIVE 工业电机控制开发套件
- TMDSHVMTRINSPIN 高压电机控制套件，支持 InstaSPIN-FOC 和 InstaSPIN-MOTION，适用于 F280049C 设备的 lab7 和 lab8

3.1.5 文档

- [提升 C2000™ 微控制器系列性能能力](#)
- [Signal Chain Benchmarking - A Demonstration of Optimized Real-time Performance of C2000™ MCU](#)
- [TMS320C28x 扩展指令集技术参考手册](#)

pC2000™ 微控制器的优化实时性能

3.2 快速片上整数除法

3.2.1 价值主张

与其他标准算术运算相比，精确的整数除法通常更复杂，并且在微控制器（MCU）中具有更高的周期计数。此外，在控制算法中，需要更线性的除法运算，例如欧几里得和取模运算，这些运算通常消耗额外的CPU开销。FID模块通过专用硬件支持这些特殊除法定义，从而使得控制算法无需这些方法通常带来的典型额外开销。最后，FID支持各种大小和有符号操作数，并具有最优的周期计数。

3.2.2 深入探讨

根据编程语言和计算机科学文献，除法和取模运算有多种定义，每种定义都提供不同的数学特性，这些特性可以在应用上下文中得到有益的应用。截断除法是许多编程语言（如C）中广泛使用的标准除法定义。在这种定义中，余数的符号始终与被除数相同。截断除法的传递函数如图3-3所示，可以看出该函数是非周期的，因为零点周围有一个“平台”。

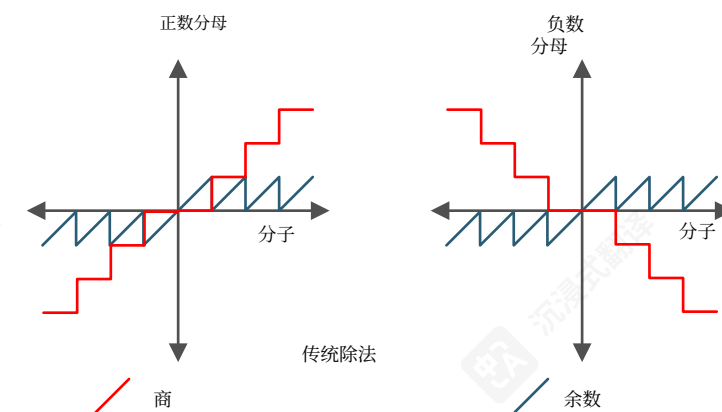


图 3-3. 截断除法函数

由于在零点附近存在非线性，该函数在控制应用中并不受欢迎。因此，有时会采用非传统的除法和模运算定义，以获得更好的线性和周期性。图3-4所示的Floor/Modulo函数和图3-5所示的欧几里得除法函数的传递函数如下所示。在模除法函数中，余数的符号始终与分母相同，因此该函数在零点附近是线性的。在欧几里得除法中，余数始终为正，因此除法函数在零点附近是线性的，模函数也是周期性的。

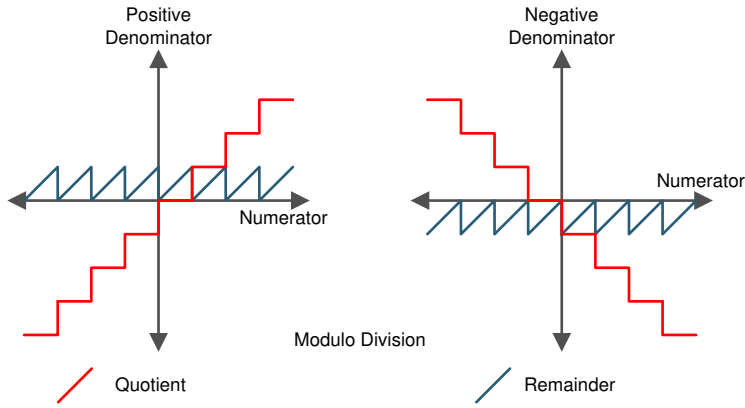


Figure 3-4. Floored Division Function

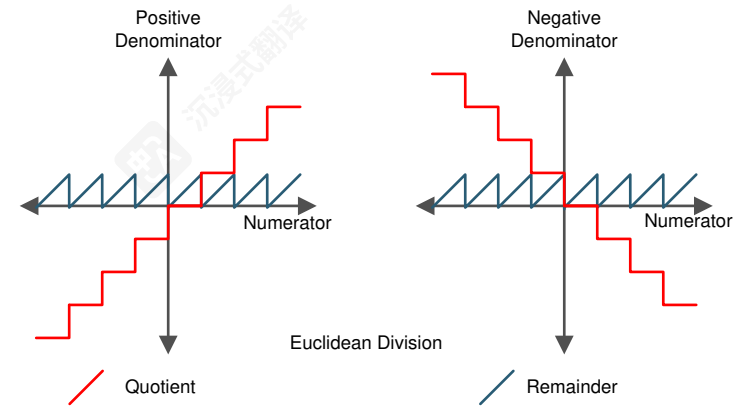


Figure 3-5. Euclidean Division Function

The C28x CPU has added specialized instructions to enable applications to implement the above division and modulo definitions efficiently in hardware. These new instructions used to enable integer division are interruptible, have very low latency and support different types of operations (ui32/ui32, i32/ui32, i64/i32, ui64/ui32, ui64/ui64, i64/i64, and so forth). The cycles count for the different types of division operations and sizes of the operands achieved using Fast Integer Division unit are listed in and are also compared with the cycles without FID module in [Table 3-2](#). As evident from the table, the FID unit provides several folds improvement in performance for different types of division operations which helps in minimizing the latency of control loop calculations.

Table 3-2. Integer Division With and Without the FID Module

Division Operation	Using C Operator '/' Without FASTINTDIV Hardware on C28x	Using Intrinsics With FASTINTDIV Hardware + C28x	Improvement Factor
i16/i16 traditional	52	16	3.3
i16/i16 Euclidean	56	14	4.0
i16/i16 Modulo	56	14	4.0
u16/u16	56	14	4.0
i32/i32 traditional	59	13	4.5
i32/i32 Euclidean	63	14	4.5
i32/i32 Modulo	63	14	4.5
i32/u32 traditional	37	14	2.6
i32/u32 Modulo	41	14	2.9
u32/u32	37	12	3.1
i32/i16 traditional	60	18	3.3
i32/i16 Euclidean	64	16	4.0
i32/i16 Modulo	64	16	4.0

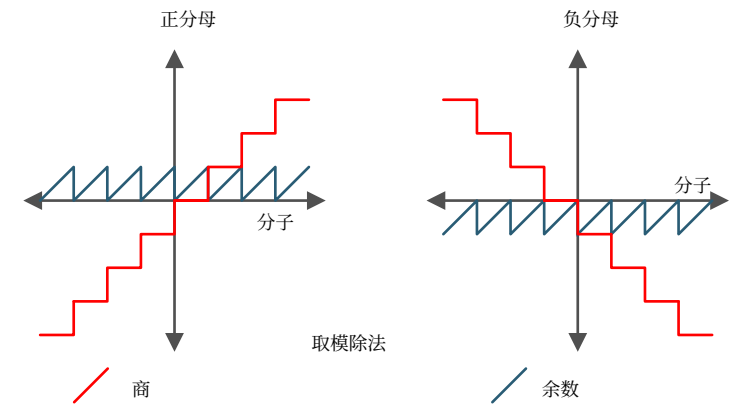


图 3-4. 向下取整除法函数

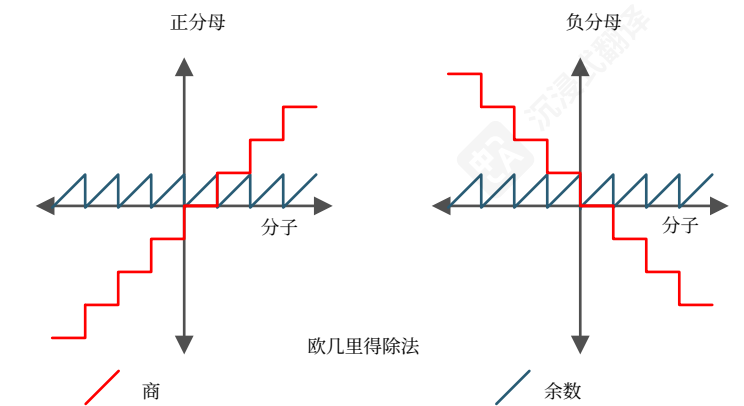


图 3-5. 欧几里得除法函数

C28x CPU增加了专用指令，以使应用程序能够在硬件中高效地实现上述除法和模运算定义。这些用于启用整数除法的新指令是可中断的，具有非常低的延迟，并支持不同类型的操作（ui32/ui32、i32/ui32、i64/i32、ui64/ui32、ui64/ui64、i64/i64等）。使用Fast Integer Division单元的不同类型除法操作和操作数大小的周期数列于其中，并与表3-2中不使用FID模块的周期数进行了比较。从表中可以看出，FID单元在不同类型的除法操作中提供了数倍的性能提升，这有助于最小化控制循环计算的延迟。

表 3-2. 带有和不带有 FID 模块的整数除法

除法运算	使用 C 运算符 '/' 无 C28x 上的 FASTINTDIV 硬件	使用 Intrinsics 与 C28x 上的 FASTINTDIV 硬件 +	改进系数
i16/i16 传统	52	16	3.3
i16/i16 欧几里得	56	14	4.0
i16/i16 取模	56	14	4.0
u16/u16	56	14	4.0
i32/i32 传统	59	13	4.5
i32/i32 欧几里得	63	14	4.5
i32/i32 取模	63	14	4.5
i32/u32 传统	37	14	2.6
i32/u32 取模	41	14	2.9
u32/u32	37	12	3.1
i32/i16 传统	60	18	3.3
i32/i16 欧几里得	64	16	4.0
i32/i16 取模	64	16	4.0

Table 3-2. Integer Division With and Without the FID Module (continued)

Division Operation	Using C Operator '/' Without FASTINTDIV Hardware on C28x	Using Intrinsics With FASTINTDIV Hardware + C28x	Improvement Factor
u32/u16	38	13	2.9
i64/i64 traditional ⁽¹⁾	78-2631	42	1.9-62.6
i64/i64 Euclidean ⁽¹⁾	82-2635	42	2.0-62.7
i64/i64 Modulo ⁽¹⁾	82-2635	42	2.0-62.7
i64/u64 traditional ⁽¹⁾	54-2605	42	1.3-62.0
i64/u64 Euclidean ⁽¹⁾	58-2609	42	1.4-62.1
i64/u6 Modulo ⁽¹⁾	58-2609	42	1.4-62.1
u64/u64/ ⁽¹⁾	53-2548	42	1.3-60.7

(1) The FASTINTDIV hardware implements 64-bit integer division with optimal fixed number of cycles for fast deterministic behavior. MCUs without such hardware acceleration, implement 64-bit integer division using generic CPU instructions that are not optimized for division or use algorithm techniques that optimize execution based on the value of the numerator and denominator. For instance, if the value of the numerator and denominator is less than 32-bits the software will execute a 32-bit division. Hence, the number of cycles can vary significantly and for large numerator and denominator values, overall cycles are much higher than achievable by the FASTINTDIV accelerator

3.2.3 Device List

- TMS320F2838xD/S
- TMS320F28003x
- TMS320F28002x

3.2.4 Hardware Platforms and Software Platforms

- F2838xD Fast Integer Division Example in C2000Ware
- F28002x Fast Integer Division Example in C2000Ware
- TMDSCNCD28388D Control CARD
- TMDSCNCD280039C Control CARD
- LAUNCHXL-F280025C LaunchPad

3.2.5 Documentation

- Fast Integer Division - A Differentiated Offering From C2000 Product Family
- TMS320C28x Optimizing C/C++ Compiler User's Guide
- TMS320C28x Assembly Language Tools User's Guide

3.3 Hardware Support for Double-Precision Floating-Point Operations

3.3.1 Value Proposition

The FPU64 module extends the computing capabilities of the C28x CPU by providing native hardware support for IEEE-754 single-precision and double-precision floating point operations. This enables customers who need double-precision floating point in their applications to implement them without seeing a significant increase in their CPU cycle load.

3.3.2 In Depth

The FPU64 module is extremely useful when 32-bit precision, such that, single-precision floating point is not sufficient for applications. In many real-time control applications, this is the case, so 64-bit precision (double-precision floating point) is needed. This normally comes at a price since native hardware on the device does not support double-precision floating point operations. Users see a significant increase in CPU cycles when running double-precision floating-point algorithms. By incorporating hardware support, this increase is avoided.

表 3-2。带和不带 FID 模块的整数除法（续）

除法操作	使用 C 运算符 '/' 不带 C28x 上的 FASTINTDIV 硬件	使用 Intrinsics 带 FASTINTDIV 硬件 + C28x	改进因子
u32/u16	38	13	2.9
i64/i64 传统 (1)	78-2631	42	1.9-62.6
i64/i64 欧几里得 (1)	82-2635	42	2.0-62.7
i64/i64 取模 (1)	82-2635	42	2.0-62.7
i64/u64 传统 (1)	54-2605	42	1.3-62.0
i64/u64 欧几里得(1)	58-2609	42	1.4-62.1
i64/u6 取余 ⁽¹⁾	58-2609	42	1.4-62.1
u64/u64/ (1)	53-2548	42	1.3-60.7

(1) FASTINTDIV 硬件实现 64 位整数除法，具有最优固定数量的周期，以实现快速确定性行为。没有此类硬件加速的 MCU，使用通用 CPU 指令实现 64 位整数除法，这些指令未针对除法进行优化，或使用基于分子和分母值优化执行算法的技术。例如，如果分子和分母的值小于 32 位，软件将执行 32 位除法。因此，周期数可能差异很大，对于较大的分子和分母值，整体周期数远高于 FASTINTDIV 加速器可实现的周期数

3.2.3 设备列表

- TMS320F2838xD/S
- TMS320F28003x
- TMS320F28002x

3.2.4 硬件平台和软件平台

- F2838xD C2000Ware中的快速整数除法示例
- F28002x C2000Ware中的快速整数除法示例
- TMDSCNCD28388D控制卡
- TMDSCNCD280039C控制卡
- LAUNCHXL-F280025C 启动板

3.2.5 文档

- 快速整数除法 - C2000产品系列的差异化产品
- TMS320C28x O ptimizin g C/C++ Com piler 用户指南
- TMS320C28x 汇编语言工具用户指南

3.3 硬件支持双精度浮点运算

3.3.1 价值主张

FPU64 模块通过提供对 IEEE-754 单精度和双精度浮点运算的原生硬件支持，扩展了 C28x CPU 的计算能力。这使得需要在应用程序中使用双精度浮点数的客户能够在 CPU 周期负载没有显著增加的情况下实现它们。

3.3.2 深入了解

FPU64 模块在需要 32 位精度的场合非常实用，因为单精度浮点数不足以满足应用需求。在许多实时控制应用中，情况正是如此，因此需要 64 位精度（双精度浮点数）。这通常需要付出代价，因为设备上的原生硬件不支持双精度浮点数运算。当运行双精度浮点算法时，用户会看到 CPU 周期显著增加。通过引入硬件支持，可以避免这种增加。

Table 3-3 compares the performance of double-precision operations on FPU64 vs FPU (single-precision floating point hardware).

Table 3-3. Cycle Comparison Between FPU64 and FPU32

Floating-Point Operations	FPU64 (cycles) -- fp_mode=relaxed	FPU (cycles) -- fp_mode=strict	FPU (cycles) -- fp_mode=relaxed FPU64 disabled	FPU (cycles) -- fp_mode=strict FPU64 disabled
32-bit Division	8	234	8	234
64-bit Division	27	27	2222	2222

The results in the above table (profiled with optimization off, code running from RAM) illustrate that the performance of double-precision floating-point division on FPU64 is slightly more expensive than single-precision floating-point division on FPU32. In the single-precision case, when the floating point mode (fp_mode) is set to relaxed, the compiler generates hardware instructions to perform single-precision division, at the slight expense of accuracy. When the floating point mode is set to strict, the compiler does not generate hardware instructions and calls into the RTS library to maintain accuracy, but at the cost of cycles.

In the double-precision case, the floating point mode does not matter, because the FPU64 implementation is accurate. As long as the FPU64 is enabled, the compiler generates hardware instructions supported by the FPU64 to perform double-precision division. If FPU64 is disabled, only then the compiler does not generate FPU64 hardware instructions and calls into the RTS library. 64-bit floating-point division cannot take advantage of 32-bit floating-point hardware, even in relaxed mode, which is why the cycles remain 2222.

Devices with the FPU64 utilize the same registers as the FPU except for the addition of eight floating-point results extension registers for double-precision floating-point operations. The FPU64 enhancements support all existing FPU single-precision floating-point instructions in addition to the 64-bit double-precision floating-point instructions. FPU64 64-bit instructions operate in one to three pipeline cycles, with some instructions also supporting a parallel move operation.

Using the FPU64 is straightforward. Users simply write C code with double-precision floating-point variables and operations, compile it with TI C28x C/C++ Compiler v18.9.0.STS (or later), and with compiler switches --float_support = fpu64. This will generate C28x native double-precision floating-point instructions. It may be helpful to refer to Table 3-4 for those unfamiliar with the size of the standard C data types on a C28x CPU. Users who want to write hand-optimized assembly for FPU64 may do so easily as well. There is roughly a one-to-one correspondence between single-precision and double-precision floating point assembly instructions. For the complete instruction set and details, see the document referenced in the documentation section. Software examples are listed in a section below that point to DSP and Math double-precision floating-point hand-optimized assembly routines that users can use for application development.

Table 3-4. Data Type Size C28x vs Arm

Data Type	C28x Bit Length (EABI)	Arm Bit Length
char	16	8
short	16	16
int	16	32
long	32	32
long long	64	64
float	32	32
double	64 ⁽¹⁾	64
long double	64	64
pointer	32	32

(1) C28x COFF treats double as 32-bits

With the advent of auto code generation tools like MATLAB's Embedded Coder, many customers are migrating to them for auto code generation. Since MATLAB uses double-precision floating-point by default, it becomes easier to port code from a simulation environment to an embedded environment without having to revalidate operating performance of the system due to a reduction in precision. This is another key benefit of using the FPU64.

表 3-3 比较了 FPU64 与 FPU（单精度浮点硬件）在双精度运算上的性能。

表 3-3. FPU64 和 FPU32 的周期比较

浮点运算	FPU64 (周期) -- fp_mode=relaxed	FPU (周期) -- fp_mode=strict	FPU (周期) -- fp_mode=relaxed FPU64 禁用	FPU (周期) -- fp_mode=strict FPU64 禁用
32-bit Division	8	234	8	234
64-bit Division	27	27	2222	2222

上述表格中的结果（未启用优化，代码从RAM运行）说明，FPU64上的双精度浮点除法比FPU32上的单精度浮点除法稍微昂贵。在单精度情况下，当浮点模式（fp_mode）设置为宽松时，编译器会生成硬件指令来执行单精度除法，略微牺牲精度。当浮点模式设置为严格时，编译器不会生成硬件指令，而是调用RTS库来保持精度，但会付出周期代价。

在双精度情况下，浮点模式无关紧要，因为FPU64的实现是准确的。只要FPU64启用，编译器就会生成FPU64支持的硬件指令来执行双精度除法。如果FPU64禁用，那么编译器才不会生成FPU64硬件指令，而是调用RTS库。64位浮点除法无法利用32位浮点硬件，即使在宽松模式下也是如此，这就是为什么周期数保持为2222。

具有 FPU64 的设备使用与 FPU 相同的寄存器，除了增加了八个用于双精度浮点运算的浮点结果扩展寄存器。FPU64 增强功能支持所有现有的 FPU 单精度浮点指令，以及 64 位双精度浮点指令。FPU64 的 64 位指令在一个到三个流水线周期内运行，其中一些指令还支持并行移动操作。

使用 FPU64 非常简单。用户只需编写包含双精度浮点变量和操作的 C 代码，使用 TI C28x C/C++ Compiler v18.9.0.STS（或更高版本）编译，并使用编译器选项 --float_support = fpu64。这将生成 C28x 本地的双精度浮点指令。对于不熟悉 C28x CPU 上标准 C 数据类型大小的用户，参考表 3-4 可能会有所帮助。想要为 FPU64 编写手优化的汇编代码的用户也可以轻松做到。单精度和双精度浮点汇编指令之间大致存在一一对应的关系。有关完整的指令集和详细信息，请参阅文档部分引用的文档。软件示例列在下面的一个部分中，这些示例指向 DSP 和 Math 双精度浮点手优化的汇编例程，用户可以使用这些例程进行应用开发。

表 3-4. 数据类型大小 C28x vs Arm

数据类型	C28x 位长度 (EABI)	Arm 位长度
char	16	8
short	16	16
int	16	32
long	32	32
long long	64	64
float	32	32
double	64 ⁽¹⁾	64
long double	64	64
pointer	32	32

(1) C28x COFF 将 double 视为 32 位

随着MATLAB的Embedded Coder等自动代码生成工具的出现，许多客户正在迁移到它们进行自动代码生成。由于MATLAB默认使用双精度浮点数，因此将代码从仿真环境移植到嵌入式环境变得更加容易，而无需由于精度降低而重新验证系统的运行性能。这是使用FPU64的另一个关键优势。

3.3.3 Device List

- [TMS320F2838xD/S](#)

3.3.4 Hardware Platforms and Software Examples

- [TMDSCNCD28388D Control CARD Evaluation Module](#)
- [64-bit floating point examples in the FPUFastRTS library in C2000Ware](#)
- [64-bit floating point examples in the DSP FPU library in C2000Ware](#)

3.3.5 Documentation

- [TMS320C28x Extended Instructions Sets Technical Reference Manual](#)

3.4 Increasing Control Loop Bandwidth With An Independent Processing Unit

3.4.1 Value Proposition

A primary concern in any control application is the time that elapses between sampling the system (sensing), applying the control function (processing), and applying the stimulus to the external system (actuation). The CLA was created specifically to address the need to minimize this time while increasing overall system throughput.

3.4.2 In Depth

The CLA found on the devices listed below is a fully parallel processor to the main C28x core. While the C28x core is a more traditional processor, executing instructions and servicing interrupts, the CLA is a task driven state machine. The CLA is a 32-bit floating point architecture.

Due to the nature of control systems, there are specific times when the sensing subsystem has new data to be processed. Advanced planning for these events with the other functions of an MCU can be difficult to time slice without introducing delay into both the system under control, but also any other functions the CPU has to perform.

As a task driven state machine, the CLA is constantly waiting in an idle state for an event, such as an ADC conversion, to process that data and actuate the system. Additionally, the CLA has full access to key control peripherals so it can fully realize the control system independent of the C28x CPU (Figure 3-6).

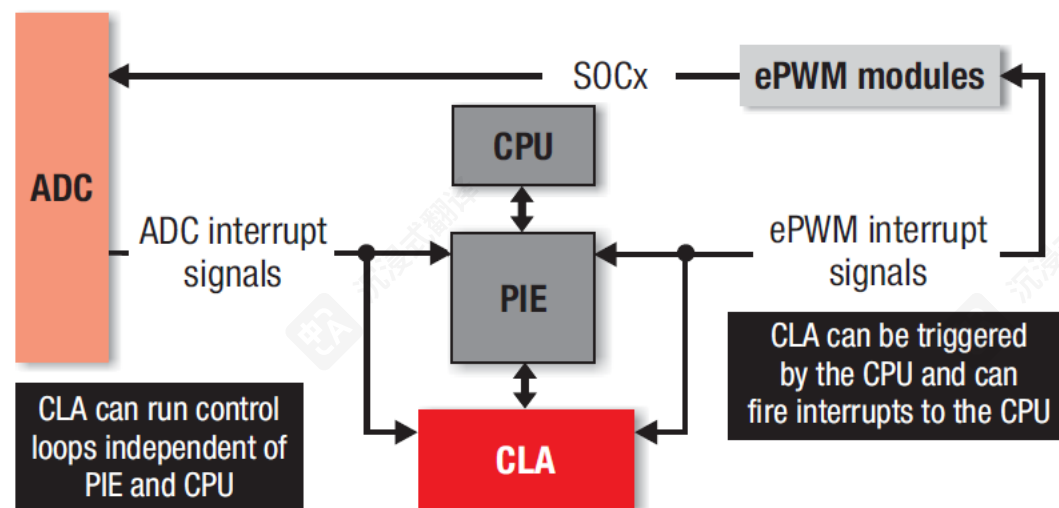


Figure 3-6. C28x and CLA Interfacing With the ADC and ePWM Modules

The above is beneficial for many reasons:

- There is little to no delay in processing the data, typically caused by the context switching of the main C28x core, and applying the new external stimulus to the system.
- There is no interruption or impact to the current C28x program execution.
- Potential to have parallel control systems running independently on the same MCU device.

3.3.3 设备列表

- [TMS320F2838xD/S](#)

3.3.4 硬件平台和软件示例

- [TMDSCNCD28388D 控制卡评估模块](#)
- [64-bit floating point examples in the FPUFastRTS library in C2000Ware](#)
- [C2000Ware 中的 DSP FPU 库中的 64 位浮点示例](#)

3.3.5 文档

- [TMS320C28x 扩展指令集技术参考手册](#)

3.4 使用独立处理单元提高控制环带宽

3.4.1 价值主张

任何控制应用中的一个主要关注点是采样系统（感知）、应用控制函数（处理）以及将刺激施加到外部系统（执行）之间经过的时间。CLA被专门创建以解决在减少此时间的同时提高整体系统吞吐量的需求。

3.4.2 深入了解

在下面列出的设备上找到的CLA是主C28x核心的一个完全并行处理器。虽然C28x核心是一个更传统的处理器，执行指令和服务中断，但CLA是一个任务驱动状态机。CLA是一个32位浮点架构。

由于控制系统的性质，在感知子系统有新数据需要处理时，使用MCU的其他功能对这些事件进行高级规划很难进行时间分配，否则会在受控系统以及其他CPU必须执行的功能中引入延迟。

作为任务驱动状态机，CLA不断处于空闲状态等待事件，例如ADC转换，以处理该数据并执行系统。此外，CLA可以完全访问关键控制外设，因此它可以独立于C28x CPU完全实现控制系统（图3-6）。

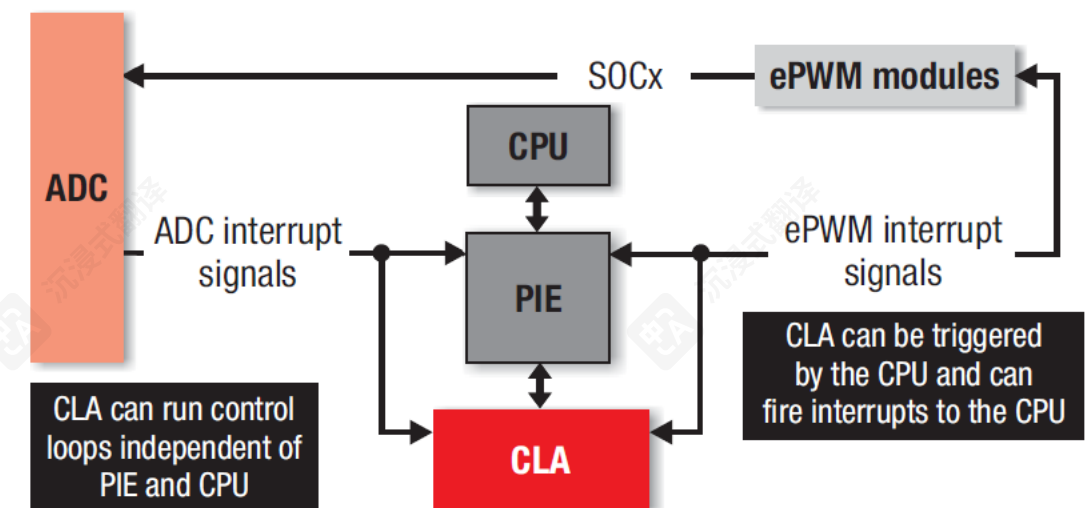


图3-6. C28x和CLA与ADC和ePWM模块的接口

以上有诸多好处：

- 数据处理几乎没有延迟，通常由主C28x核心的上下文切换引起核心，以及 pplying 新的外部刺激到s Y词干。
- 当前C28x程序执行没有中断或影响。
- 具有并行控制系统在相同的MCU设备上独立运行的能力。

The CLA has its own dedicated memory region for its code and shared memory for passing information between it and a C28x CPU in the system. The CLA is supported in Code Composer Studio™ by its own C compiler.

3.4.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)

3.4.4 Hardware Platforms and Software Examples

- [Valley switching boost power factor correction \(PFC\) reference design](#)
- [C2000 DesignDRIVE Development Kit for Industrial Motor Control](#)

3.4.5 Documentation

- [Software Examples to Showcase Unique Capabilities of TI's C2000™ CLA](#)
- [CLA Hands On Workshop](#)
- [CLA Usage in Valley Switching Boost Power Factor Correction \(PFC\) Reference Design](#)
- [CLA FAQ on E2E](#)
- [Enhancing the Performance Capabilities of the C2000™ MCU Family](#)
- [Software Examples to Showcase Unique Capabilities of TI's C2000™ CLA](#)

3.5 Flexible System Interconnect: C2000 X-Bar

3.5.1 Value Proposition

Three on-chip signal crossbars (X-bars): Input, Output, and ePWM provide the necessary mechanism in hardware to efficiently connect multiple subsystems across different control system implementations. Lower system latency, simpler PCB routing, and consistent timing are all key benefits.

3.5.2 In Depth

The inter-dependence of the Sensing and Actuation subsystems in a real-time control MCU is an obvious one. Whether it is incoming signals into the device, signals generated by on-chip logic (comparator, SDFM, ADC, and so forth), or outgoing signals; routing these signals in the system can be challenging at best. The on-chip X-bars provide a flexible mechanism to do that in hardware inside the MCU. There are three key benefits that these modules provide to the system:

- Simple routing of external signals into the chip: Any GPIO can go to multiple modules on-chip ([Figure 3-7](#)). For example, this allows the eCAP module to choose from any input pin as its source, or any input pin to go to the CPU/CLA as an external interrupt. This also gives flexibility in signal routing and layout of the Printed Circuit Board (PCB) since pins are not hard configured for a set group of functions.

CLA 拥有自己专用的代码内存区域，以及用于在它与系统中的 C28x CPU 之间传递信息的共享内存。CLA 在 Code Composer Studio™ 中由其自己的 C 编译器支持。

3.4.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)

3.4.4 硬件平台和软件示例

- [山谷切换式升压功率因数校正 \(PFC\) 参考设计](#)
- [C2000 DesignDRIVE 工业电机控制开发套件](#)

3.4.5 文档

- [展示 TI C2000™ CLA 独特功能的软件示例](#)
- [CLA 手把手工作坊](#)
- [CLA 在谷值切换式升压功率因数校正 \(PFC\) 参考设计中使用](#)
- [CLA 在 E2E 上的常见问题解答](#)
- [提升 C2000™ MCU 系列的性能能力](#)
- [展示 TI 的 C2000™ CLA 独特功能的软件示例](#)

3.5 灵活的系统互连：C2000 X-Bar

3.5.1 价值主张

三个片上信号交叉开关（X-bar）：输入、输出和ePWM在硬件中提供必要的机制，以高效地连接不同控制系统实现中的多个子系统。低系统延迟、简单的PCB布线以及一致的时间性都是其关键优势。

3.5.2 深入解析

实时控制MCU中感知和驱动子系统之间的相互依赖性显而易见。无论是设备中的输入信号、片上逻辑（比较器、SDFM、ADC等）生成的信号，还是输出信号；在系统中路由这些信号在最好情况下也可能是具有挑战性的。片上X-bar提供了一种灵活的机制，可以在MCU内部硬件中完成这一操作。这些模块为系统提供了三个主要优势：

- 外部信号简单布线到芯片：任何GPIO都可以连接到芯片上的多个模块（图3-7）。例如，这允许eCAP模块从任何输入引脚选择其源，或任何输入引脚作为外部中断连接到CPU/CLA。这也为信号布线和印刷电路板（PCB）的布局提供了灵活性，因为引脚没有被硬配置为特定功能组。

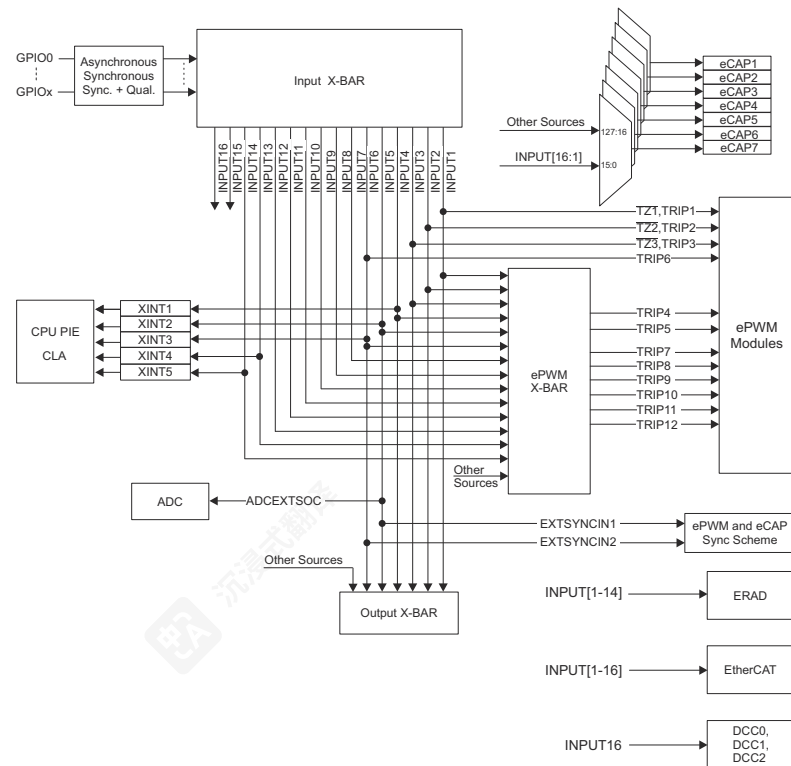


Figure 3-7. Input X-Bar on the TMS320F2837xD MCU

- Lower system latency: For simple point-to-point transactions, there is no need for the main C28x CPU or the CLA to spend cycles routing signals from one domain to another when using the X-bars. Inside of each X-bar, there is also a simple logical OR (Figure 3-8) that allows combination of any of the inputs to the X-bar in hardware. This not only saves cycles for the processors, but allows flexibility of the system since the mux selections are all controlled in software and can be changed dynamically as needed.

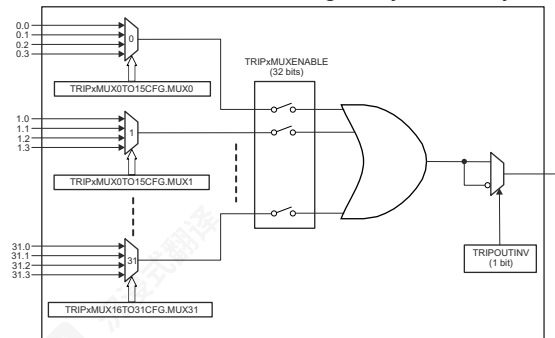


Figure 3-8. Local Mux and Logical OR on the TMS320F2837xD MCU

- Consistent timing: Similar to the above, since there is no CPU involvement, the signal propagates at the system clock in real time as it happens and is not contingent on another block to allow the signal to pass (Figure 3-9). This results in more predictable and repeatable system behavior. Given that the nature of many of the signals is to pass into the actuation sub-system timing is especially critical.

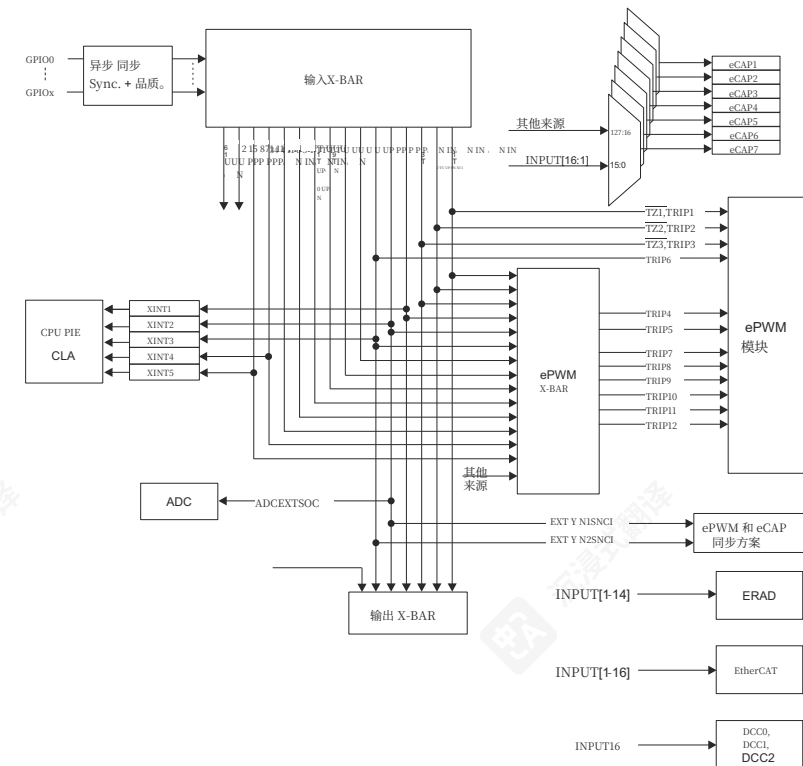


图 3-7. TMS320F2837xD 微控制器的输入 X-Bar

- 降低系统延迟：对于简单的点对点事务，在使用X-bar时，主C28x CPU或CLA无需花费周期来路由信号从一个域到另一个域。在每个X-bar内部，还有一个简单的逻辑或（图3-8），允许在硬件中组合X-bar的任何输入。这不仅为处理器节省了周期，还提高了系统的灵活性，因为多路复用选择全部由软件控制，可以根据需要动态更改。

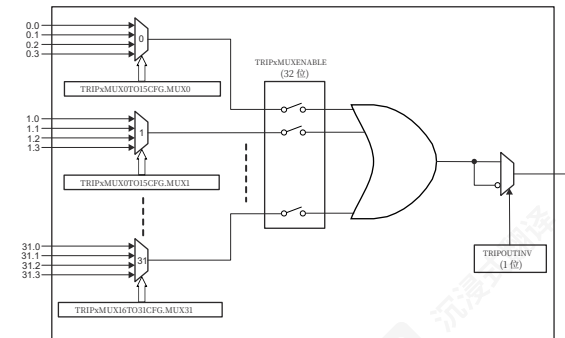


图 3-8. TMS320F2837xD 微控制器的本地 Mux 和逻辑 OR

- 保持时间一致：与上述情况类似，由于没有CPU参与，信号以系统时钟实时传播，不会依赖于另一个模块允许信号通过（图3-9）。这导致系统行为更加可预测和可重复。考虑到许多信号的性质是传递到执行子系统，因此时间尤其关键。

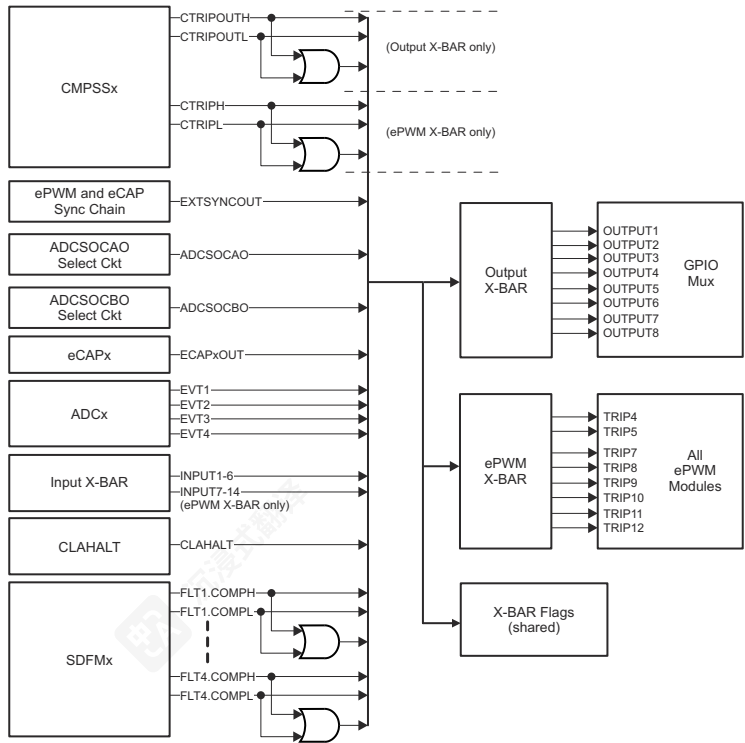


Figure 3-9. X-Bar Sources and Destinations on the TMS320F2837xD MCU

3.5.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

3.5.4 Hardware Platforms and Software Examples

- [C2000 DesignDRIVE Development Kit for Industrial Motor Control](#)
- [Use TI SysConfig Tool to Simplify X-bar Setup](#)

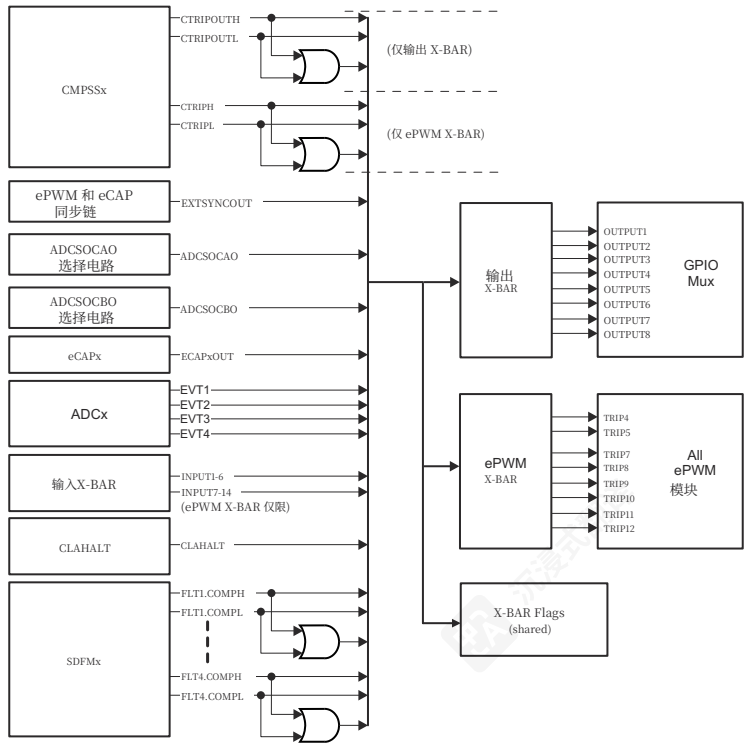
3.5.5 Documentation

- [TMS320F2838xD Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#)

3.6 Improving Control Performance With Nonlinear PID Control

3.6.1 Value Proposition

The nonlinear PID (NLPID) provides the ability to improve control loop performance beyond that normally available with linear controllers. The controller is available in the [Digital Control Library \(DCL\)](#), making it easy to integrate with existing C2000 code.



3.5.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

3.5.4 硬件平台和软件示例

- [C2000 DesignDRIVE 工业电机控制开发套件](#)
- [使用 TI SysConfig 工具简化 X-bar 设置](#)

3.5.5 文档

- [TMS320F2838xD 实时微控制器技术参考手册](#)
- [TMS320F2837xD 实时微控制器技术参考手册](#)
- [TMS320F28004x 实时微控制器技术参考手册](#)
- [TMS320F28003x 实时微控制器技术参考手册](#)
- [TMS320F28002x 实时微控制器技术参考手册](#)
- [TMS320F280013x 实时微控制器技术参考手册](#)
- [TMS320F280015x 实时微控制器技术参考手册](#)

3.6 使用非线性PID控制改进控制性能

3.6.1 价值主张

非线性PID（NLPID）能够提供超越线性控制器通常可用的控制回路性能提升能力。该控制器可在数字控制库（DCL）中获取，便于与现有的C2000代码集成。

3.6.2 In Depth

Conventional linear controllers such as the Proportional, Integral, Derivative (PID) are widely used with digital power applications, including motor control and motion control. The nonlinear PID (DCL Training Video - Non-linear Control) provided with the DCL extends the performance of its linear counterpart by shaping the loop error using a nonlinear law. A nonlinear shaping block is introduced in series with each of the three controller paths as shown below.

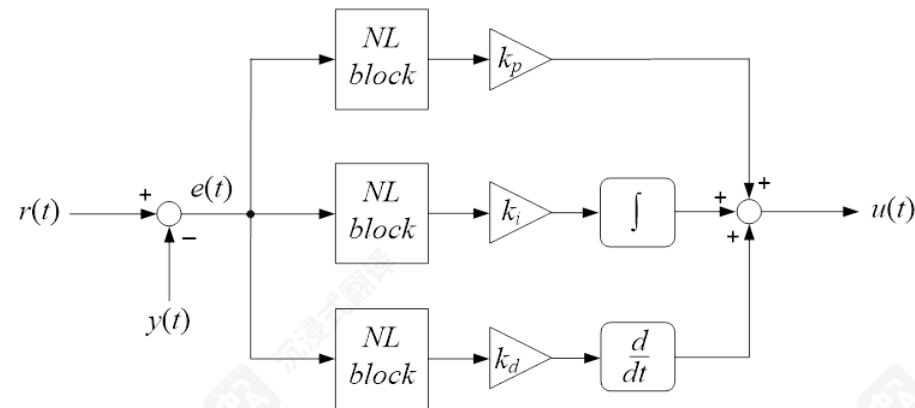


Figure 3-10. Non-Linear PID Block Diagram

The shape and aggressiveness of nonlinear action are configurable via six additional controller parameters (two in each nonlinear block), which are typically tuned in an iterative fashion to optimize a transient response. Like other controllers in the DCL, the NLPID parameters can be updated safely using a shadow parameter set and an update function.

Figure 3-11 shows an example of the potential improvement in step response available from the use of nonlinear control action.

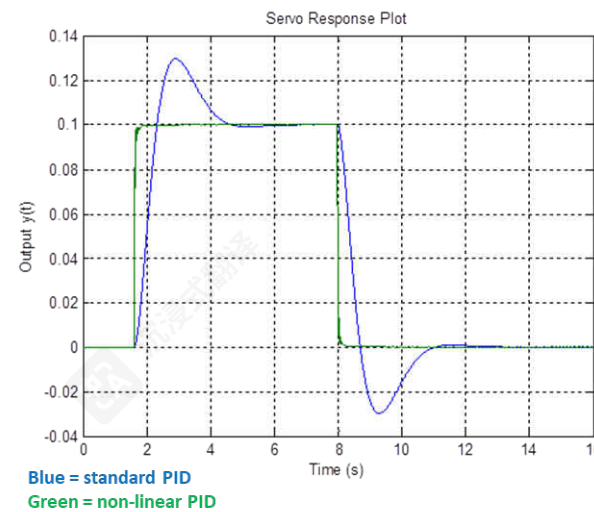


Figure 3-11. Comparison of Response Time Between Linear and Non-linear PID

The NLPID executes with highest efficiency on devices equipped with the type 1 TMU (see Section 3.1), such as the F280025 device. These devices have CPU instructions that allow the nonlinear controller to be executed in 117 cycles compared with around 3,300 cycles without such instructions. This cycle efficiency allows the nonlinear PID to be used in high frequency applications such as switching power supplies and current control loops.

The DCL is packaged in C2000Ware, which is available for free download by C2000 users. The library includes a PID controller tuning guide to help users get the most from the NLPID controller.

3.6.2 深入探讨

传统的线性控制器，如比例-积分-微分（PID）控制器，在数字电源应用中广泛使用，包括电机控制和运动控制。DCL提供的非线性PID（DCL培训视频 - 非线性控制）通过使用非线性法则塑造回路误差，扩展了其线性对应物的性能。每个控制器路径中串联引入了一个非线性整形模块，如下所示。

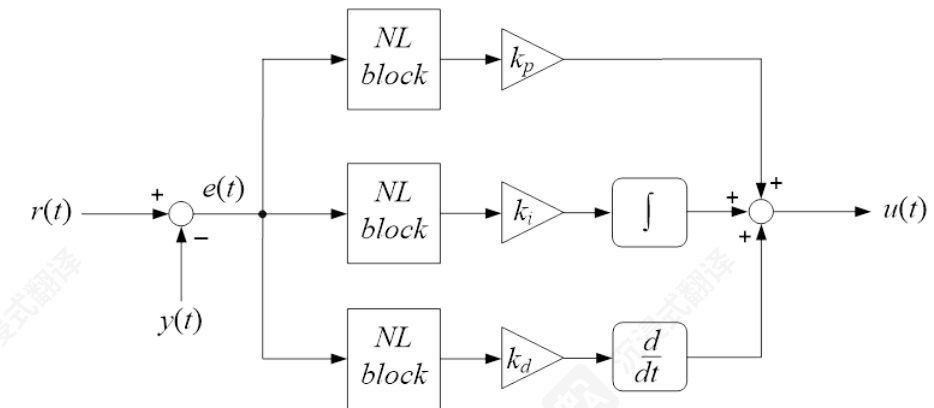


图3-10. 非线性PID框图

非线性动作的形状和侵略性可以通过六个额外的控制器参数（每个非线性模块中有两个）进行配置，这些参数通常以迭代方式调整以优化瞬态响应。与其他DCL中的控制器一样，NLPID参数可以使用影子参数集和更新函数安全地更新。

图3-11显示了使用非线性控制动作可获得的阶跃响应潜在改进的示例。

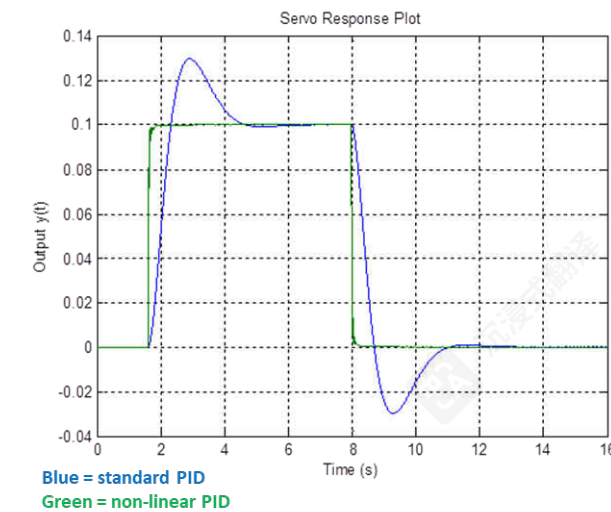


图3-11. 线性PID与非线性PID的响应时间比较

NLPID 在配备类型 1 TMU 的设备上（参见第 3.1 节），例如 F280025 设备，执行效率最高。这些设备具有 CPU 指令，允许非线性控制器在 117 个周期内执行，而没有这些指令则需要大约 3,300 个周期。这种周期效率使得非线性 PID 能够用于高频应用，例如开关电源和电流控制回路。

DCL 打包在 C2000Ware 中，C2000 用户可以免费下载。库中包含一个 PID 控制器调谐指南，以帮助用户充分利用 NLPID 控制器。

3.6.3 Device List

- [TMS320F28003x](#)
- [TMS320F28002x](#)

3.6.4 Hardware Platforms and Software Examples

- [Digital Control Library in C2000Ware](#)
- [F280039C controlCARD](#)
- [F280025 controlCARD](#)

3.6.5 Documentation

- [DCL Training Video - Non-linear Control](#)

3.7 Understanding Flash Memory Performance In Real-Time Control Applications

3.7.1 Value Proposition

In spite of the wait state penalty for flash memory accesses, the 128-bit-wide prefetch logic in association with the pipeline buffer makes sequential code execution performance equal to that of 0 wait state (WS) RAM. Allowing for common code discontinuities, most applications will run with an efficiency of approximately 80% relative to the code executing from RAM. In addition, there is a 128-bit cache on the data bus as well to improve data read performance. All of this performance enhancement comes with zero cycle-cost Error Code Correction (ECC) evaluation.

3.7.2 In Depth

Flash memory is a non-volatile memory that provides the advantage of retaining its content even after a power cycle. However, due to its physical construction Flash memory is typically not as fast as volatile memories (SRAM, DRAM, and so forth). As a result, wait states are used in order to scale the MCU clock rate while accessing Flash memory, which can impact CPU performance. In order to greatly reduce this impact on performance, C2000 FMC's (Flash Module Controller) read interface provides a prefetch mode (Figure 3-12). This mode significantly improves the performance of linear code, which typically makes up the majority of application code in real-time control systems.

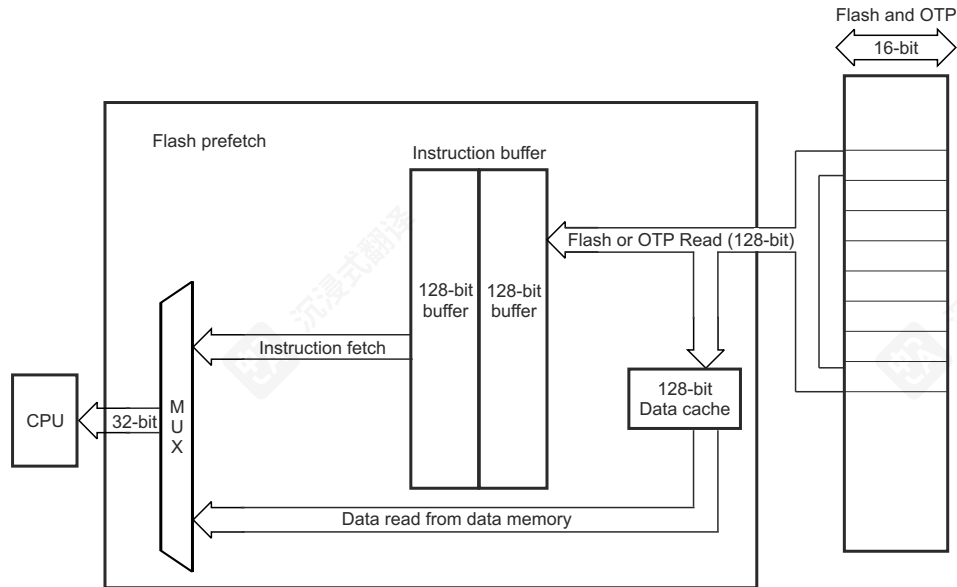


Figure 3-12. C2000 Flash Prefetch Module

3.6.3 设备列表

- [TMS320F28003x](#)
- [TMS320F28002x](#)

3.6.4 硬件平台和软件示例

- [Digital Control Library 在 C2000Ware 中](#)
- [F280039C controlCARD](#)
- [F280025 controlCARD](#)

3.6.5 文档

- [DCL 培训视频 - 非线性控制](#)

3.7 理解实时控制应用中的闪存性能

3.7.1 价值主张

尽管闪存访问存在等待状态惩罚，但与流水线缓冲区配合的128位宽预取逻辑使顺序代码执行性能等同于0等待状态（WS）RAM。考虑到常见代码不连续性，大多数应用程序相对于从RAM执行的代码，效率约为80%。此外，数据总线上还有一个128位缓存以提升数据读取性能。所有这些性能提升都伴随着零周期成本的错误码纠正（ECC）评估。

3.7.2 深入解析

闪存是一种非易失性存储器，即使在电源循环后也能保持其内容。然而，由于其物理结构，闪存通常比易失性存储器（SRAM、DRAM 等等）慢。因此，在访问闪存时，会使用等待状态来扩展 MCU 时钟频率，这会影 CPU 性能。为了大大减少这种对性能的影响，C2000 FMC 的（闪存模块控制器）读接口提供了一个预取模式（图 3-12）。这种模式显著提高了线性代码的性能，而线性代码通常构成实时控制系统应用代码的大部分。

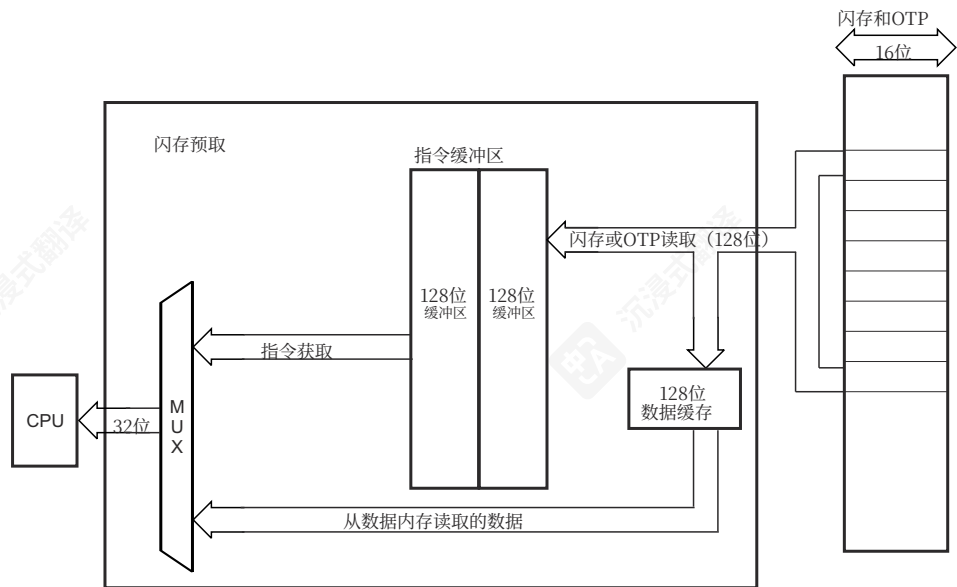


图3-12。C2000闪存预取模块

When enabled on C2000 MCUs, the module does a look-ahead prefetch (128-bit aligned) on linear address increments starting from the last instruction fetch address and stores it in a 128-bit wide by 2-level deep instruction prefetch buffer. This buffer can hold up to sixteen 16-bit instructions and will be continuously filled in the background by the prefetch mechanism as the CPU continues to use the already fetched instructions in the buffer. As a result, no wait states are incurred for each opcode fetch, which is a significant performance boost when compared to a one time wait-stated fetch. Wait states are incurred only when there is a program counter (PC) discontinuity such as a branch, function call, and so forth.

Table 3-5 provides some real-world examples of the code efficiency that can be expected on two different classes of C2000 flash devices.

Table 3-5. Effective Flash Access Times With Prefetch Enable

Part Number	Device Properties	32-Bit Float Math	16-Bit If-Then-Else	ACI Motor Signal Chain
TMS320F2838xD/S TMS320F2837xD/S	200 MHz CPU Clock 50 MHz Flash Speed 3 Wait States	- Flash Access Efficiency: 93% - Effective Performance: 188 MHz	- Flash Access Efficiency: 87% - Effective Performance: 174 MHz	- Flash Access Efficiency: 92% - Effective Performance: 184 MHz
TMS320F28004x TMS320F28002x	100 MHz CPU Clock 20 MHz Flash Speed 4 Wait States	- Flash Access Efficiency: 84% - Effective Performance: 84 MHz	- Flash Access Efficiency: 84% - Effective Performance: 84 MHz	- Flash Access Efficiency: 82% - Effective Performance: 82 MHz
TMS320F28003x	120 MHz CPU Clock 20 MHz Flash Speed 5 Wait States	- Flash Access Efficiency: 70% - Effective Performance: 84 MHz	- Flash Access Efficiency: 78% - Effective Performance: 93 MHz	- Flash Access Efficiency: 73% - Effective Performance: 87 MHz
TMS320F280013x TMS320F280015x	120 MHz CPU Clock 40 MHz Flash Speed 2 Wait States	- Flash Access Efficiency: 96% - Effective Performance: 115 MHz	- Flash Access Efficiency: 91% - Effective Performance: 109 MHz	- Flash Access Efficiency: 95% - Effective Performance: 114 MHz

Note

The number of flash wait states is always 1 less than the access time.

- FLASH_wait_states = FLASH_access_cycles - 1
- FLASH_access_cycles = round_up(CPU MHz/FLASH MHz) For Example: = round_up(165/50) = round_up(3.3) = 4

While the efficiency of program code executing from flash has been considered thus far, there also exists a 128-bit data cache. Users can enable this to increase the flash data read performance. When the CPU requests data from a flash address, the flash wrapper will store entire 128-bits of Flash data (aligned) in this cache instead of simply providing the requested address's data to CPU. The CPU can access the remaining data in this cache without incurring any wait states. This data cache gets flushed and refilled when there is a cache miss.

Finally, there is also an Error Correction Code (ECC) value for each 64-bits of flash memory. The ECC is such that it provides for single bit error correction and dual bit error detection per 64-bits. The ECC is evaluated for correctness before the data is placed into the prefetch buffer with no impact to the access times/latencies mentioned previously. If an uncorrectable error is detected, a non-maskable interrupt is generated to halt normal code execution in parallel to the normal code execution of the CPU. You have the ability to set a threshold for correctable errors to trigger an interrupt to the C28x core as well.

在 C2000 MCU 上启用时，该模块会对从最后一次指令获取地址开始的线性地址增量执行预取（128 位对齐），并将其存储在一个 128 位宽、2 级深的指令预取缓冲区中。该缓冲区最多可以容纳十六个 16 位指令，并且会由预取机制在背景中不断填充，同时 CPU 继续使用缓冲区中已经获取的指令。因此，每个操作码获取都不会产生等待状态，与一次性等待状态的获取相比，这是一个显著的性能提升。只有在程序计数器（PC）出现不连续（如分支、函数调用等等）时才会产生等待状态。

表3-5提供了在两种不同类别的C2000闪存上可以预期的代码效率的一些实际示例。

表3-5。带预取启用的有效闪存访问时间

部件编号	设备属性	32位浮点数学	16位If-Then-Else	ACI Motor Signal Chain
TMS320F2838xD/S TMS320F2837xD/S	200 MHz CPU 时钟 50 MHz Flash 速度 3 等待状态	- 闪存访问效率：93% - 有效性能：188 MHz	- 闪存访问效率：87% - 有效性能：174 MHz	- 闪存访问效率：92% - 有效性能：184 MHz
TMS320F28004x TMS320F28002x	100 MHz CPU 时钟 20 MHz 闪存速度 4 等待状态	- 闪存访问效率：84% - 有效性能：84 MHz	- 闪存访问效率：84% - 有效性能：84 MHz	- 闪存访问效率：82% - 有效性能：82 MHz
TMS320F28003x	120 MHz CPU 时钟 20 MHz 闪存速度 5 等待状态	- 闪存访问效率：70% - 有效性能：84 MHz	- 闪存访问效率：78% - 有效性能：93 MHz	- 闪存访问效率：73% - 有效性能：87 MHz
TMS320F280013x TMS320F280015x	120 MHz CPU 时钟 40 MHz 闪存速度 2 等待状态	- 闪存访问效率：96% - 有效性能：115 MHz	- 闪存访问效率：91% - 有效性能：109 MHz	- 闪存访问效率：95% - 有效性能：114 MHz

Note

闪存等待状态的数量总是比访问时间少1。

- FLASH_wait_状态 = FLASH_访问_周期 - 1
- FLASH_访问_周期 = 轮_向上(单位：CPU MHz/FLASH MHz) 例如：= 轮_向上(165/50) = 轮_向上(3.3) = 4

虽然到目前为止已经考虑了从闪存执行程序代码的效率，但还存在一个128位数据缓存。用户可以启用此缓存以提高闪存数据读取性能。当CPU从闪存地址请求数据时，闪存包装器会将整个128位的Flash数据（对齐）存储在此缓存中，而不是仅向CPU提供请求地址的数据。CPU可以无等待状态地访问此缓存中的剩余数据。当发生缓存未命中时，此数据缓存会被刷新并重新填充。

最后，每个64位的闪存也有一个错误校正码（ECC）值。ECC的作用是每64位提供单比特错误校正和双比特错误检测。在数据被放入预取缓冲区之前，会评估ECC的正确性，而不会影响之前提到的访问时间/延迟。如果检测到不可纠正的错误，会生成一个不可屏蔽中断，以并行方式暂停CPU的正常代码执行。您还可以设置一个可纠正错误的阈值，以触发对C28x核心的中断。

3.7.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

3.7.4 Hardware Platforms and Software Examples

All code examples for the supported devices enable the prefetch buffer and ECC logic as part of their initialization routines inside the SysCtrl.c file.

3.7.5 Documentation

- [Signal Chain Benchmarking - A Demonstration of Optimized Real-time Performance of C2000™ MCU](#)
- [TMS320F2838x Real-Time Microcontrollers With Connectivity Manager Data Sheet](#) (for more information, see the *Flash Parameters* section)
- [TMS320F2837xD Dual-Core Microcontrollers Data Sheet](#) (for more information, see the *Flash Parameters* section)
- [TMS320F28004x Microcontrollers Data Sheet](#) (for more information, see the *Flash Parameters* section)
- [TMS320F28003x Real-Time Microcontrollers Data Sheet](#) (for more information, see the *Flash Parameters* section)
- [TMS320F28002x Real-Time Microcontrollers Data Sheet](#) (for more information, see the *Flash Parameters* section)
- [TMS320F280013x Real-Time Microcontrollers Data Sheet](#) (for more information, see the *Flash Parameters* section)
- [TMS320F280015x Real-Time Microcontrollers Data Sheet](#) (for more information, see the *Flash Parameters* section)

3.8 Deterministic Program Execution With the C28x DSP Core

3.8.1 Value Proposition

One important aspect of any real-time control system is the consistency of program execution over time. Whether it is performance across the system update period or from multiple power ups over time, optimization of real-time systems relies on critical system events taking place at known points in time. The well defined 8 stage CPU pipeline of the C28x MCU, as well as the complementary behavior of its interrupt handling logic, provides this level of determinism.

3.8.2 In Depth

A processor that is not consistent in its behavior over time can cause perturbations to a real-time control system, either by not actuating the system in a timely manner or by sampling the state of the system at the incorrect point in time. Furthermore, real-time control systems are largely interrupt driven in their program flow. For this reason, a cache memory is undesirable since it would need to be discarded often whenever a program discontinuity occurs (in this case quite often).

Instead, large amounts of fast memory are desirable for program execution as well as a processor with a instruction pipeline that is deep enough to parallelize instructions, but also shallow enough to not incur large time penalties when discontinuities occur. The C28x CPU employees an 8 stage pipeline as shown in [Figure 3-13](#). Once an instruction has entered the D2 phase of the pipeline, it cannot be stopped from full execution by an interrupt. Conversely, any instruction that is in a pre-D2 phase of the pipeline will be flushed when the incoming interrupt is received by the C28x core. Upon returning from the interrupt program, execution begins again with the F1 fetch stage. It is beyond the scope of this article to go deeper into the nuances of the pipeline, but those details are covered in the [TMS320C28x DSP CPU and Instruction Set Reference Guide](#).

3.7.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

3.7.4 硬件平台和软件示例

所有支持的设备的代码示例在其初始化例程中启用预取缓冲区和ECC逻辑，这些例程位于SysCtrl.c文件内。

3.7.5 文档

- [信号链基准测试 - C2000™微控制器的优化实时性能演示](#)
- [TMS320F2838x 实时微控制器带连接管理器数据表](#)（更多信息，请参阅闪存参数部分）
- [TMS320F2837xD 双核微控制器数据表](#)（更多信息，请参阅闪存参数部分）
- [TMS320F28004x 微控制器数据表](#)（更多信息，请参阅闪存参数部分）
- [TMS320F28003x 实时微控制器数据表](#)（更多信息，请参阅闪存参数部分）
- [TMS320F28002x 实时微控制器数据表](#)（更多信息，请参阅闪存参数部分）
- [TMS320F280013x 实时微控制器数据表](#)（更多信息，请参阅闪存参数部分）
- [TMS320F280015x 实时微控制器数据手册](#)（更多信息，请参阅闪存参数部分）

3.8 使用 C28x DSP 内核的确定性程序执行

3.8.1 价值主张

任何实时控制系统的一个重要方面是程序执行随时间的稳定性。无论是系统更新期间的性能还是多次上电随时间的性能，实时系统的优化都依赖于关键系统事件在已知的时间点发生。C28x MCU 的 8 级 CPU 管道以及其中断处理逻辑的补充行为提供了这种级别的确定性。

3.8.2 深入探讨

一个处理器如果随时间的行为不一致，可能会对实时控制系统造成扰动，要么不能及时地驱动系统，要么在错误的时间点采样系统的状态。此外，实时控制系统的程序流在很大程度上是由中断驱动的。因此，缓存内存是不受欢迎的，因为它在程序不连续发生时（在这种情况下相当频繁）需要经常被丢弃。

相反，程序执行需要大量快速内存，同时还需要一个指令流水线深度足够以并行化指令，但又足够浅以避免在出现不连续时产生大的时间惩罚的处理器。C28x CPU 采用 8 级流水线，如图 3-13 所示。一旦指令进入流水线的 D2 阶段，中断就无法阻止其完整执行。相反，任何处于流水线 D2 阶段之前的指令在 C28x 核收到中断时都会被刷新。从中断程序返回后，执行将从 F1 取指阶段重新开始。本文的范围之外是更深入地探讨流水线的细节，但那些细节在《TMS320C28x DSP CPU 和指令集参考指南》中有所涵盖。

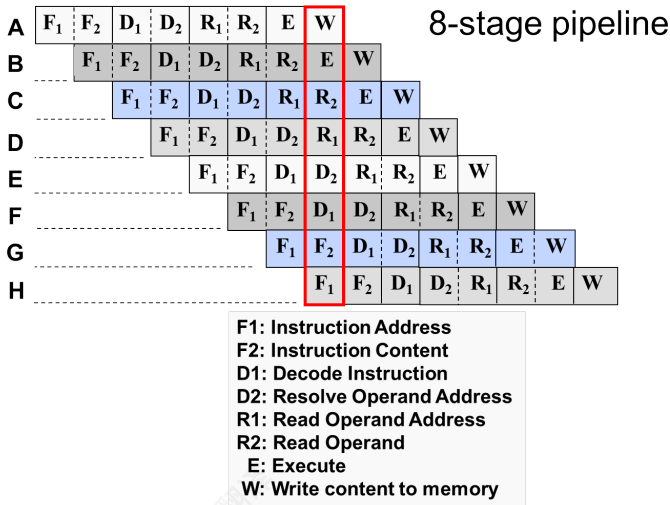


Figure 3-13. C28x Pipeline Visualization

While you can see that the behavior of the program execution is repeatable over time, what about the behavior of the incoming interrupt? Figure 3-14 shows that this, too, is deterministic. Note that unless manually altered all other interrupt requests will pend until the current interrupt has been fully serviced. This is important in order to keep servicing of any interrupt consistent in the time domain once it begins. Both of these components of the C28x core help maintain deterministic code execution for the system.

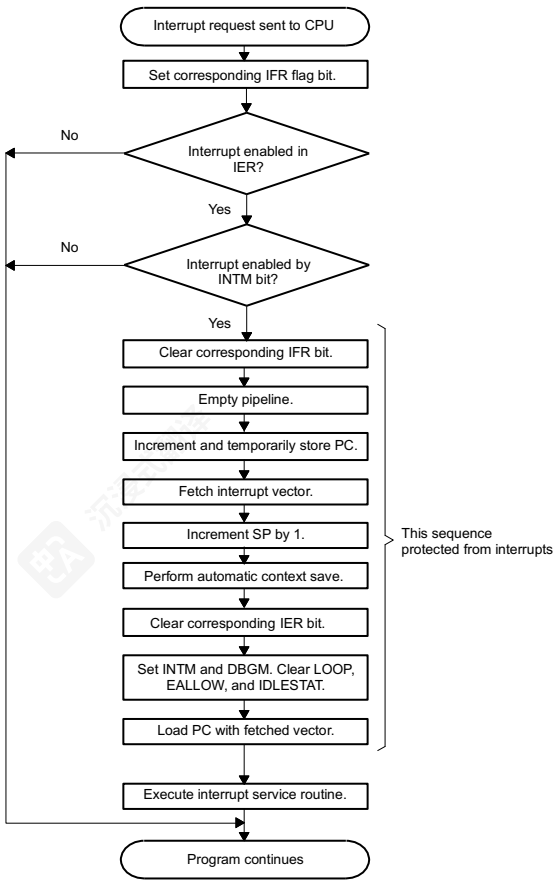


Figure 3-14. Standard Operation for a C28x CPU Maskable Interrupt

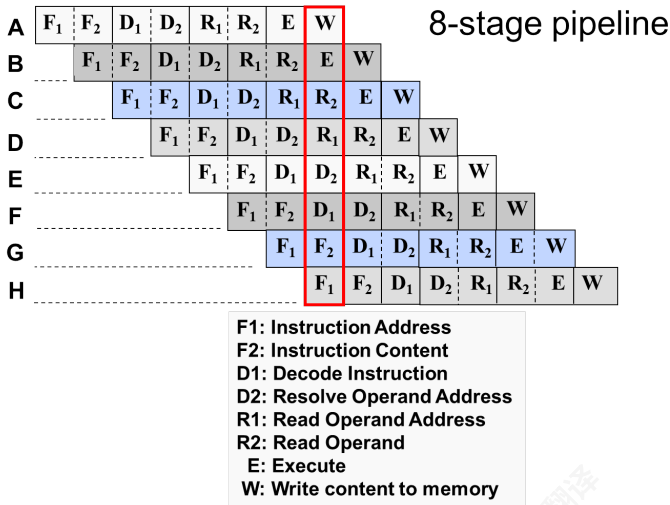


图 3-13。C28x 管道可视化

虽然你可以看到程序执行的行为在时间上是可重复的，那么中断的行为呢？图 3-14 显示，这也具有确定性。请注意，除非手动更改，否则所有其他中断请求都会等到当前中断完全服务完成后才挂起。这对于在开始后保持任何中断在时间域内的一致性服务非常重要。C28x 核的这两个组件都有助于系统保持确定性代码执行。

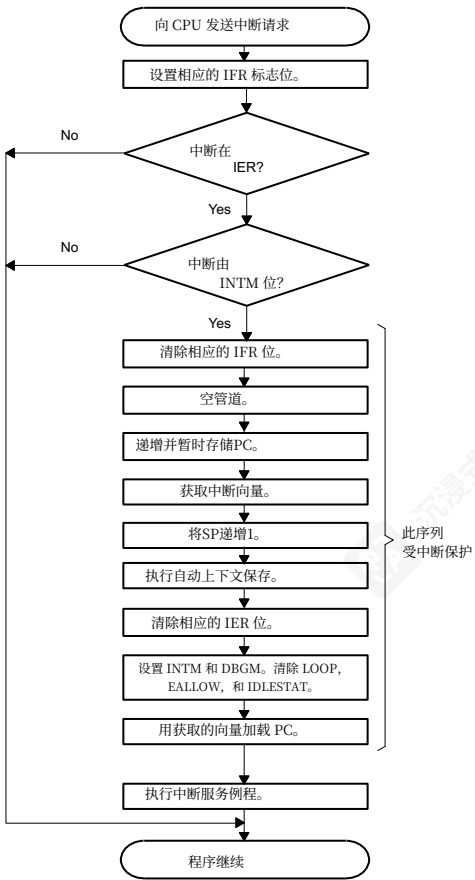


图 3-14。C28x CPU 可屏蔽中断的标准操作

3.8.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

3.8.4 Hardware Platforms and Software Examples

- [F28388D controlCARD evaluation module](#)
- [C2000 MCU F28379D LaunchPad™ development kit](#)
- [C2000 MCU F280049C LaunchPad™ development kit](#)
- [F280039C controlCARD evaluation module](#)
- [F280025 controlCARD evaluation module](#)
- [F2800137 controlCARD evaluation module](#)
- [F2800157 controlCARD evaluation module](#)

3.8.5 Documentation

- [TMS320C28x CPU and Instruction Set Reference Guide](#)
- [C2000™ Academy Overview](#)

3.9 Efficient Live Firmware Updates (LFU) and Firmware Over-The-Air (FOTA) updates

3.9.1 Value Proposition

Industrial end equipment like Server Power Supply Units (PSU) require LFU support, so that a firmware update does not take the system offline. Automotive markets require FOTA support, where the firmware of any supported ECU (Electronic Control Unit) within the vehicle can be remotely updated. C2000 real-time MCUs have hardware features like multi flash bank support, interrupt vector table swapping, RAM block swapping and software examples, to efficiently perform both LFU and FOTA.

3.9.2 In Depth

Terminology:

It is important to understand the terminology since there is no single standard that defines this, at least for LFU. The following definitions are used:

- Installation – transfer of the firmware application or “image” from a host controller to the target (on which the firmware will execute).

This also includes programming the new firmware on the Flash memory of the target device.
- Activation – as the name suggests, activating the new firmware application. In some of the LFU documentation, the term "Switchover" is also used
- LFU – installation and activation of new firmware occurs without a device reset
- FOTA – installation of new firmware occurs while the old firmware is running, but activation of new firmware requires a device reset
- Swapping – ability to map more than one physical hardware blocks to the same address space. Hardware could refer to flash banks, RAM blocks, interrupt vector tables, and so forth.

Building blocks:

A number of Software and Hardware pieces are available to enable system level features like LFU and FOTA:

- Multiple flash banks – for example, one flash bank containing the active firmware, and the others containing inactive firmware.
- Hardware features to enable LFU – for example, interrupt vector table swapping and RAM block swapping, which enable fast activation for LFU.

3.8.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

3.8.4 硬件平台和软件示例

- [F28388D controlCARD 评估模块](#)
- [C2000 MCU F28379D LaunchPad™ 开发套件](#)
- [C2000 MCU F280049C LaunchPad™ 开发套件](#)
- [F280039C controlCARD 评估模块](#)
- [F280025 controlCARD 评估模块](#)
- [F2800137 controlCARD 评估模块](#)
- [F2800157 controlCARD 评估模块](#)

3.8.5 文档

- [TMS320C28x CPU 和指令集参考指南](#)
- [C2000™ 学院概览](#)

3.9 高效的实时固件更新 (LFU) 和固件空中下载 (FOTA) 更新

3.9.1 价值主张

工业终端设备如服务器电源单元 (PSU) 需要LFU支持，以便固件更新不会使系统离线。汽车市场需要FOTA支持，其中车辆内任何受支持的ECU（电子控制单元）的固件都可以远程更新。C2000实时MCU具有多闪存银行支持、中断向量表切换、RAM块切换和软件示例等硬件特性，以高效地执行LFU和FOTA。

3.9.2 深入了解

术语：

理解术语很重要，因为没有单一的规范来定义这一点，至少对于LFU来说是这样。以下定义被使用：

- 安装 – 将固件应用程序或“镜像”从主机控制器转移到目标（固件将在其上执行）。

这也包括在目标设备的闪存上编程新的固件。
- 激活 – 顾名思义，激活新的固件应用程序。在一些LFU文档中，术语 "Switchover" 也被使用
- LFU – 新固件的安装和激活无需设备重置
- FOTA – 新固件的安装发生在旧固件运行时发生，但新固件的激活需要设备重置

- Swapping – 能够将多个物理硬件块映射到同一地址空间。硬件可以指闪存组、RAM 块、中断向量表等。

构建模块：

一些软件和硬件组件可用于启用系统级功能，如LFU和FOTA：

- 多个闪存组 – 例如，一个闪存组包含活动固件，其他闪存组包含非活动固件。
- 硬件特性以启用LFU——例如，中断向量表交换和RAM块交换，这些特性可快速激活LFU。

- Compiler – an LFU aware compiler, which makes it easier for the user to integrate LFU support into their solution. For LFU, the compiler uses the existing firmware executable as a reference to generate the new firmware executable, which allows activation to be fast and efficient by optimizing the LFU initialization routine.
- Flash kernels – these refer to bootloaders that are provided as examples, reside in Flash, and support LFU and FOTA functionality. They can interact with a host to install firmware, and contain bank selection logic to determine, after a device reset, whether a flash bank contains valid firmware, and which firmware needs to be executed. Bank selection logic is also now built into the Boot ROM of a C2000 MCU and available as a boot mode.
- LFU software examples – simple examples as well as a reference design and a detailed user guide that walks through all the building blocks and LFU flow, enabling them to quickly integrate LFU into their system and achieve optimal performance. This is illustrated with examples of LFU on both the C28x CPU as well as the CLA. With a few changes, these examples can be repurposed for FOTA as well.

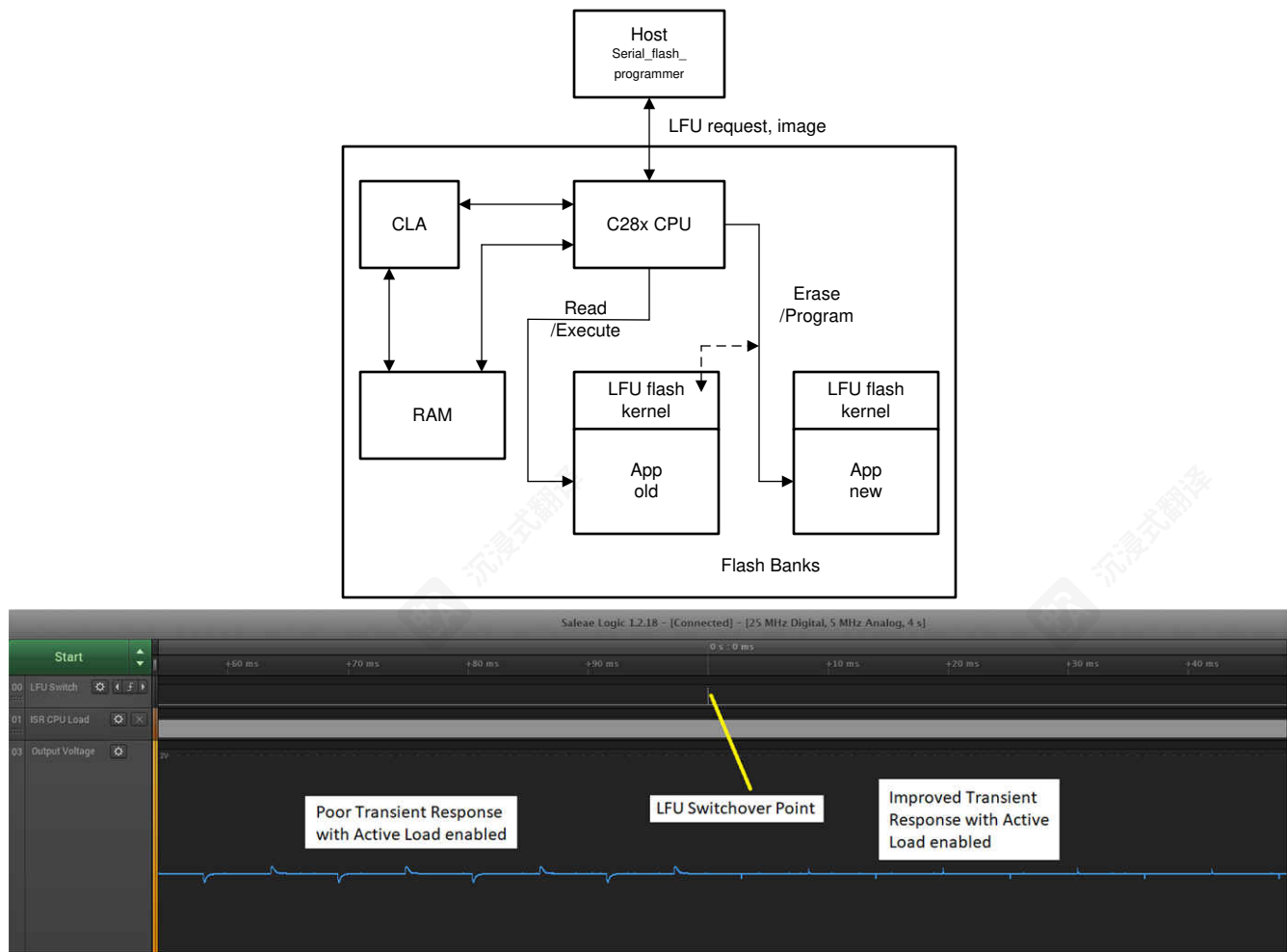
There are two flows defined for a LFU:

- At production:
 - Program Flash kernel, firmware in one or more flash banks
- In the field:
 - Option 1 - with this approach, you need to know which flash bank the firmware update is targeted for:
 - Developer creates new image (using previous image as reference image and Compiler with LFU support) knowing which flash bank it is targeted to
 - Host initiates LFU – downloads image corresponding to inactive bank
 - Flash kernel installs image to inactive bank
 - Old App (with LFU software/hardware support) activates new App without device reset (if previous step was successful)
 - Presence of old App allows fallback option in case of failure
 - Option 2 – with this approach, you **do not** need to know which Flash bank the firmware update is targeted for:
 - Developer creates two new images (using previous image as reference image and Compiler with LFU support) without knowing which flash bank it is targeted to. For example, if 2 Flash banks are used, and the new version is vN, the user would create vN built for Bank 1 using as reference vN-1 built for Bank 0. The user would also create vN built for Bank 0 using as reference vN-1 built for Bank 1.
 - Host initiates LFU – transfers **both images** to the target device
 - Flash kernel installs only the image targeted to the inactive bank, ignores the image targeted to the active bank
 - Old App (with LFU software/hardware support) activates new App without device reset (if previous step was successful)
 - Presence of old App allows fallback option in case of failure

- 编译器 – 一个感知LFU的编译器，它使用户更容易将LFU支持集成到他们的解决方案。对于LFU，com piler使用现有的 g 固件可执行文件作为参考 g生成新的固件可执行文件，这允许通过优化LFU初始化例程来快速高效地激活
- 闪存内核——这些是指作为示例提供的引导加载程序，驻留在闪存中，并支持LFU和FOTA功能的引导加载程序。它们可以与主机交互来安装固件，并包含银行选择逻辑，以确定在设备重置后闪存银行是否包含有效固件，以及需要执行的固件。银行选择逻辑现在也构建在C2000微控制器的引导ROM中，并作为引导模式提供。
- LFU软件示例——简单的示例、参考设计和详细用户指南，涵盖了所有构建块和LFU流程，使他们能够快速将LFU集成到他们的系统中并实现最佳性能。这通过C28x CPU和CLA上的LFU示例来说明。通过一些更改，这些示例也可以用于FOTA。

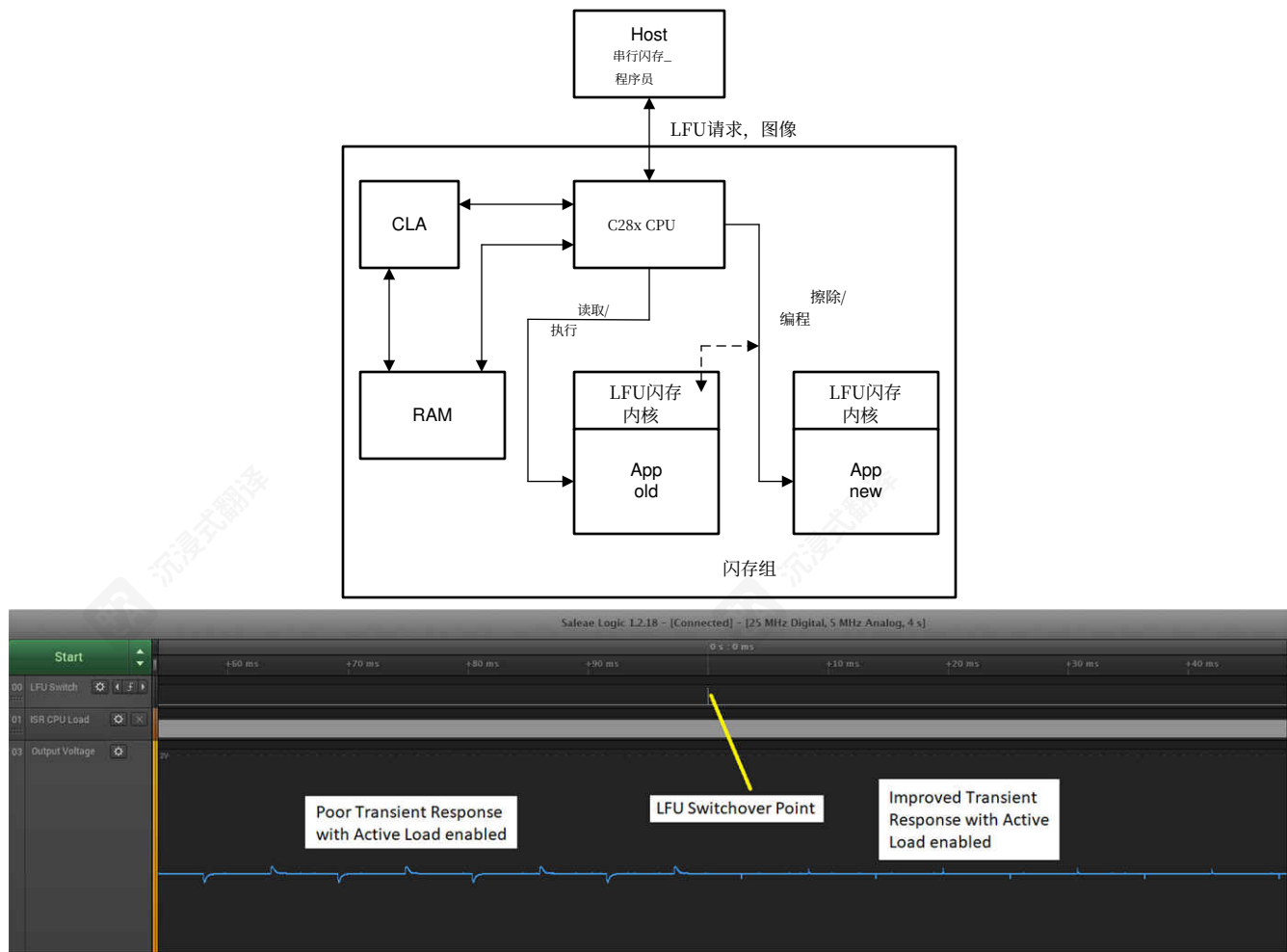
为LFU定义了两种流程：

- 在生产中：
 - 程序闪存内核，固件在一个或多个闪存银行中
- 在现场：
 - 选项 1 - 使用此方法，您需要知道固件更新针对的是哪个闪存组：
 - 开发者创建新镜像（使用先前镜像作为参考镜像并使用支持 LFU 的编译器），知道其针对的闪存组
 - 主机启动 LFU – 下载对应于非活动组的镜像
 - 闪存内核将镜像安装到非活动组
 - 旧应用程序（支持 LFU 软件/硬件）在不重置设备的情况下激活新应用程序（如果上一步成功）
 - 选项 2 – 使用这种方法，您不需要知道固件更新针对哪个闪存组 for:
 - 开发者创建两个新镜像（使用先前镜像作为参考镜像和带 LFU 支持的编译器），而不知道它针对哪个闪存组。例如，如果使用 2 个闪存组，并且新版本是 vN，用户会使用 vN-1（为 Bank 0 构建）作为参考镜像，为 Bank 1 创建 vN。用户还会使用 vN-1（为 Bank 1 构建）作为参考镜像，为 Bank 0 创建 vN。
 - 主机启动 LFU – 将两个镜像传输到目标设备
 - 闪存内核仅安装针对非活动闪存组的镜像，忽略针对活动闪存组的镜像
 - 旧应用程序（带 LFU 软件或硬件支持）在不重置设备的情况下激活新应用程序（如果前一步成功）
 - 旧应用程序的存在允许在失败时回退选项



There are two flows applicable to FOTA as well:

- At production:
 - Program Flash kernel, firmware in one or more flash banks
- In the field:
 - Without hardware support for Flash bank swapping, the factory would need to know which Flash Bank (Bank1 or Bank0) to build the firmware for. This approach is not acceptable because with automotive FOTA, you may freely update from any version to any other version, that is, skip versions, so it is impossible to pose a constraint like this
 - So a viable approach would be to always build the firmware to be Loaded to Bank1, and Run from Bank0. Since swapping is not available, the activation process involves copying the image from Bank1 to Bank0. The problem here is once the firmware is copied from Bank1 to Bank0, both Banks have the same new firmware now, without a valid backup present in case a rollback is needed. Since a rollback is an essential requirement with FOTA, a 3rd partition, or Flash bank, is needed
 - Developer creates new firmware to be loaded to Bank 1 and run from Bank 0
 - Host (FOTA Master ECU) initiates FOTA – transfers image to target device
 - Flash kernel installs image to Bank 1
 - Host initiates device reset. The old image in Bank 0 is copied by the Flash kernel to Bank 2 to be available in case a rollback is needed. Then the new image in Bank 1 is copied by the Flash kernel to Bank 0. Then the regular image activation steps occur



FOTA 适用两种流程：

- 在生产时：
 - 程序闪存内核，固件在单个或多个闪存组中
- 在领域：
 - 没有硬件支持闪存组切换，工厂需要知道哪个闪存组 (Bank1 或 Bank0) 来构建固件。这是一个 pp 蟑螂不是接受 p 表格 因为 使用汽车 FOTA，你可以自由地从任何版本更新到任何其他版本，也就是说，可以跳过版本，所以不可能提出这种约束
 - 因此，一个可行的方案是始终构建要加载到Bank1的固件，并从Bank0运行。由于交换不可用，激活过程涉及将镜像从Bank1复制到Bank0。这里的问题是，一旦固件从Bank1复制到Bank0，两个Bank现在都有相同的新固件，在没有有效备份的情况下，如果需要回滚，就无法提供。由于回滚是FOTA的一个基本要求，因此需要一个第3分区，或Flash bank
 - 开发者创建新的固件以加载到Bank 1并从Bank 0运行
 - 主机（FOTA主ECU）启动FOTA——将镜像传输到目标设备
 - Flash内核将镜像安装到Bank 1
 - 主机启动设备重置。Bank 0中的旧镜像被Flash内核复制到Bank 2以备使用如果需要回滚。然后Bank 1中的新镜像会被Flash内核复制到Bank 0。然后常规镜像激活步骤发生

- Constraints:
 - FOTA without Flash bank swapping requires three Flash banks or partitions
 - In general, with LFU, firmware updates cannot be skipped since the reference image is needed for activation to be fast and efficient.
 - With FOTA, firmware updates can be skipped since there is no constraint on activation time, and activation occurs after a device reset, so a reference image is not needed

3.9.3 Device List

- [TMS320F2837xS](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)

3.9.4 Hardware Platforms and Software Examples

- [Live firmware update reference design with C2000 real-time MCUs](#)
- Flash kernel example illustrating LFU with device reset on F28002x (single flash bank)
- C2000Ware Flash kernel examples with LFU build configurations: [F28003x](#), [F28004x](#)

3.9.5 Documentation

- [Live Firmware Update With Device Reset on C2000™ MCUs](#)
- [Live Firmware Update Without Device Reset on C2000™ MCUs](#)
- [Serial Flash Programming of C2000 Microcontrollers](#)
- [TMS320F28003x Real-Time Microcontrollers](#)

4 Control Key Technologies

4.1 Reducing Limit Cycling in Control Systems With C2000 HRPWMs

4.1.1 Value Proposition

Limit cycling in a PWM controlled system refers to the in-ability for the PWM output to physically converge on the mathematical solution to the control law. This causes the PWM output to cycle about the true solution, resulting in instability in the control system. The High Resolution PWM (HRPWM) module on the C2000 MCU has the ability to modulate the PWM edge in 150 ps increments. This represents a 60 fold improvement over traditional PWM creation techniques based off the system clock rate ([Figure 4-1](#)) and can be used to realize a higher order of accuracy in PWM edge placement. A waveform's period phase relationship to its complement, as well as deadband insertion time can all realize this high resolution benefit.

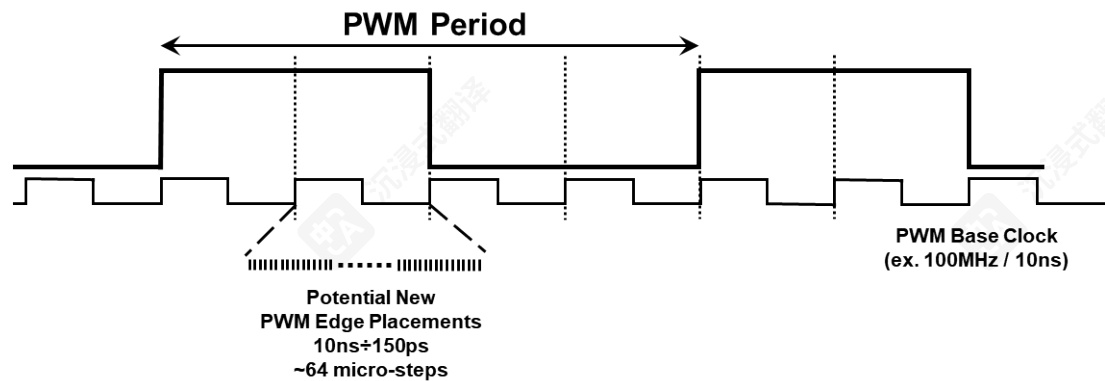


Figure 4-1. HRPWM Capability vs Traditional PWM Generation Methods

- 约束条件:
 - 无需闪存切换的FOTA需要三个闪存或分区
 - 通常情况下，使用LFU，固件更新不能被跳过，因为需要参考镜像才能快速高效地激活。
 - 使用FOTA，固件更新可以被跳过，因为没有激活时间的限制，激活发生在设备重置之后，因此不需要参考镜像

3.9.3 设备列表

- [TMS320F2837xS](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)

3.9.4 硬件平台和软件示例

- [C2000 实时 MCU 的实时固件更新参考设计](#)
- 闪存内核示例 [ple illustrating F28002x 上的设备重置 LFU](#) (single flash bank)
- C2000Ware 闪存内核示例，使用LFU构建配置：[F28003x](#)、[F28004x](#)

3.9.5 文档

- [在C2000™ MCU上执行设备重置的固件更新](#)
- [在C2000™微控制器上进行无需设备重置的固件更新](#)
- [Serial Flash Programming of C2000 Microcontrollers](#)
- [TMS320F28003x 实时微控制器](#)

4 控制关键技术

4.1 使用 C2000 HRPWM 降低控制系统中的极限循环

4.1.1 价值主张

在PWM控制系统中，限制循环是指PWM输出无法在控制律的数学解上物理收敛。这导致PWM输出在真实解周围循环，从而引起控制系统的不稳定。C2000微控制器的HRPWM（高分辨率PWM）模块能够在150 ps的增量下调制PWM边缘。这比基于系统时钟频率的传统PWM创建技术提高了60倍（图4-1），并可用于实现PWM边缘放置的更高精度。波形周期相位与其补相位的关系，以及死区插入时间都可以实现这种高分辨率优势。

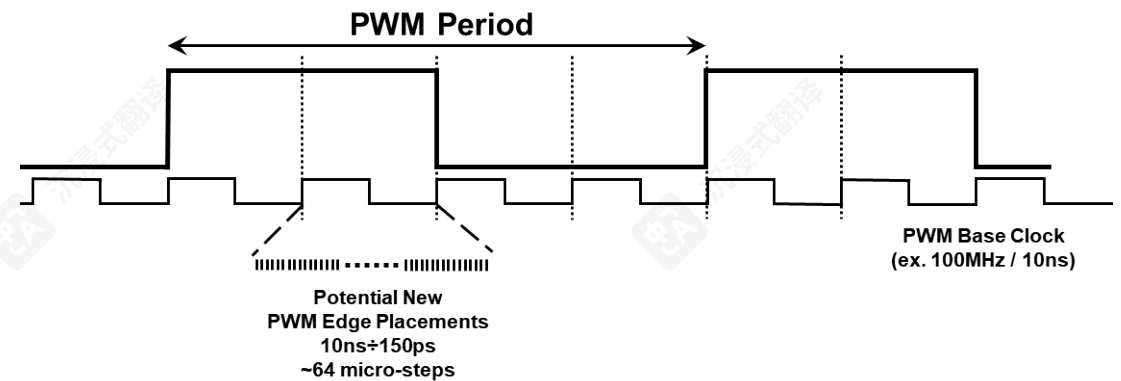


图 4-1. HRPWM 能力与传统 PWM 生成方法对比

4.1.2 In Depth

All PWM-controlled power topologies are inherently bandwidth limited by the ability of the controller to place the PWM edge as close as possible to the mathematical solution of the control law. Whatever error exists from “rounding” the solution created in the form of an output PWM signal dictates the maximum efficiency that can be realized in the system.

In this sense, it may be helpful to think of the PWM as a type of DAC with a fixed resolution. Any error that results in the selection of the next available PWM edge placement would then be equivalent to quantization error term that is inherent to any DAC. Therefore, the minimum time step that can be achieved by a PWM module can be translated into “bits” of resolution of this equivalent DAC.

As shown in [Table 4-1](#), the increase of resolution of the C2000 MCU HRPWM vs a traditional PWM is very apparent, increasing the effective resolution by approximately 6 bits.

Table 4-1. Resolution for PWM vs HRPWM

PWM Freq (kHz)	Regular Resolution (PWM) 100 MHz EPWMCLK		High Resolution (HRPWM)	
	Bits	% Error	Bits	% Error
20	12.3	0.02	18.1	0.000
50	11	0.05	16.8	0.001
100	10	0.1	15.8	0.002
150	9.4	0.15	15.2	0.003
200	9	0.2	14.8	0.004
250	8.6	0.25	14.4	0.005
500	7.6	0.5	13.4	0.009
1000	6.6	1	12.4	0.018
1500	6.1	1.5	11.9	0.027
2000	5.6	2	11.4	0.036

4.1.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- <https://www.ti.com/product/TMS320F2800157>

4.1.4 Hardware Platforms and Software Examples

- [TIDM-02002 Bidirectional CLLLC resonant dual active bridge for HEV/EV onboard charger](#)
- [TIDA-00961 Highly Efficient 1.6kW High Density GaN Based 1MHz CrM Totem-pole PFC Converter](#)
- [TIDA-010054 Bi-directional, dual active bridge reference design for level 3 electric vehicle charging stations](#)
- [C2000Ware HRPWM Example for TMS320F28388D](#)
- [C2000Ware HRPWM Example for TMS320F28379D](#)
- [C2000Ware HRPWM Example for TMS320F280049C](#)
- [C2000Ware HRPWM Example for TMS320F280039C](#)
- [C2000Ware HRPWM Example for TMS320F280025](#)
- [C2000Ware HRPWM Example for TMS320F2800137](#)
- [C2000Ware HRPWM Example for TMS320F2800157](#)

4.1.2 深入解析

所有PWM控制功率拓扑结构本质上都受限于控制器将PWM边缘尽可能接近控制律数学解的能力。从“四舍五入”形式输出的PWM信号所形成的解中存在的任何误差，决定了系统能够实现的最高效率。

在这个意义上，将PWM视为一种具有固定分辨率的DAC可能会有所帮助。任何导致选择下一个可用PWM边缘位置的错误，然后等同于任何DAC固有的量化误差项。因此，PWM模块可以实现的最小时间步长可以转换为等效DAC的“位”分辨率。

如表4-1所示，C2000 MCU HRPWM与传统PWM的分辨率提升非常明显，有效分辨率提高了约6位。

表4-1. PWM与HRPWM的分辨率

PWM频率 (kHz)	常规分辨率(PWM) 100 MHz EPWMCLK		高分辨率 (HRPWM)	
	Bits	% 误差	Bits	% 误差
20	12.3	0.02	18.1	0.000
50	11	0.05	16.8	0.001
100	10	0.1	15.8	0.002
150	9.4	0.15	15.2	0.003
200	9	0.2	14.8	0.004
250	8.6	0.25	14.4	0.005
500	7.6	0.5	13.4	0.009
1000	6.6	1	12.4	0.018
1500	6.1	1.5	11.9	0.027
2000	5.6	2	11.4	0.036

4.1.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- <https://www.ti.com/product/TMS320F2800157>

4.1.4 硬件平台和软件示例

- [TIDM-02002 双向 CLLLC 谐振双有源桥式电路，用于 HEV/EV 车载充电器](#)
- [TIDA-00961 高效 1.6kW 高密度氮化镓 1MHz CrM 总线型 PFC 转换器](#)
- [TIDA-010054 双向、双有源桥式电路参考设计，用于三级电动汽车充电站](#)
- [C2000Ware HRPWM 示例，适用于 TMS320F28388D](#)
- [C2000Ware HRPWM 示例 for TMS320F28379D](#)
- [C2000Ware HRPWM 示例 for TMS320F280049C](#)
- [C2000Ware HRPWM 示例 for TMS320F280039C](#)
- [C2000Ware HRPWM 示例 for TMS320F280025](#)
- [C2000Ware HRPWM 示例 for TMS320F2800137](#)
- [C2000Ware HRPWM 示例 for TMS320F2800157](#)

4.1.5 Documentation

- [TMS320F2838xD Dual-Core Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F2837xD Dual-Core Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#)
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#)
- [C2000 F2837xD Microcontroller 1-Day Workshop Section 1.6: Control Peripherals](#) skip to 14:38 for HRPWM

4.2 Shoot Through Prevention for Current Control Topologies With Configurable Deadband

4.2.1 Value Proposition

C2000 MCUs have the ability to implement current control techniques, such as Peak Current Mode Control (PCMC), in hardware using the on-chip comparators to control the PWM duty cycle. Variable deadband insertion has been added to the Type 4 PWM, enabling the ability to tune out potential shoot through, without the need for any CPU overhead.

4.2.2 In Depth

For better power efficiency, many DC-DC systems implement a synchronous boost controller, where the secondary switch replaces the feed forward diode that exists in a regular boost controller (Figure 4-2). Peak Current Mode Control is one of the more common methods used to control this topology and the C2000 MCU has some unique features that allow it to implement this type of control very efficiently.

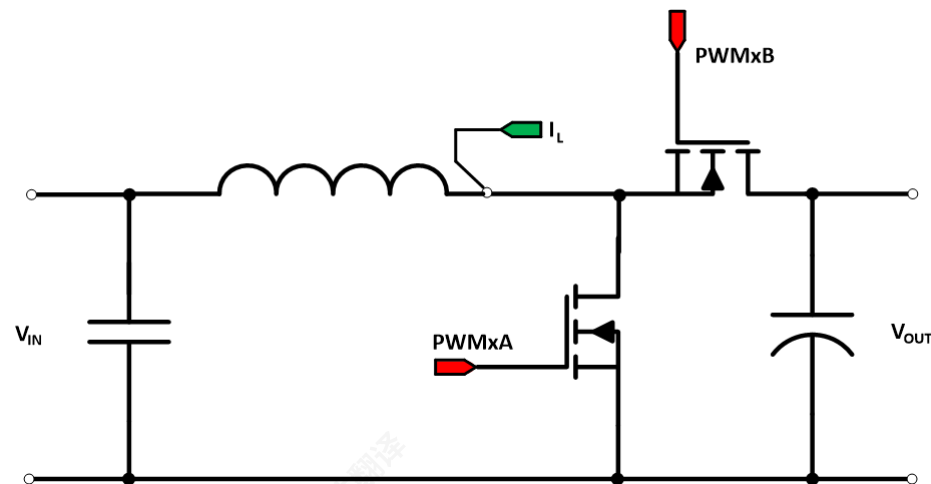


Figure 4-2. Synchronous Boost Controller

The addition of the second FET demands precise control of the ON/OFF time of the primary and secondary switch relative to one another. If the switches are in the “ON” state at the same time, there is a direct path to ground for the active current to flow, which is not only in-efficient but also potentially harmful to the lifetime of the FET switches.

An accurate way to implement this type of system is to have the comparator monitor the inductor current and actuate the FETs when the current exceeds a predefined threshold. Ideally, when one FET is switched ON, the other FET can be switched OFF at the same time. However, due to switch mismatch and board propagation delays, simultaneous switching from the controller likely will not result in simultaneous switching at the FETs, which creates the shoot through mentioned earlier. While techniques in software can help hold off the switching of the secondary switch to avoid this condition, these can be challenging to implement with the various hardware interdependencies coupled with the time constraints of the control loop.

4.1.5 文档

- [TMS320F2838xD双核实时微控制器技术参考手册](#)
- [TMS320F2837xD 双核实时微控制器技术参考手册](#)
- [TMS320F28004x 实时微控制器技术参考手册](#)
- [TMS320F28003x 实时微控制器技术参考手册](#)
- [TMS320F28002x 实时微控制器技术参考手册](#)
- [TMS320F280013x 实时微控制器技术参考手册](#)
- [TMS320F280015x 实时微控制器技术参考手册](#)
- [C2000 F2837xD 微控制器 1 天研讨会 第 1.6 节：控制外设 跳转到 14:38 查看 HRPWM](#)

4.2 带可配置死区的电流控制拓扑的直通保护

4.2.1 价值主张

C2000 微控制器能够在硬件中通过片上比较器实现电流控制技术，例如峰值电流模式控制 (PCMC)，以控制 PWM 占空比。类型 4 PWM 已添加可变死区插入功能，能够在不增加任何 CPU 开销的情况下消除潜在的直通现象。

4.2.2 深入探讨

为了更好的电源效率，许多 DC-DC 系统采用同步升压控制器，其中次级开关取代了常规升压控制器中存在的续流二极管（图 4-2）。峰值电流模式控制是用于控制该拓扑结构的一种常见方法，而 C2000 MCU 具有一些独特的特性，使其能够非常高效地实现这种控制。

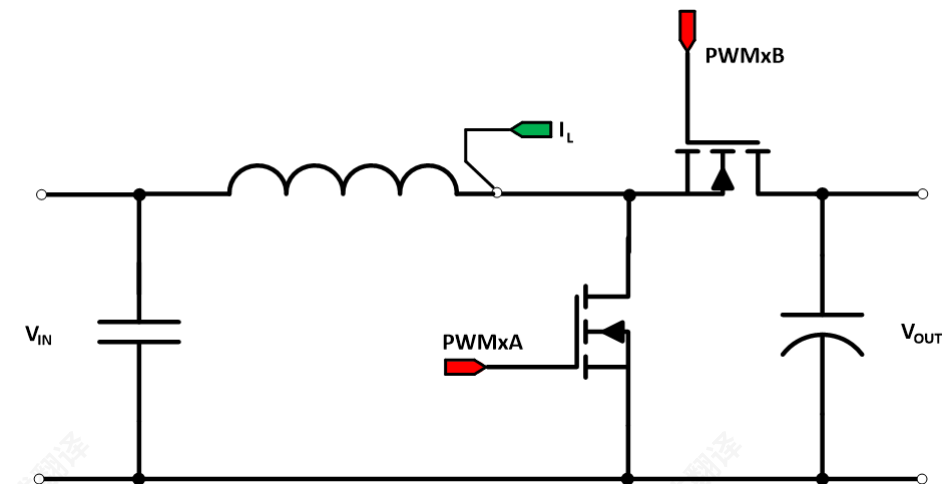


图4-2。同步升压控制器

增加第二个 FET 需要精确控制主、次级开关的 ON/OFF 时间相对于彼此。如果开关同时处于 “ON” 状态，则主动电流会有一个直接到地的路径，这不仅效率低下，还可能对 FET 开关的寿命造成损害。

实现此类系统的准确方法是由比较器监控电感电流，并在电流超过预定义阈值时驱动 FETs。理想情况下，当一个 FET 被切换 ON 时，另一个 FET 可以同时被切换 OFF。然而，由于开关不匹配和电路板传播延迟，控制器从同一时间进行切换很可能不会导致 FETs 同时切换，这会引发前面提到的直通现象。虽然软件技术可以帮助延迟次级开关的切换以避免这种情况，但这些技术在各种硬件相互依赖和时间受限的控制循环中可能难以实现。

The C2000 MCU has implemented programmable deadband control, derived from the comparator output itself, to prevent this condition while keeping the C28x CPU unloaded (Figure 4-3). This allows a complete PCMC solution to be realized outside of the CPU domain once initialized. This logic exists on all PWM modules on a given device, allowing multiple stages to have different deadbands, such as a Phase Shifted Full Bridge, where there are multiple switching pairs.

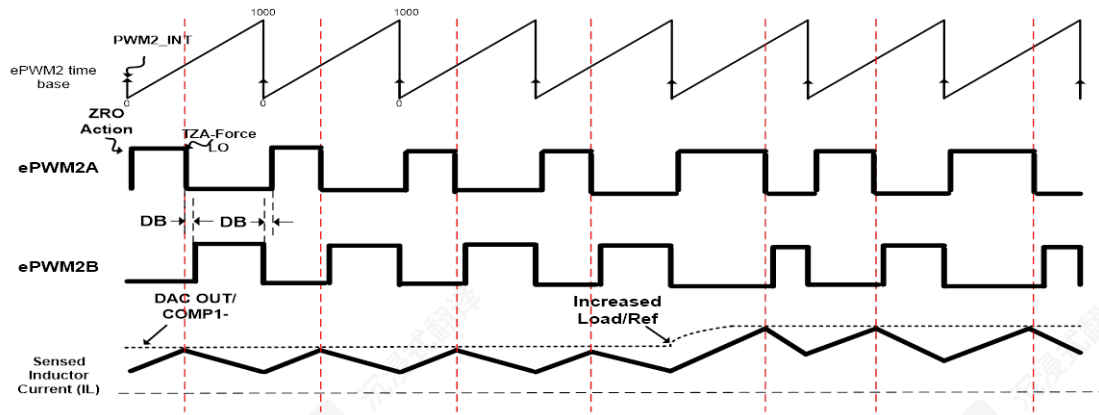


Figure 4-3. Cycle by Cycle Trip Action of the COMP Module With Configurable Deadband

4.2.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

4.2.4 Documentation

- [TMS320F2838xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the Action-Qualifier (AQ) Submodule and Dead-Band Generator (DB) Submodule sections of the Enhanced Pulse Width Modulator (ePWM) chapter)
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the Action-Qualifier (AQ) Submodule and Dead-Band Generator (DB) Submodule sections of the Enhanced Pulse Width Modulator (ePWM) chapter)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the Action-Qualifier (AQ) Submodule and Dead-Band Generator (DB) Submodule sections of the Enhanced Pulse Width Modulator (ePWM) chapter)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the Action-Qualifier (AQ) Submodule and Dead-Band Generator (DB) Submodule sections of the Enhanced Pulse Width Modulator (ePWM) chapter)
- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the Action-Qualifier (AQ) Submodule and Dead-Band Generator (DB) Submodule sections of the Enhanced Pulse Width Modulator (ePWM) chapter)
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the Action-Qualifier (AQ) Submodule and Dead-Band Generator (DB) Submodule sections of the Enhanced Pulse Width Modulator (ePWM) chapter)
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the Action-Qualifier (AQ) Submodule and Dead-Band Generator (DB) Submodule sections of the Enhanced Pulse Width Modulator (ePWM) chapter)
- [C2000 Academy ePWM Module](#)

C2000微控制器已实现了基于比较器输出的可编程死区控制，以防止这种情况，同时保持C28x CPU空闲（图4-3）。这允许在初始化后，在CPU域之外实现完整的PCMC解决方案。这种逻辑存在于给定设备上所有PWM模块中，允许多个阶段具有不同的死区，例如相移全桥，其中存在多个开关对。

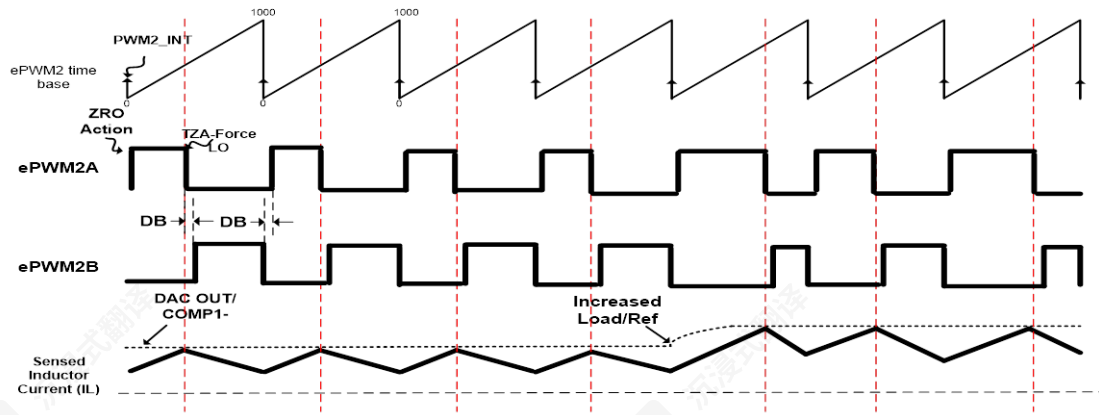


图4-3。具有可配置死区的COMP模块的循环行程动作

4.2.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

4.2.4 文档

- [TMS320F2838xD 实时微控制器技术参考手册](#)（更多信息，请参阅增强型脉宽调制器（ePWM）章节中的动作限定符（AQ）子模块和死区发生器（DB）子模块部分）
- [TMS320F2837xD 实时微控制器技术参考手册](#)（更多信息，请参阅增强型脉宽调制器（ePWM）章节中的动作限定符（AQ）子模块和死区发生器（DB）子模块部分）
- [TMS320F28004x 实时微控制器技术参考手册](#)（更多信息，请参阅增强型脉宽调制器（ePWM）章节中的动作限定符（AQ）子模块和死区发生器（DB）子模块部分）
- [TMS320F28003x 实时微控制器技术参考手册](#)（更多信息，请参阅增强型脉宽调制器（ePWM）章节中的动作限定符（AQ）子模块和死区发生器（DB）子模块部分）
- [TMS320F28002x 实时微控制器技术参考手册](#)（更多信息，请参阅增强型脉宽调制器（ePWM）章节中的动作限定符（AQ）子模块和死区发生器（DB）子模块部分）
- [TMS320F280013x 实时微控制器技术参考手册](#)（更多信息，请参阅增强型脉宽调制器（ePWM）章节中的动作限定符（AQ）子模块和死区发生器（DB）子模块部分）
- [TMS320F280015x 实时微控制器技术参考手册](#)（更多信息，请参阅增强型脉宽调制器（ePWM）章节中的动作限定符（AQ）子模块和死区发生器（DB）子模块部分）
- [C2000 Academy ePWM 模块](#)

4.3 On-Chip Hardware Customization Using the C2000 Configurable Logic Block

4.3.1 Value Proposition

The Configurable Logic Blocks (CLB) provide a mechanism for creating custom logic inside the chip. Enhancing existing subsystems, creating new subsystem and replacing off-chip logic circuits are all possible.

4.3.2 In Depth

The on-chip CLB provide a flexible mechanism to add personalized logic customization in hardware inside the MCU. Whether it is modifying an existing subsystem to satisfy an application specific need (Realizing Rotary Sensing Solutions [Section 2.5](#)), creating a completely new subsystem, or replacing external logic circuits, the CLB can accomplish all these tasks.

There are three key benefits that the CLB modules provide to the system:

1. Enhancing existing on-chip subsystems: The custom logic implemented inside the CLB modules can be inserted inside other on-chip subsystem, such as ePWM subsystem, to enhance the capability of the peripheral. Examples of this include creating a T-format absolute encoder ([List item.referenceTitle](#)) interface or Pulse Train Output ([List item.referenceTitle](#)).

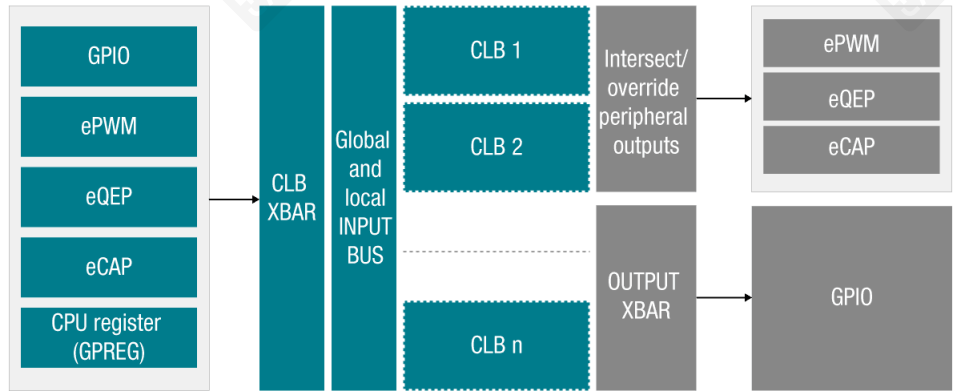


Figure 4-4. CLB Integration in C2000 MCU Architecture

2. Creating new subsystems: The CLB modules can be combined to form new subsystems, which could be a completely new peripheral that does not exist in C2000 MCUs, or replicate a C2000 MCU peripheral and create an extra subsystem, for example:
 - a. [Creating an auxiliary PWM module SW example](#)
 - b. [Designing With the C2000 Configurable Logic Block](#) as shown in [CLB State Machine SW example](#)
3. Replacing external logic: In some cases the CLB modules can be used to absorb external devices such as FPGA or CPLD, which implements application specific customized logic. The CLB modules can also be used to replace external logic circuits. Examples of this are:
 - a. [Migrating from FPGA/CPLD to CLB](#)
 - b. [Replacing external PWM protection circuit with custom logic in the CLB SW Example](#)

4.3 使用C2000可配置逻辑块进行片上硬件定制

4.3.1 价值主张

可配置逻辑块 (CLB) 提供了一种在芯片内部创建自定义逻辑的机制。增强现有子系统、创建新子系统以及替换片外逻辑电路都是可能的。

4.3.2 深入解析

片上CLB提供了一种灵活的机制，用于在MCU内部的硬件中添加个性化的逻辑定制。无论是修改现有子系统以满足特定应用需求（第2.5节 实现旋转传感解决方案），创建全新的子系统，还是替换外部逻辑电路，CLB都能完成所有这些任务。

CLB模块为系统提供了三个关键优势：

1. 增强现有的片上子系统：CLB模块内部实现的定制逻辑可以被插入到其他片上子系统内部，例如ePWM子系统，以增强外设的功能。这方面的例子包括创建T格式绝对编码器（[List item.referenceTitle](#)）接口或脉冲序列输出（[List item.referenceTitle](#)）。

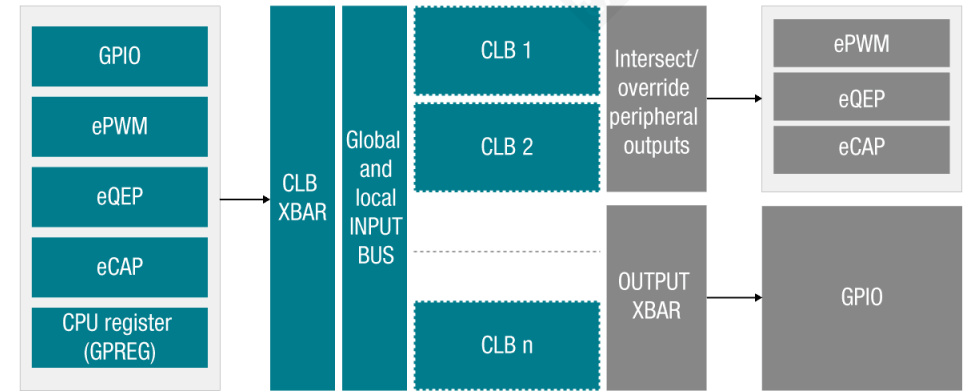


图4-4。C2000 MCU架构中的CLB集成

2. Creating new subsystems: CLB模块可以组合形成新的subsystems，这可能是一个完全新的外设，该外设不在C2000 MCUs中存在，或复制一个C2000 MCU外设并创建一个额外的子系统，例如：
 - a. [创建辅助 PWM 模块 SW 示例](#)
 - b. [使用 C2000 可配置逻辑块进行设计，如 CLB 状态机 SW 示例所示](#)
3. 替换外部逻辑：在某些情况下，CLB 模块可用于吸收外部设备，如 FPGA 或 CPLD，这些设备实现了特定应用的定制逻辑。CLB 模块也可用于替换外部逻辑电路。这方面的示例包括：
 - a. [从 FPGA/CPLD 迁移到 CLB](#)
 - b. [使用 CLB 中的自定义逻辑替换外部 PWM 保护电路，如 CLB SW 示例所示](#)

While the physical implementation of the custom logic is controlled through uploaded memory values into the C2000 MCU, TI provides several GUI tools such as the CLB Configuration Tool ([Figure 4-5](#)) to both realize the logical implementation as well as verify the operation of the logic in simulation before it is used in the system.

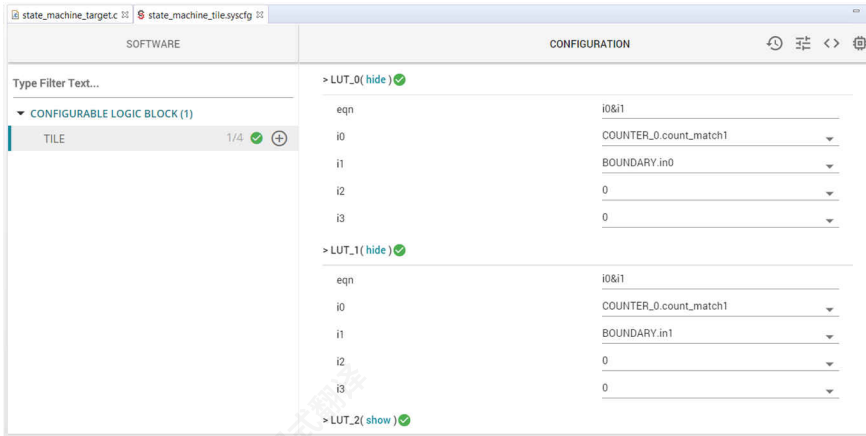


Figure 4-5. CLB Configuration Tool in SysConfig

4.3.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

4.3.4 Hardware Platforms and Software Examples

- [F28388D controlCARD evaluation module](#)
- [LAUNCHXL-F28379D](#)
- [LAUNCHXL-F280049C](#)
- [F280039C controlCARD evaluation module](#)
- [LAUNCHXL-F280025C](#)
- [Creating an auxiliary PWM module SW example](#)
- [CLB State Machine SW example](#)
- [Replacing external PWM protection circuit with custom logic in the CLB SW Example](#)

4.3.5 Documentation

- [CLB Programming Tool Training\(Video\)](#)
- [Designing with the C2000™ Configurable Logic Block](#)
- [How to Migrate Custom Logic From an FPGA/CPLD to C2000™ Microcontrollers](#)
- [CLB Tool User's Guide](#)

虽然自定义逻辑的物理实现是通过上传的内存值到 C2000 MCU 控制的，但 TI 提供了几个 GUI 工具，例如 CLB 配置工具（图 4-5），以实现逻辑实现并在系统使用前在仿真中验证逻辑的操作。

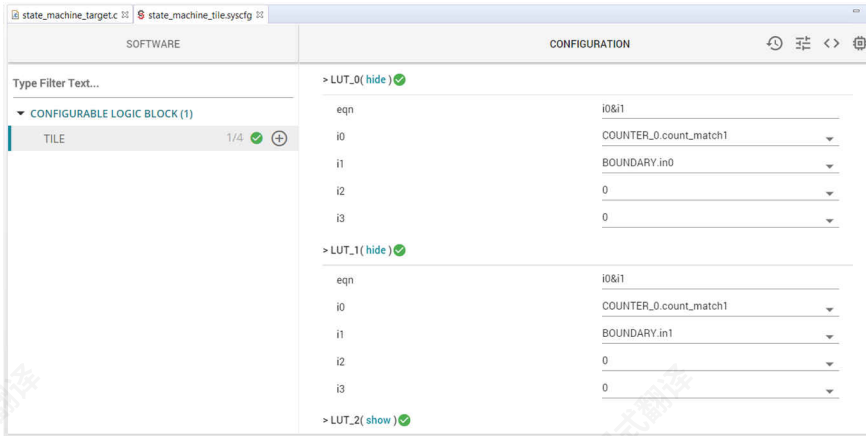


图4-5。SysConfig中的CLB配置工具

4.3.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

4.3.4 硬件平台和软件示例

- [F28388D 控制CARD 评估模块](#)
- [LAUNCHXL-F28379D](#)
- [LAUNCHXL-F280049C](#)
- [F280039C 控制CARD 评估模块](#)
- [LAUNCHXL-F280025C](#)
- [创建辅助 PWM 模块 SW 示例](#)
- [CLB 状态机 SW 示例](#)
- [在 CLB SW 示例中用自定义逻辑替换外部 PWM 保护电路](#)

4.3.5 文档

- [CLB 编程工具培训（视频）](#)
- [使用 C2000™ 可配置逻辑块进行设计](#)
- [如何将自定义逻辑从 FPGA/CPLD 迁移到 C2000™ 微控制器](#)
- [CLB 工具用户指南](#)

4.4 Fast Detection of Over and Under Currents and Voltages

4.4.1 Value Proposition

Every control system can experience random events that can cause damage to the system. Fast detection and reaction to these events is critical to keeping the system safe and in good working condition. The on-chip comparators can detect and react to these events in a fraction of the time that it would take for an ADC and processor.

4.4.2 In Depth

Fault detection and reaction is important in most systems, not only for avoiding an undefined output, but for preventing damage to components both on and off the main Printed Circuit Board (PCB). The speed at which the fault detection takes place, as well as the final FET output state change, is critical to the system. A dedicated subsystem that ties together the analog and digital domains has been implemented on the C2000 MCU for handling this requirement: the Comparator Subsystem or CMPSS (Figure 4-6).

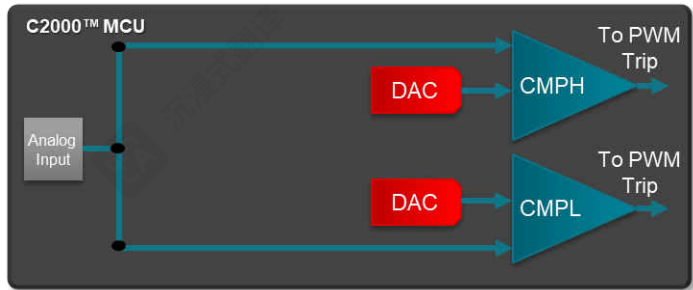


Figure 4-6. CMPSS Visualization

There are up to eight CMPSS modules on each C2000 MCU, with internal DACs that give the inverting/ comparison detection level for the line that is being monitored. As shown in Figure 4-6, each CMPSS module has two comparators for simultaneous high and low detection. Using the CMPSS has several advantages over using the ADC for fault detection:

- **System Overhead:** Using the CMPSS to monitor a pin is an essentially zero overhead operation after the initial setup. The pin is always monitored against the comparison value until disabled. Other techniques would require periodic ADC conversions and threshold checking.
- **Latency:** While the ADC sampling rate can be simply factored into the period of the control loop, there is not a deterministic constant to a fault condition. As such there will be an inherent delay to detect the fail, both from a point of sample to the conversion time of the ADC itself. The comparator has no such trigger requirement or sample time, it is continuously monitoring the analog signal.
- **Dedicated PWM Trip Zone input:** The output of every CMPSS module can be tied directly into the Trip Zone of any PWM, and the action when the signal is received is configurable in software. This means there is no software overhead as there would be in processing an ADC ISR to then create the action to the PWM in software.
- **No clock dependence:** Since by definition a comparator is a purely analog domain circuit there are not clock dependencies to the changing state of its output based on the input. The C2000 MCU has carried this forward to give a asynchronous path from the comparator to the PWM. This allows for the fastest possible time from fault detection to pin state change, in addition to removing any clock dependence (Table 4-2).

Table 4-2. Comparison of Fault Detection and Trip Methods

Sampling Method	Sample Time (min)	Result Ready (min)	Latch and Change PWM Pin (@200 MHz SysClk)	Total Time From Fault To Trip
12-bit ADC	75 ns	260 ns	approximately 100 ns(inc ISR)	435 ns
12-bit ADC w/PPB	75 ns	260 ns	10 ns	355 ns
CMPSS	NA	NA	NA	60 ns

4.4 快速检测过流和欠流电压

4.4.1 价值主张

每个控制系统都可能遇到随机事件，这些事件可能导致系统损坏。快速检测和响应这些事件对于保持系统安全和正常运行至关重要。片上比较器可以在 ADC 和处理器所需时间的几分之一内检测和响应这些事件。

4.4.2 深入探讨

故障检测和响应在大多数系统中都很重要，不仅是为了避免未定义的输出，还是为了防止主印刷电路板 (PCB) 上和下组件的损坏。故障检测的速度以及最终的 FET 输出状态变化对系统至关重要。C2000 MCU 上实现了一个专门的子系统，将模拟和数字域连接起来，以处理此要求：比较器子系统或 CMPSS（图 4-6）。

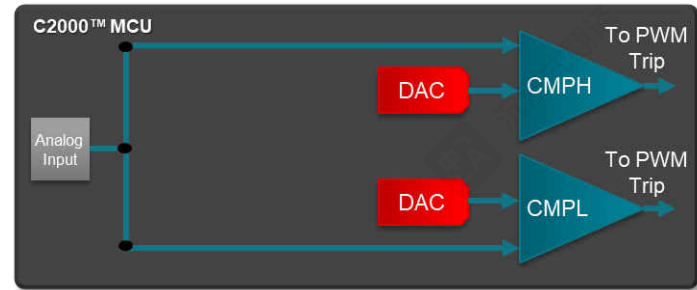


图 4-6. CMPSS 可视化

每个 C2000 MCU 上最多有八个 CMPSS 模块，内部 DAC 提供被监控线路的反相/比较检测电平。如图 4-6 所示，每个 CMPSS 模块有两个比较器，用于同时进行高低检测。使用 CMPSS 进行故障检测比使用 ADC 有几个优点：

- **System Overhead:** Using the CMPSS to monitor a pin is an essentially zero overhead operation after the initial setup. The pin is always monitored against the comparison value until disabled. Other techniques would require periodic ADC conversions and threshold checking.
- **延迟:** 虽然 ADC 采样率可以简单地 factored 到控制循环的周期中，但故障条件没有确定的常数。因此，在检测到故障时会有固有的延迟，这既来自从采样点到 ADC 本身的转换时间。比较器没有这样的触发要求或采样时间，它持续监控模拟信号。
- **专用 PWM 跳转区输入:** 每个 CMPSS 模块的输出可以直接连接到任何 PWM 的跳转区，并且当接收到信号时的动作可以在软件中配置。这意味着没有像处理 ADC ISR 然后创建软件中 PWM 动作那样的软件开销。
- **无时钟依赖:** 由于比较器在定义上是一个纯粹的模拟域电路，因此其输出状态的变化没有时钟依赖。C2000 MCU 将这一点继承下来，为比较器到 PWM 提供了一条异步路径。这允许从故障检测到引脚状态变化的最快时间，此外还消除了任何时钟依赖（表 4-2）。

表4-2. 故障检测和跳闸方法的比较

采样方法	采样时间 (分钟)	结果就绪 (分钟)	锁存和更改 PWM 引脚 (@200 MHz SysClk)	从故障到跳闸的总时间
12 位 ADC	75 ns	260 ns	大约 100 ns(含 ISR)	435 ns
12 位 ADC w/PPB	75 ns	260 ns	10 ns	355 ns
CMPSS	NA	NA	NA	60 ns

- Simultaneous high and low detection: Each input to a CMPSS module routes the signal to two physically independent comparators that give the ability to detect both overshoot and undershoot at the same time.

4.4.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

4.4.4 Hardware Platforms and Software Examples

The following kits implement the CMPSS to detect out of range current/voltage events

- [TIDM-02002 Bi-Directional CLLLC Resonant Dual Active Bridge Reference Design for HEV/EV Onboard Charger](#)
- [TIDM-1022 Valley Switching Boost Power Factor Corrector](#)
- [TMDXIDDK379D C2000 Design DRIVE Development Kit for Industrial Motor Control](#)

4.4.5 Documentation

- [TMS320F2838x Real-Time Microcontrollers With Connectivity Manager Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F2837xD Dual-Core Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F28004x Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F28003x Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F28002x Real-Time Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F280013x Real-Time Microcontrollers Data Sheet](#) (for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F280015x Real-Time Microcontrollers Data Sheet](#)(for more information, see the *Comparator Subsystem (CMPSS)* section)
- [TMS320F2838xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#) (see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#) (see the *Comparator Subsystem (CMPSS)* chapter)
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#)(see the *Comparator Subsystem (CMPSS)* chapter)

- 同时高低检测：每个输入到CMPSS模块的信号都会被路由到两个物理上独立的比较器，这使得能够同时检测过冲和欠冲。

4.4.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

4.4.4 硬件平台和软件示例

以下套件实现了CMPSS以检测超出范围的电流/电压事件

- [TIDM-02002 双向 CLLLC 谐振双有源桥参考设计，适用于 HEV/EV 车载充电器](#)

- [TIDM-1022 峰谷切换升压功率因数校正器](#)
- [TMDXIDDK379D C2000 设计 驱动 开发套件 用于工业电机控制](#)

4.4.5 文档

- [TMS320F2838x 实时微控制器 带连接管理器 数据表](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F2837xD 双核微控制器 数据表](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F28004x 微控制器 数据表](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F28003x 微控制器数据手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F28002x 实时微控制器数据手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F280013x 实时微控制器数据手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F280015x 实时微控制器数据手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 部分）
- [TMS320F2838xD 实时微控制器技术参考手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F2837xD 实时微控制器技术参考手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F28004x 实时微控制器技术参考手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F28003x 实时微控制器技术参考手册](#)（更多信息，请参阅比较器子系统 (CMPSS) 章节）
- [TMS320F28002x 实时微控制器技术参考手册](#)（参见比较器子系统 (CMPSS) 章节）
- [TMS320F280013x 实时微控制器技术参考手册](#)（参见比较器子系统 (CMPSS) 章节）
- [TMS320F280015x 实时微控制器技术参考手册](#)(参见比较器子系统 (CMPSS) 章节)

4.5 Improving System Power Density With High Resolution Phase Control

4.5.1 Value Proposition

In typical charging systems used for EV such as On-Board Chargers or Off Board DC Chargers only 1% of ripple is allowed at the output voltage and current. Dual Active Bridge (Figure 4-7) is a popular topology to implement the DC-DC converter in these charging systems as it is able to achieve a wide operating range of voltages. The ability of the C2000 HRPWM to control phase shift within 150 ps allows for reducing component size while still meeting output ripple requirements.

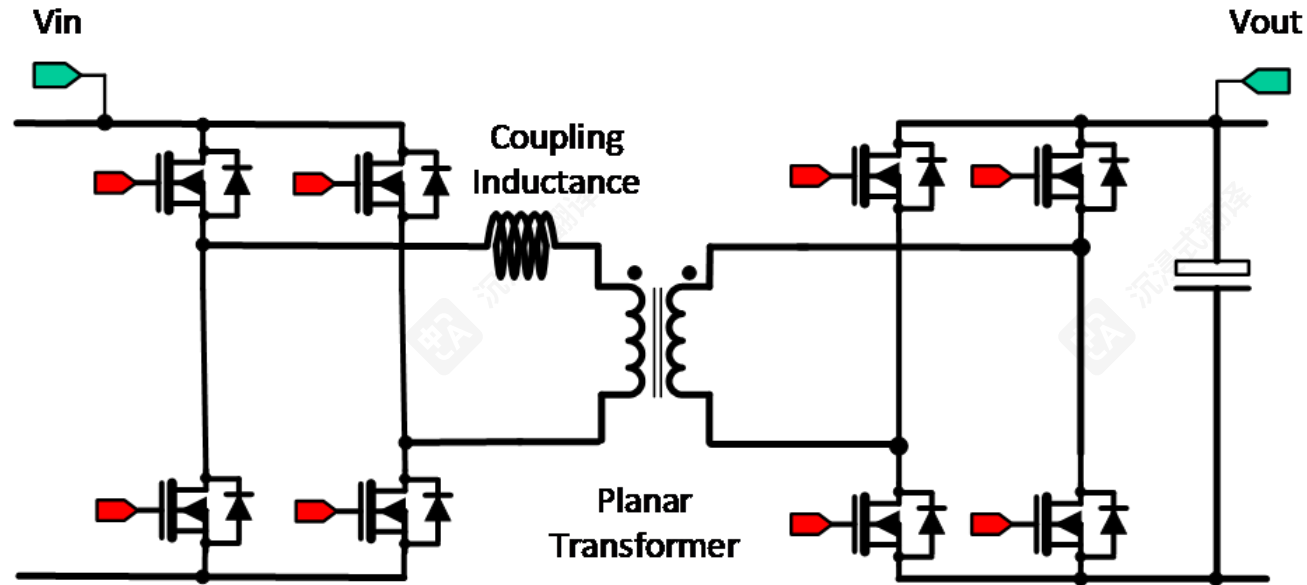


Figure 4-7. Dual Active Bridge Block Diagram

4.5.2 In Depth

Power transfer in single phase shift DAB is related to the coupling inductance (L) value and the switching frequency (Fs) and the phase (θ).

$$P_{output} = \frac{nV_{in}V_{out}\theta(\pi - \theta)}{2\pi^2 F_s L}$$

where

- n is number of turns of the transformer
- Vin is voltage input to the DAB
- Vout is voltage output from the DAB
- θ is the phase
- Fs is the switching frequency
- L is the inductance

For improved power density (smaller overall physical dimensions of the design), both lower inductance and higher switching frequency are necessary. A reference example is TIDA-010054, which is a 10kW DC-DC DAB converter: 100 kHz PWM switching frequency is used along with a 35 μH for the leakage inductor. A three phase PFC with 800 V bus such as Vienna Rectifier (TIDM-1000), or T-Type PFC (TIDA-010039) is used as the front end. Finally, for an output voltage of 400V at 10kW we can then determine the resultant output steps in the voltage the phase shift causes and check if it allows it to be within the ± 1% of our ripple requirement.

4.5 通过高分辨率相控技术提升系统功率密度 4.5.1 价值主张

在用于电动汽车的典型充电系统中（如车载充电器或离板直流充电器），输出电压和电流中仅允许1%的纹波。双有源桥（图4-7）是这些充电系统中实现DC-DC转换器的常用拓扑，因为它能够实现宽电压工作范围。C2000 HRPWM在150 ps内控制相移的能力允许在仍然满足输出纹波要求的同时减小组件尺寸。

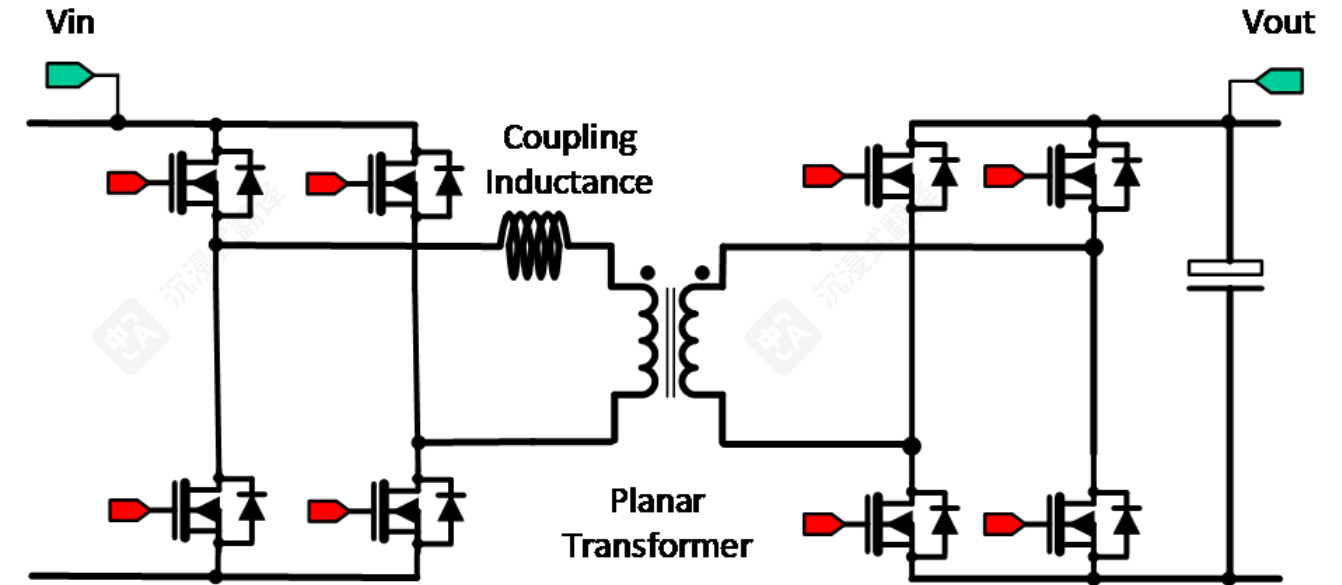


图 4-7. 双有源桥电路图

4.5.2 深入解析

单相移相 DAB 的功率传输与耦合电感 (L) 值、开关频率 (Fs) 以及相 (θ) 相关。

$$P_{output} = \frac{nV_{in}V_{out}\theta(\pi - \theta)}{2\pi^2 F_s L}$$

其中

- n 是变压器的绕组数
- Vin 是 DAB 的电压输入
- Vout 是 DAB 的电压输出
- θ 是相位
- Fs 是开关频率
- L 是电感

为了提高功率密度（设计整体物理尺寸更小），需要较低的电感和较高的开关频率。一个参考示例是 TIDA-010054，它是一个 10kW DC-DC DAB 转换器：使用 100 kHz PWM 开关频率，并使用 35 μH 作为漏感。前端使用 800 V 总线的三相 PFC，例如维也纳整流器（TIDM-1000），或 T 型 PFC（TIDA-010039）。最后，对于 400V、10kW 的输出电压，我们可以确定相位偏移引起的电压输出步进，并检查它是否在 ± 1% 的纹波要求范围内。

As shown in [Table 4-3](#), the voltage cannot be regulated to 1% accuracy about 400 V by using typical PWM generation logic that is clocked at MCU clock rate. Assuming a 100 MHz CPU clock, would result in 10 ns step changes that is too large for our output ripple tolerance. Using the HRPWM module it is possible to modulate the phase shift to within 150 ps, regardless of the main MCU clock. More so, exceeding the phase step requirement represents the ability to modulate at a 13x step size within those bounds providing capability to find tune the output to more closely match the ideal.

Table 4-3. Phase Shift Requirements to Meet 1% Output Tolerance in a DAB Topology

Vin=800V, Pout=10kW, Fs=100kHz	Vout (V)	Required Phase Shift (ns)
θ = 19.032 deg	400	528.66
θ = 18.9249 deg	402	525.69
θ = 18.7662 deg	405	521.11
θ = 18.5077 deg	408	517.10

4.5.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

4.5.4 Hardware Platforms and Software Examples

- [TIDA-010054 Bi-directional, dual active bridge reference design for level 3 electric vehicle charging stations](#)
- [TIDM-1000 Vienna Rectifier-Based Three Phase Power Factor Correction Reference Design](#)
- [TIDM-1000 Reference Software Project in Digital Power SDK](#)
- [TIDA-010039 Three-level, three-phase SiC AC-to-DC converter reference design](#)

4.5.5 Documentation

- [TMS320F2838xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [HRPWM](#) chapter)
- [TMS320F2837xD Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [HRPWM](#) chapter)
- [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [HRPWM](#) chapter)
- [TMS320F28003x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [HRPWM](#) chapter)
- [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [HRPWM](#) chapter)
- [TMS320F280013x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [HRPWM](#) chapter)
- [TMS320F280015x Real-Time Microcontrollers Technical Reference Manual](#) (for more information, see the [HRPWM](#) chapter)

如表 4-3 所示，使用以 MCU 时钟频率时钟的典型 PWM 生成逻辑无法将电压调节到 400 V 附近的 1% 精度。假设 100 MHz CPU 时钟，将导致 10 ns 的步进变化，这对于我们的输出纹波容差来说太大。使用 HRPWM 模块，可以将相位偏移调制在 150 ps 以内，而不管主 MCU 时钟如何。更重要的是，超过相位步进要求表示能够在这些范围内以 13 倍的步长进行调制，从而提供能力微调输出，使其更接近理想值。

表4-3. 满足DAB拓扑中1%输出容差的相移要求

Vin=800V, Pout=10kW, Fs=100kHz	Vout (V)	所需相移 (ns)
θ = 19.032 deg	400	528.66
θ = 18.9249 deg	402	525.69
θ = 18.7662 度	405	521.11
θ = 18.5077 度	408	517.10

4.5.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

4.5.4 硬件平台和软件示例

- [TIDA-010054 双向双有源桥参考设计，适用于三级电动汽车充电站](#)
- [TIDM-1000 维也纳整流器式三相功率因数校正参考设计](#)
- [TIDM-1000 数字电源 SDK 中的参考软件项目](#)
- [TIDA-010039 三电平、三相 SiC 交流到直流转换器参考设计](#)

4.5.5 文档

- [TMS320F2838xD 实时微控制器技术参考手册（更多信息，请参阅 \[HRPWM\]\(#\) 章节）](#)
- [TMS320F2837xD 实时微控制器技术参考手册（更多信息，请参阅 \[HRPWM\]\(#\) 章节）](#)
- [TMS320F28004x 实时微控制器技术参考手册（更多信息，请参阅 \[HRPWM\]\(#\) 章节）](#)
- [TMS320F28003x 实时微控制器技术参考手册（更多信息，请参阅 \[HRPWM\]\(#\) 章节）](#)
- [TMS320F28002x 实时微控制器技术参考手册（更多信息，请参阅 \[HRPWM\]\(#\) 章节）](#)
- [TMS320F280013x 实时微控制器技术参考手册（更多信息，请参阅 \[HRPWM\]\(#\) 章节）](#)
- [TMS320F280015x 实时微控制器技术参考手册（更多信息，请参阅 \[HRPWM\]\(#\) 章节）](#)

4.6 Safe and Optimized PWM Updates in High-Frequency, Multi-Phase and Variable Frequency Topologies

4.6.1 Value Proposition

The global load and global link features simplify applications that require active management of multiple PWM parameters, such as variable frequency, and multi-phase power topologies. More specifically, the global load feature helps eliminate PWM glitches by providing a single strobe signal to load PWM parameters from the shadow to active registers either within a single PWM module, or across multiple modules. The global link feature reduces CPU overhead by allowing simultaneous updates to linked registers across multiple PWM modules.

4.6.2 In Depth

Shadow registers are used as a temporary location for active registers until a load event occurs that enables the transfer of its content into the active registers at a strategic point within the PWM cycle. This shadow to active load feature prevents corruption and spurious operation due to registers being asynchronously modified by software. Traditionally, the load events that cause a transfer of content from shadow to active registers are configured for each register individually. However, when global load mode is enabled (GLDCTL[GLD] =1), the transfer of contents from shadow register to active register occurs at the same event (GLDCTL[GLDMODE]) for all registers with this mode enabled. Critical PWM registers for submodules such as the Time-Base, Action Qualifier, and Deadband can be enabled to use the global load feature through the GLDCFG register. Additionally, the number of load strobe events that need to occur before the active registers are updated GLDCTL[GLDPRD] can also be configured to meet application needs.

If the control ISR is asynchronous to the PWM switching frequency, the one-shot load mode feature is required to ensure all the registers within a PWM module are updated only once all of the required PWM parameters have been updated. The one-shot load mode is enabled through the GLDCTL[OSHTMODE] register. When GLDCTL2[OSHTLD] is set to one, the transfer of contents from shadow to active registers, for registers that are configured to use the global load mechanism, will occur on the next event selected by GLDCTL[GLDMODE]. After the shadow to active load event happens anymore global load events will be blocked until GLDCTL2[OSHTLD] is set to one again. [Figure 4-8](#) is an example of the one-shot load mode action.

For variable frequency applications, there is a need for simultaneous writes of period and compare registers between ePWM modules. The EPWMXLINK register provides the capability to update the period and compare registers across multiple PWM modules by creating links between them. To further expand upon this, the global load mode register GLDCTL2 which allows for a reload or force of the one-shot load mode can also be linked between modules. This creates a scheme in which the period and compare registers can all be updated at the same time with the same value across multiple PWM modules.

4.6 安全和优化的PWM更新在高频、多相和可变频率拓扑中

4.6.1 价值主张

全局负载和全局链路功能简化了需要主动管理多个PWM参数的应用，例如变频和多相电源拓扑。更具体地说，全局负载功能通过提供一个单次脉冲信号来消除PWM毛刺，该信号用于从阴影寄存器加载PWM参数到活动寄存器，无论是在单个PWM模块内，还是跨多个模块。全局链路功能通过允许跨多个PWM模块同时更新链接寄存器来降低CPU开销。

4.6.2 深入探讨

影子寄存器用作活动寄存器的临时位置，直到发生加载事件，该事件使内容传输到PWM周期中的战略点处的活动寄存器。这种影子到活动的加载功能防止了由于软件异步修改寄存器而导致的损坏和异常操作。传统上，导致内容从影子寄存器传输到活动寄存器的加载事件是针对每个寄存器单独配置的。然而，当全局加载模式启用时（GLDCTL[GLD] =1），具有此模式启用的所有寄存器的内容从影子寄存器到活动寄存器的传输发生在同一事件（GLDCTL[GLDMODE]）。子模块的关键PWM寄存器，如时间基准、动作限定符和死区，可以通过GLDCFG寄存器启用以使用全局加载功能。此外，需要在活动寄存器更新之前发生的加载脉冲事件的数量（GLDCTL[GLDPRD]）也可以配置以满足应用需求。

如果控制ISR与PWM开关频率异步，则需要使用单次加载模式功能，以确保PWM模块内的所有寄存器仅在所有所需的PWM参数更新后更新一次。单次加载模式通过GLDCTL[OSHTMODE] 寄存器启用。当GLDCTL2[OSHTLD] 设置为1时，对于配置为使用全局加载机制的寄存器，从阴影寄存器到活动寄存器的数据传输将在GLDCTL[GLDMODE]选择的下一个事件上发生。阴影到活动加载事件发生后，任何全局加载事件都将被阻塞，直到GLDCTL2[OSHTLD] 再次设置为1。图4-8是单次加载模式动作的示例。

对于变频应用，需要在ePWM模块之间同时写入周期和比较寄存器。EPWMXLINK寄存器通过在它们之间创建链接，提供了跨多个PWM模块更新周期和比较寄存器的功能。为了进一步扩展这一功能，允许重新加载或强制单次加载模式的全局加载模式寄存器GLDCTL2也可以在模块之间进行链接。这创建了一种方案，即周期和比较寄存器可以在多个PWM模块中以相同的值同时更新。

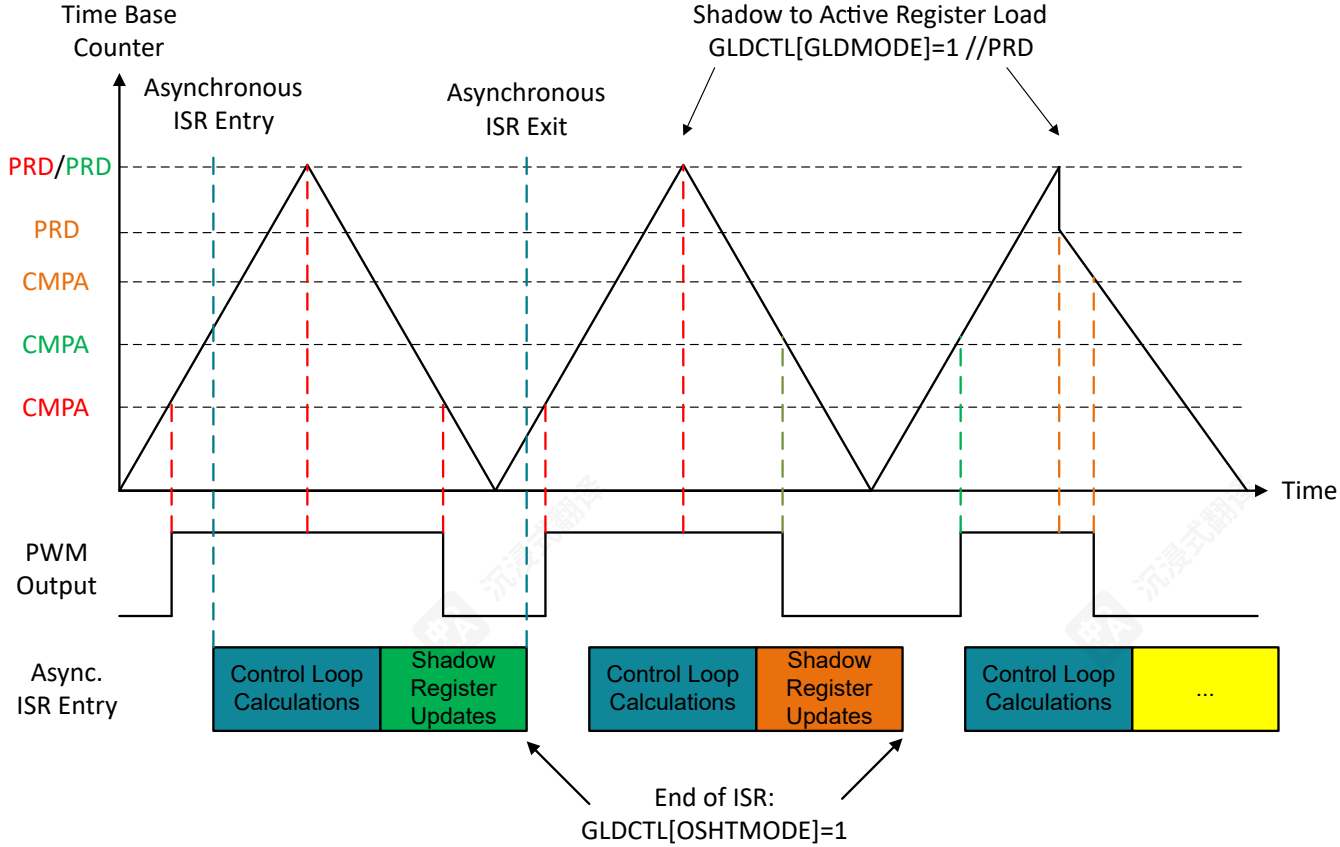


Figure 4-8. Shadow to Active Load Action

4.6.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

4.6.4 Hardware Platforms and Software Examples

- [TIDM-1001 Two Phase Interleaved LLC Resonant Converter Reference Design Using C2000 MCUs \(Global Load\)](#)
- [TIDM-02002 Bidirectional CLLLC Resonant Dual Active Bridge Ref Design HEV/EV Onboard \(Global Link\)](#)

4.6.5 Documentation

- [Leverage New Type ePWM Features for Multiple Phase Control](#)
- 'Global Load' section of ePWM 'Time-Base (TB) Submodule section of TRM

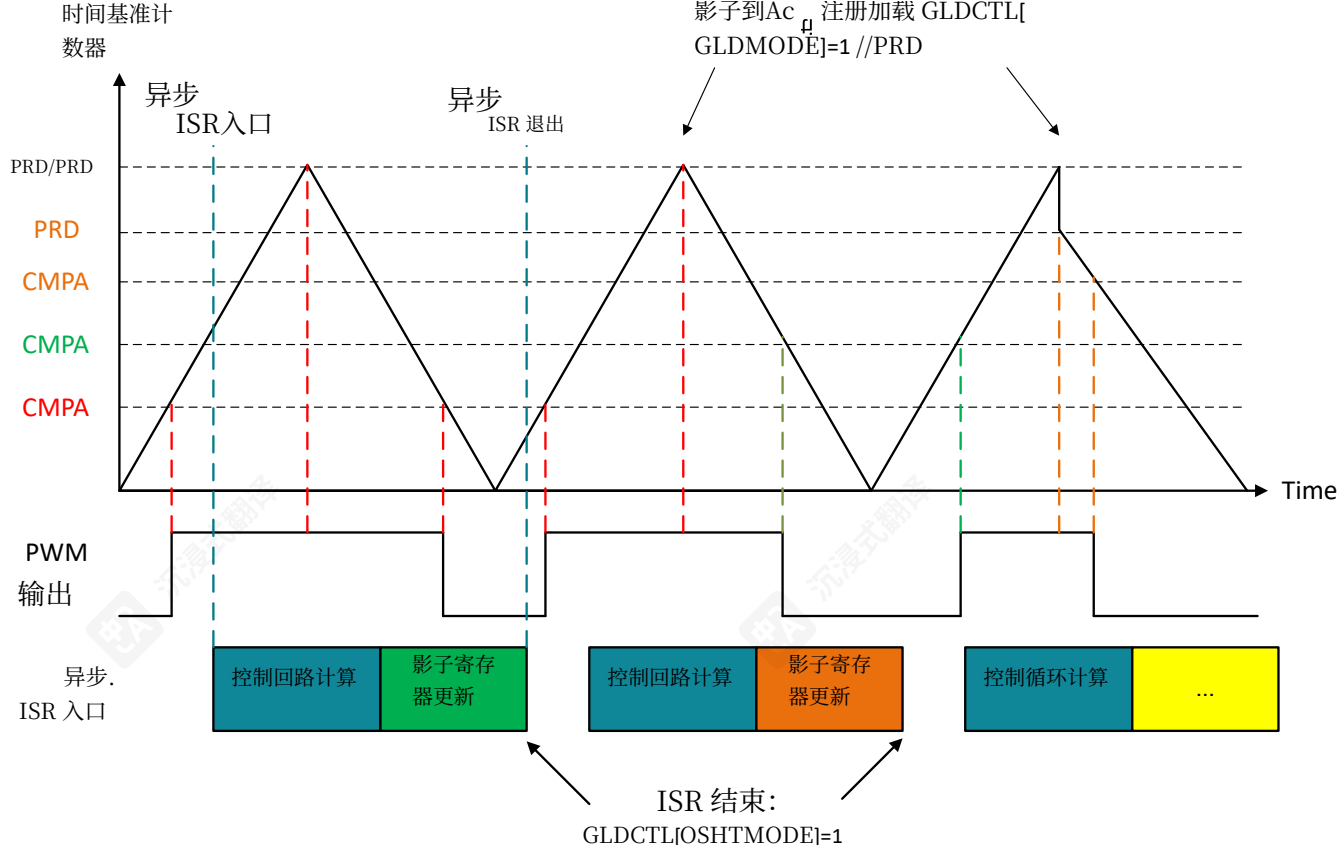


图 4-8. Shadow to Active Load 动作

4.6.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

4.6.4 硬件平台和软件示例

- [TIDM-1001 使用 C2000 MCU 的两相交错 LLC 谐振转换器参考设计 \(全球 Load\)](#)
- [TIDM-02002 双向 CLLLC 谐振双主动桥参考设计 HEV/EV 车载 \(全球链\)](#)

4.6.5 文档

- [利用新型ePWM功能实现多相控制](#)
- '全局加载'部分ePWM '时间基准 (TB) 子模块部分TRM

4.7 Solving Event Synchronization Across Multiple Controllers in Decentralized Control Systems

4.7.1 Value Proposition

Decentralized architectures used in industrial and automotive real-time control applications need a way to synchronize certain events with a minimal amount of trigger latency and jitter between devices. FSI is a high-throughput, low latency, wired communications peripheral that can achieve sub ~100ns time synchronized event triggering in multi-device network topologies. Using an already existing communication port, like FSI, for system event synchronization eliminates the need for additional Sync signals between devices saving device resources and additional design/BOM cost.

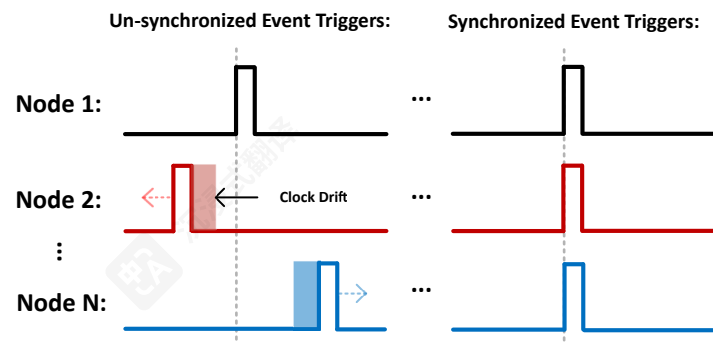


Figure 4-9. Un-synchronized vs Synchronized Event Triggers

4.7.2 In Depth

The communication interfaces between devices in decentralized architectures can vary in network topology, number of node devices, physical distance between the nodes, and more. Devices can also have slight deviations in their local clock during operation due to manufacturing uncertainties, thermal effects, aging, and so forth. Regardless of these variations it is often important for specific events to occur at the same time across all devices in a system, such as ADC start of conversions or rising/falling edges of PWM signals. Synchronous events between C2000 devices can be implemented using the FSI module and custom CLB logic.

With the goal of synchronizing equivalent PWM signals across devices as an example, the implementation can be explored further and is depicted in Figure 4-10. The lead device in a network will periodically send PWM sync requests, as FSI PING frames, to all node devices. In a daisy-chain topology each node device will forward the sync request frame to the next device in the chain. When a node device receives the sync request frame the CLB module will internally route the Ping Packet Received (PING_PKT_RCVD) signal from the FSI RX module through a configurable delay before connecting to the ePWM's EPWMSYNCIN signal. The configurable delay would be calibrated such that all node device's EPWMSYNCIN signals get triggered at the same time and match the lead device's TBCTR equals zero or PRD event. The implementation is completely hardware based and does not require software intervention after initialization. In a daisy-chain topology, nodes at the beginning of the chain will need a longer configurable delay compared to those at the end, since it takes time for the sync request frame to reach all devices. This will ensure the lead and all node devices PWM signals remain in-sync during operation.

4.7 解决分布式控制系统中多个控制器的事件同步

4.7.1 价值主张

工业和汽车实时控制应用中使用的分布式架构需要一种方法来同步某些事件，并在设备之间实现极低的触发延迟和抖动。FSI是一种高吞吐量、低延迟的有线通信外设，可以在多设备网络拓扑中实现亚~100纳秒时间同步的事件触发。使用现有的通信端口（如FSI）进行系统事件同步，可以消除设备之间额外的Sync信号需求，节省设备资源和额外的设计/BOM成本。

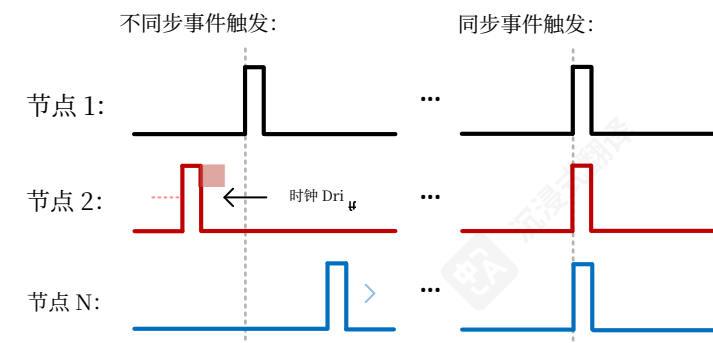


图4-9. 不同步与同步的事件触发

4.7.2 深入探讨

分布式架构中设备之间的通信接口在网络拓扑、节点设备数量、节点之间的物理距离等方面可能有所不同。由于制造不确定性、热效应、老化等原因，设备在运行期间其本地时钟也可能存在轻微偏差。无论这些变化如何，对于系统中的所有设备来说，在特定事件同时发生往往非常重要，例如ADC转换启动或PWM信号的上升/下降边缘。C2000设备之间的同步事件可以使用FSI模块和自定义CLB逻辑实现。

以同步等效PWM信号跨设备为例，可进一步探索其实现方式，如图4-10所示。网络中的主设备将周期性地以FSI PING帧的形式向所有节点设备发送PWM同步请求。在菊花链拓扑中，每个节点设备会将同步请求帧转发到链中的下一个设备。当节点设备接收到同步请求帧时，CLB模块将内部将FSI RX模块的Ping数据包接收（PING_PKT_RCVD）信号通过可配置延迟连接到ePWM的EPWMSYNCIN信号。可配置延迟将被校准，以便所有节点设备的EPWMSYNCIN信号同时触发，并与主设备的TBCTR等于零或PRD事件匹配。该实现完全基于硬件，初始化后无需软件干预。在菊花链拓扑中，链头端的节点需要比链尾端的节点更长的可配置延迟，因为同步请求帧到达所有设备需要时间。这将确保主设备和所有节点设备的PWM信号在运行期间保持同步。

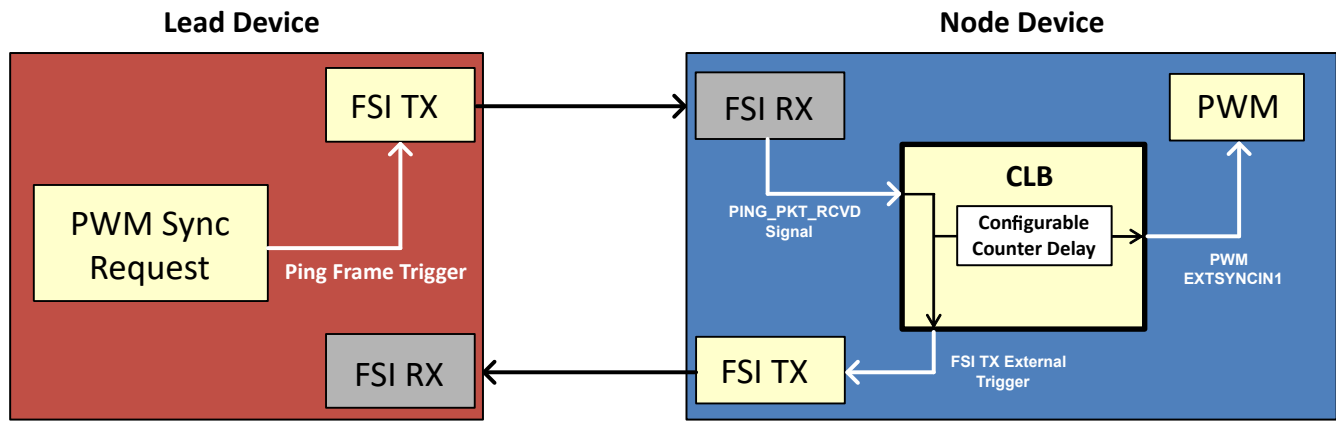


Figure 4-10. Lead and Node Device Implementation

Figure 4-11 demonstrates the PWM sync sequence and timing for an N node daisy-chain network. Some minimal jitter will be experienced on the PWM edge of each node due to device and physical layer uncertainties, but it will increase with the number of node devices in a daisy-chain network. Other network topologies, such as star or bus, that don't require frame forwarding would experience even less jitter.

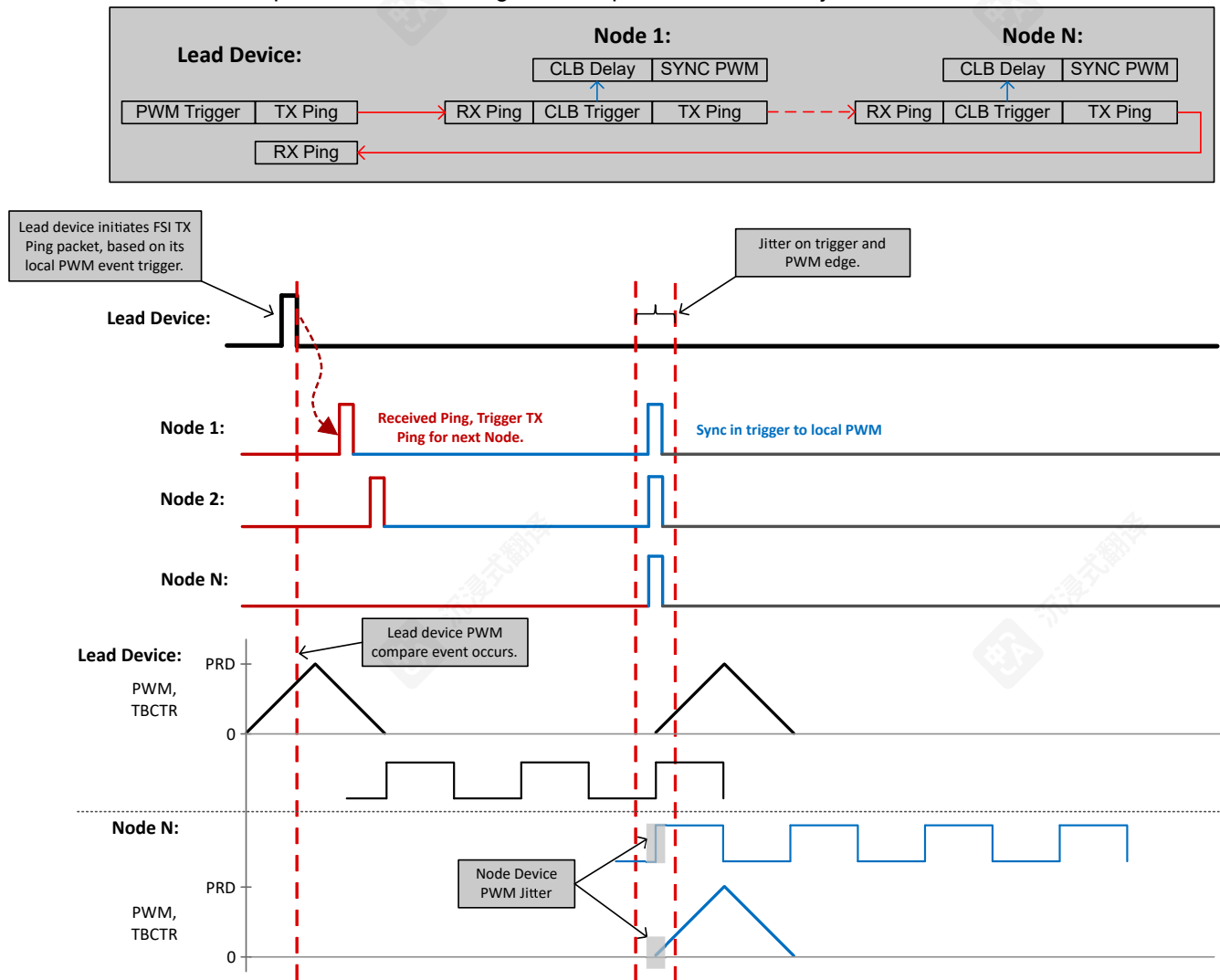


Figure 4-11. Daisy Chain Network Synchronization and Timing

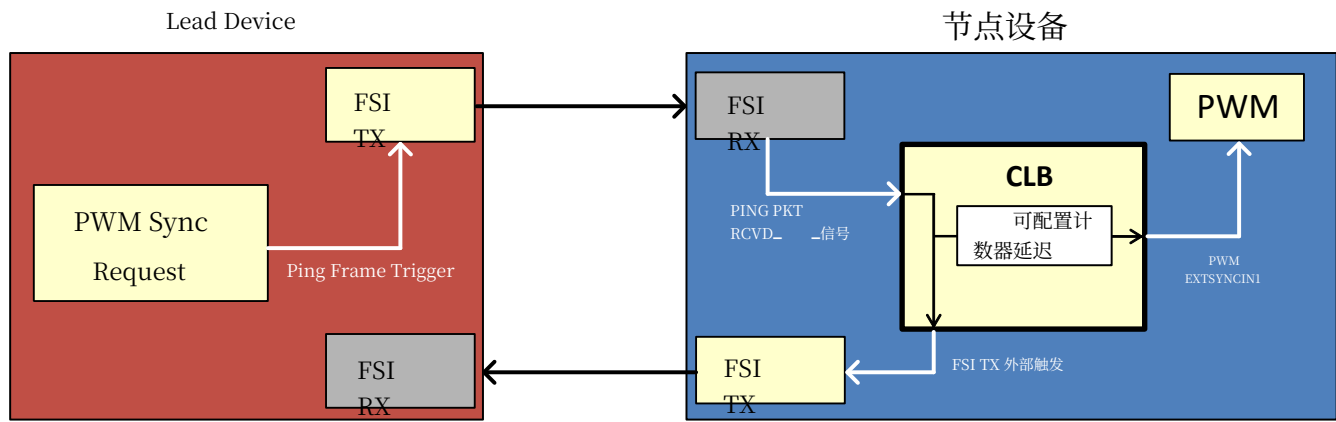


图 4-10。引脚和 Node Device 实现

图4-11展示了N节点菊花链网络中PWM同步序列和时序。由于设备和物理层的不确定性，每个节点的PWM边缘会经历一些最小的抖动，但抖动会随着菊花链网络中节点设备数量的增加而增加。其他不需要帧转发的网络拓扑，如星型或总线拓扑，将经历更小的抖动。

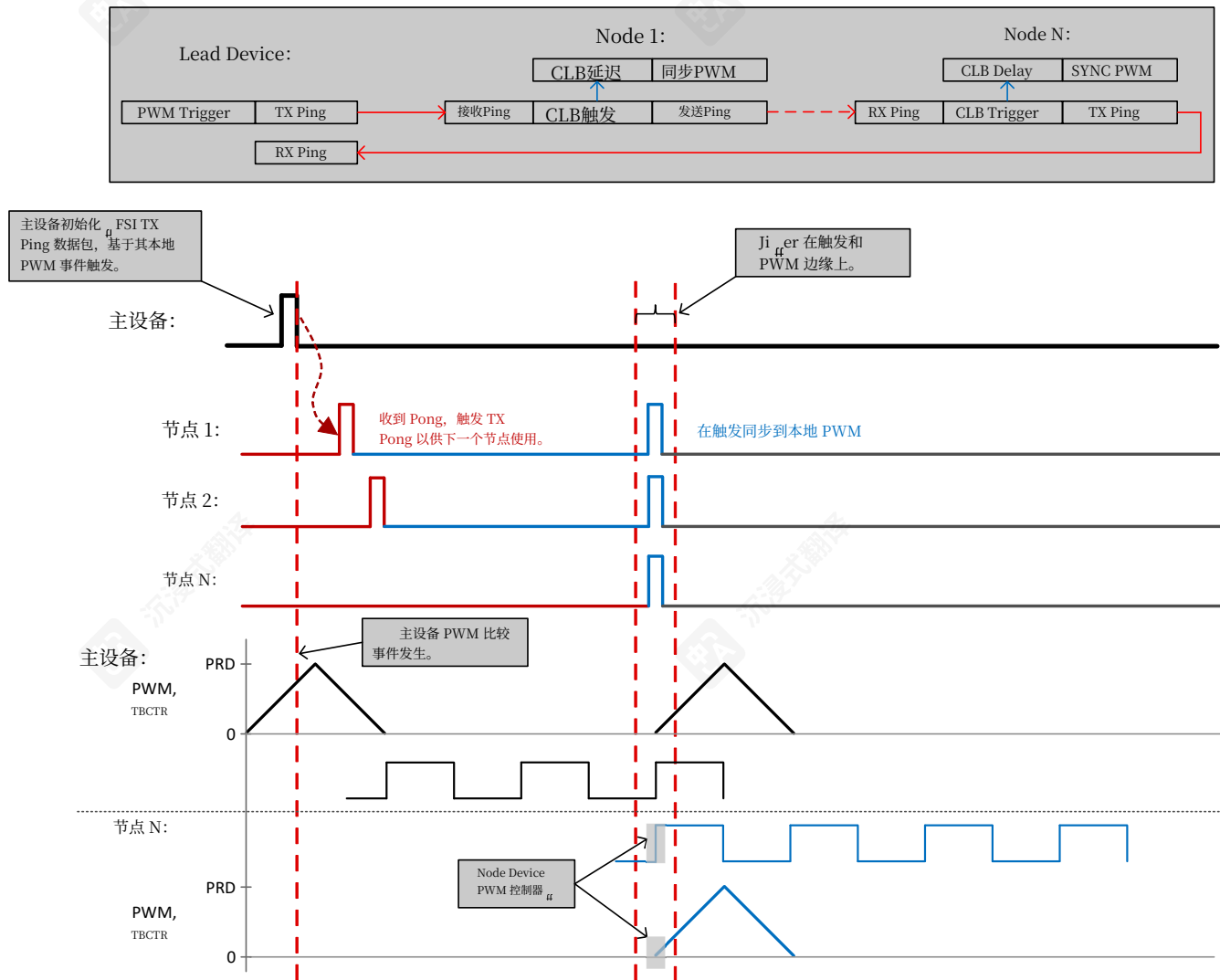


图 4-11。菊花链网络同步和定时

For a more detailed explanation of the event synchronization over FSI along with test results, see the **Event Synchronization Over FSI** section of the [Using the Fast Serial Interface \(FSI\) With Multiple Devices in an Application](#).

4.7.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

4.7.4 Hardware Platforms and Software Examples

- [TMDSFSIADAPEVM](#)
- [FSI Daisychain ePWM Sync Lead](#)
- [FSI Daisychain ePWM Sync Node](#)

4.7.5 Documentation

- [Using the Fast Serial Interface \(FSI\) With Multiple Devices in an Application](#)

5 Interface Key Technologies

5.1 Direct Host Control of C2000 Peripherals

5.1.1 Value Proposition

Many industrial control systems implement a host controller that oversees multiple aspects of the system, including interfacing with the C2000 real-time MCU as it implements the direct control loop. In other cases there may be a desire to add the differentiated IP of the C2000 to an existing system. The Host Interface Controller (HIC) logic inside the C2000 real-time MCU allows control of various peripherals using a common asynchronous interface.

Through the HIC, the host can use the C2000 device as a peripheral expander via the ASRAM port, by taking advantage of the C2000's differentiated features such as Fast Serial Interface (FSI), Configurable Logic Block (CLB), and others.

5.1.2 In Depth

The HIC supports a variety of configurations with which the host can be connected. Each configuration has certain advantages and implications with respect to latency, Input/Output (IO) pin overheads and so on.

The HIC allows access to many common C2000 MCU peripherals such as but not limited to: Analog to Digital Converters, Comparators, PWMs, CAN controller, Fast Serial Interface (FSI) communications, and the Configurable Logic Block (CLB).

For a full overview of the HIC features, see the [Design Guide for Enabling Peripheral Expansion Applications Using the HIC](#).

The following section details few practical applications of HIC using the FSI for isolated communications and CLB to implement an absolute encoder.

有关通过 FSI 进行事件同步的更详细说明以及测试结果，请参阅《在应用程序中使用快速串行接口 (FSI) 与多个设备》中的事件同步通过 FSI 部分。

4.7.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

4.7.4 硬件平台和软件示例

- [TMDSFSIADAPEVM](#)
- [FSI Daisychain ePWM Sync Lead](#)
- [FSI Daisychain ePWM Sync Node](#)

4.7.5 文档

- [在应用程序中使用快速串行接口（FSI）与多个设备](#)

5 接口关键技术

5.1 直接控制C2000外设

5.1.1 价值主张

许多工业控制系统实现了一个主机控制器，该控制器负责系统的多个方面，包括与C2000实时MCU接口，以实现直接控制循环。在其他情况下，可能希望将C2000的独特IP添加到现有系统中。C2000实时MCU中的主机接口控制器（HIC）逻辑允许使用通用异步接口控制各种外设。

通过 HIC，主机可以通过 ASRAM 端口利用 C2000 的差异化特性（如快速串行接口 (FSI)、可配置逻辑块 (CLB) 等）将 C2000 设备用作外设扩展器。

5.1.2 深入了解

HIC 支持多种配置，主机可以通过这些配置进行连接。每种配置在延迟、输入/输出 (IO) 引脚开销等方面都有一定的优势和影响。

HIC 可以访问许多常见的 C2000 MCU 外设，包括但不限于：模数转换器、比较器、PWM、CAN 控制器、快速串行接口 (FSI) 通信以及可配置逻辑块 (CLB)。

有关 HIC 功能的完整概述，请参阅《使用 HIC 实现外设扩展应用的设计指南》。

以下部分详细介绍了使用 FSI 进行隔离通信和使用 CLB 实现绝对编码器的 HIC 的实际应用。

5.1.2.1 HIC Bridge for FSI Applications

The Fast Serial Interface (FSI) is capable of supporting low-latency and robust high-speed communication across isolation in a system. Figure 5-1 shows how the host with an Asynchronous RAM interface can use the C2000 device as a peripheral bridge for high-speed communication across isolation using the FSI.

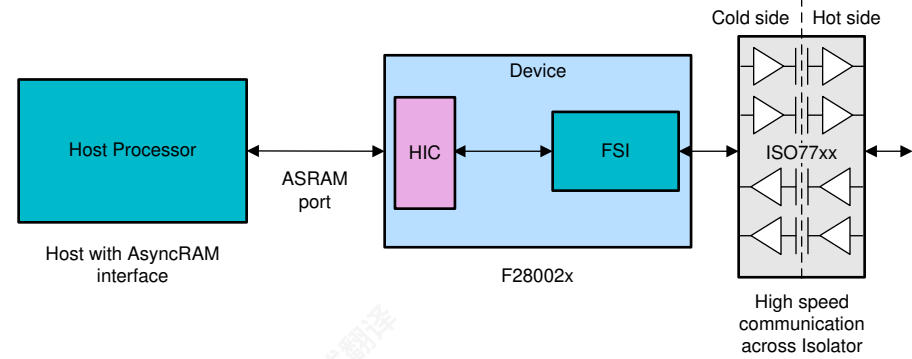


Figure 5-1. HIC Bridge for FSI Applications

5.1.2.2 HIC Bridge for Position Encoder Applications Using CLB

The CLB enables custom logic implementation and augments the existing C2000 peripheral set, thereby eliminating or reducing the need for FPGA, CPLD, or external logic components to achieve the same results. There are numerous solutions that can be implemented with the CLB, one of which is the absolute encoder protocol implementation to interface with the position sensors in an industrial drive control system (Figure 5-2). For the encoder protocols that are supported by the CLB, see the [Position Manager Technology Section on the C2000 Website](#). The HIC can be used as a bridge for the host to interface with the position sensors as shown in Figure 5-2.

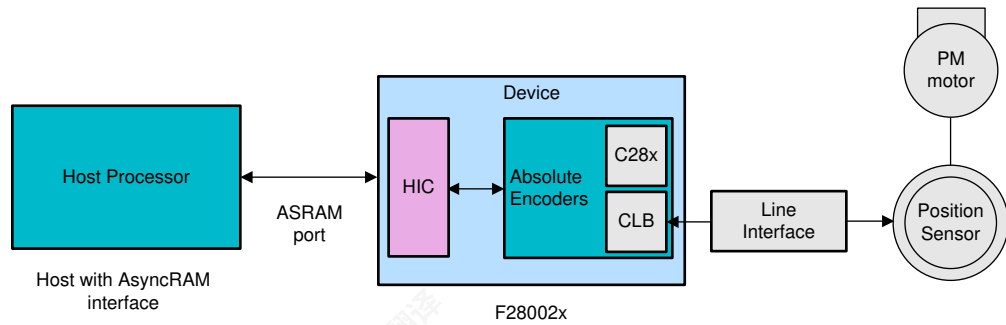


Figure 5-2. HIC Bridge for Position Encoder Applications

5.1.3 Device List

- [TMS320F28003x](#)
- [TMS320F28002x](#)

5.1.4 Hardware Platforms and Software Examples

- [16 Bit FSI Bridge Example in C2000Ware](#)
- [16 Bit EMIF Example to access HIC](#)
- [8 Bit Example in C2000Ware](#)
- [8 Bit EMIF Example to access ADC using HIC](#)

5.1.5 Documentation

- [Design Guide for Enabling Peripheral Expansion Applications Using the HIC](#)
- [TMS320F28003x Microcontrollers Technical Reference Manual](#)
- [TMS320F28003x Microcontrollers Data Sheet](#)
- [TMS320F28002x Microcontrollers Technical Reference Manual](#)
- [TMS320F28002x Microcontrollers Data Sheet](#)

5.1.2.1 FSI应用的HIC桥

快速串行接口 (FSI) 能够在系统中支持隔离的低延迟和鲁棒的高速通信。图 5-1 显示了带有异步 RAM 接口的主机如何使用 C2000 设备作为 FSI 的隔离高速通信外设桥接器。

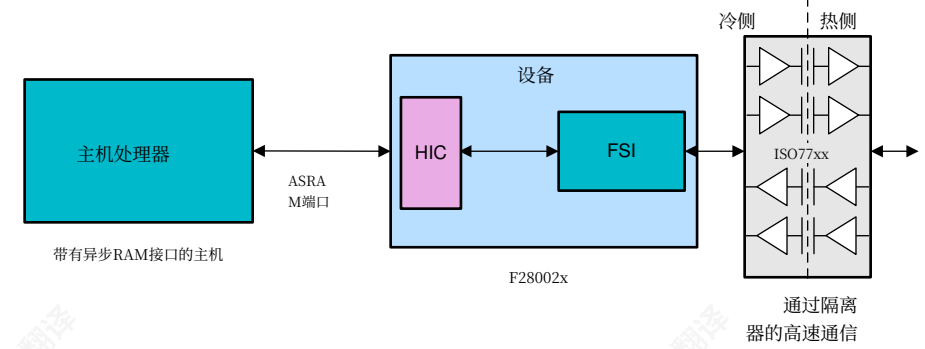


图5-1. FSI应用的HIC桥

5.1.2.2 使用CLB的用于位置编码器应用的HIC桥

CLB支持自定义逻辑实现，并扩展了现有的C2000外设集，从而消除了或减少了FPGA、CPLD或外部逻辑组件的需求，以实现相同的结果。CLB可以实现多种解决方案，其中之一是绝对编码器协议实现，用于与工业驱动控制系统中的位置传感器接口（图 5-2）。有关 CLB 支持的编码器协议，请参阅 C2000 网站上的位置管理技术部分。HIC 可用作主机与位置传感器接口的桥梁，如图 5-2 所示。

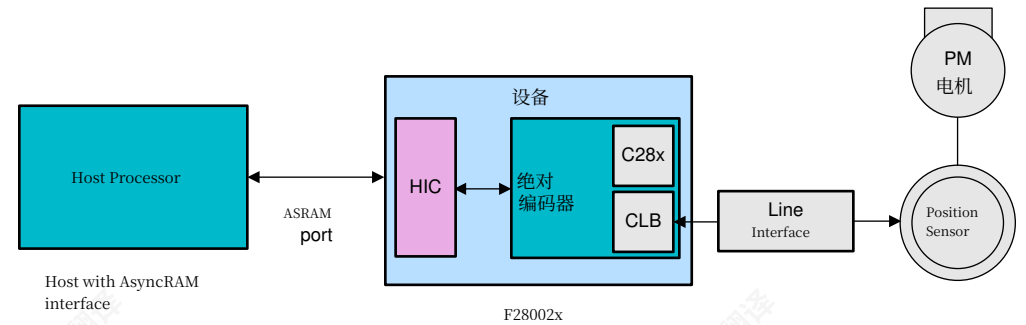


图5-2. 用于位置编码器应用的HIC桥

5.1.3 设备列表

- [TMS320F28003x](#)
- [TMS320F28002x](#)

5.1.4 硬件平台和软件示例

- [C2000Ware 中的 16 位 FSI 桥接器示例](#)
- [用于访问 HIC 的 16 位 EMIF 示例](#)
- [C2000Ware 中的 8 位示例](#)
- [8 位 EMIF 示例，使用 HIC 访问 ADC](#)

5.1.5 文档

- [使用 HIC 启用外设扩展应用程序的设计指南](#)
- [TMS320F28003x 微控制器技术参考手册](#)
- [TMS320F28003x 微控制器数据表](#)
- [TMS320F28002x 微控制器技术参考手册](#)
- [TMS320F28002x 微控制器数据手册](#)

5.2 Securing External Communications and Firmware Updates With an AES Engine

5.2.1 Value Proposition

Integrity and confidentiality of incoming commands, such as firmware updates, are essential to securing your system. Performing cryptographic algorithms in software is slow and requires a certain level of expertise. TI C2000 products have implemented a dedicated Advanced Encryption Standard (AES) engine in hardware to reduce the overhead associated with the encryption, decryption, tagging, and authentication of messages and data.

5.2.2 In Depth

The Advanced Encryption Standard (AES) hardware accelerator found on TI C2000 products is capable of operating in the most common modes for encryption/decryption, authentication, and authenticated encryption with associated data (AEAD). It supports the key sizes of 128 bits, 192 bits, and 256 bits, with a throughput of up to 4 bits/cycle, depending on the operation and key size. Utilizing a DMA, the engine requires very little CPU intervention, allowing you to manage the security of your communications while performing other vital tasks with the CPU.

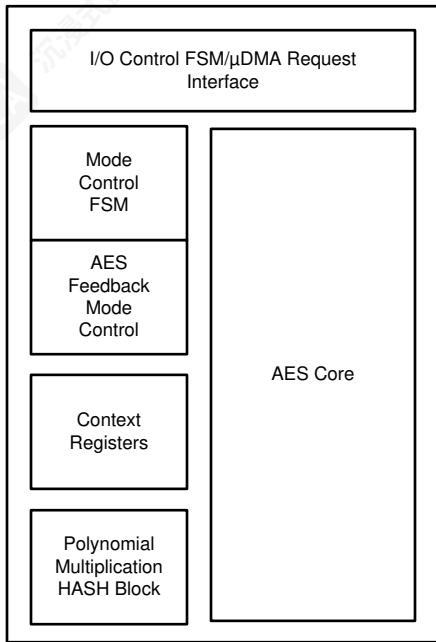


Figure 5-3. AES Block Diagram

For message confidentiality alone, the following encryption/decryption modes are supported:

- Electronic Code Book (ECB)
- Cipher Block Chaining (CBC)
- Counter (CTR) or Integer Counter (ICM)
- Cipher Feedback (CFB)
- Cipher Text Stealing (XTS)
- F8

For message integrity alone, the following authentication modes are supported:

- Cipher Block Chaining Message Authentication Code (CBC_MAC)
- F9

If both confidentiality and integrity of messages are required, the following authenticated encryption with associated data (AEAD) modes are supported:

- Galois Counter Mode (GCM)
- Counter Mode with CBC-MAC (CCM)

5.2 使用 AES 引擎保障外部通信和固件更新 5.2.1 价值主张

传入命令（如固件更新）的完整性和机密性对于保障您的系统至关重要。在软件中执行加密算法速度慢，并且需要一定程度的专业知识。TI C2000 产品在硬件中实现了一个专用的高级加密标准（AES）引擎，以减少与消息和数据的加密、解密、标记和认证相关的开销。

5.2.2 深入探讨

TI C2000 产品上集成的先进加密标准 (AES) 硬件加速器能够在加密/解密、认证以及带关联数据的认证加密 (AEAD) 的最常用模式下运行。它支持 128 位、192 位和 256 位的密钥长度，吞吐量最高可达每周期 4 位，具体取决于操作和密钥长度。利用 DMA，该引擎需要极少的 CPU 干预，使您能够在 CPU 执行其他关键任务的同时管理通信安全。

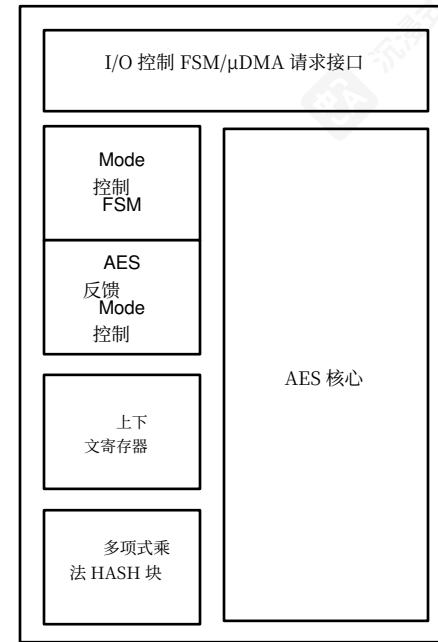


图 5-3. AES 块图

仅用于消息机密性，支持以下加密/解密模式：

- 电子密码本 (ECB)
- 密码分组链接 (CBC)
- 计数器 (CTR) 或 整数计数器 (ICM)
- 密码反馈 (CFB)
- 密码文窃取 (XTS)
- F8

仅就消息完整性而言，支持以下认证模式：

- 密码分组链接消息认证码 (CBC_MAC)
- F9

如果需要消息的机密性和完整性，则支持以下与关联数据一起使用的认证加密（AEAD）模式：

- Galois 计数模式 (GCM)
- 带 CBC-MAC 的计数模式 (CCM)

5.2.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F28003x](#)

5.2.4 Hardware Platforms and Software Examples

- [TMDSCNCD28388D Control CARD](#)
- [AES ECB Encrypt](#)
- [AES ECB Decrypt](#)
- [AES GCM Encrypt](#)
- [AES GCM Decrypt](#)

5.2.5 Documentation

- [TMS320F2838x Microcontrollers Technical Reference Manual](#) (for more information, see the AES chapter)
- [TMS320F28003x Microcontrollers Technical Reference Manual](#) (for more information, see the AES chapter)

5.3 Distributed Real-Time Control Across an Isolation Boundary

5.3.1 Value Proposition

Many real-time control systems rely on external communications for vital system information either as inputs into the control loop or as monitors to the system. Due to the time critical nature of any operation in these systems both the absolute speed as well as the integrity of the data is important. C2000 MCU's Fast Serial Interface (FSI) fulfills this need offering higher throughput compared to other serial communication peripherals, up to 200 Mbps, along with additional features in HW that add data integrity using only a few wires.

5.3.2 In Depth

There are a number of serial communication peripherals to choose from when designing a multi-device real-time control system. With processors needing to pass critical data between each other within very short periods of time, latency is a primary concern for the system designer.

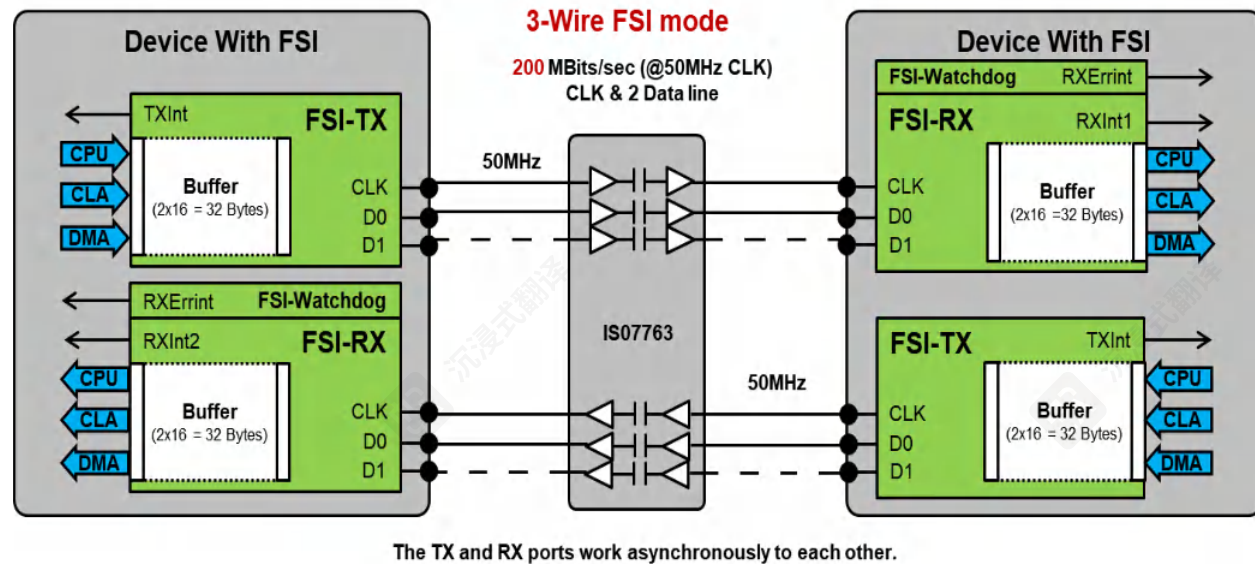


Figure 5-4. Full Duplex 3-Wire FSI Implementation

Note

Single data (D0) and clock (CLK) is minimum set of signals for FSI communications at 100Mbps with lowest signal count and isolation cost. Figure 5-4 shows that signal D1 is an optional data line used to achieve the full 200Mbps data rate of the FSI.

5.2.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F28003x](#)

5.2.4 硬件平台和软件示例

- [TMDSCNCD28388D 控制卡](#)
- [AES ECB 加密](#)
- [AES ECB 解密](#)
- [AES GCM 加密](#)
- [AES GCM 解密](#)

5.2.5 文档

- TMS320F2838x 微控制器技术参考手册（更多信息，请参阅 AES 章节）
- TMS320F28003x 微控制器技术参考手册（更多信息，请参阅 AES 章节）

5.3 跨隔离边界分布式实时控制

5.3.1 价值主张

许多实时控制系统依赖于外部通信来获取关键系统信息，这些信息要么作为控制回路的输入，要么作为系统监控器。由于这些系统中任何操作都具有时间关键性，因此绝对速度和数据完整性都很重要。C2000微控制器的快速串行接口（FSI）满足这一需求，与其他串行通信外设相比，其吞吐量更高，最高可达200 Mbps，并且通过硬件中的附加功能，仅使用少量线路即可增强数据完整性。

5.3.2 深入探讨

在设计多设备实时控制系统时，有多个串行通信外设可供选择。由于处理器需要在非常短的时间内相互传递关键数据，因此延迟是系统设计者需要考虑的主要问题。

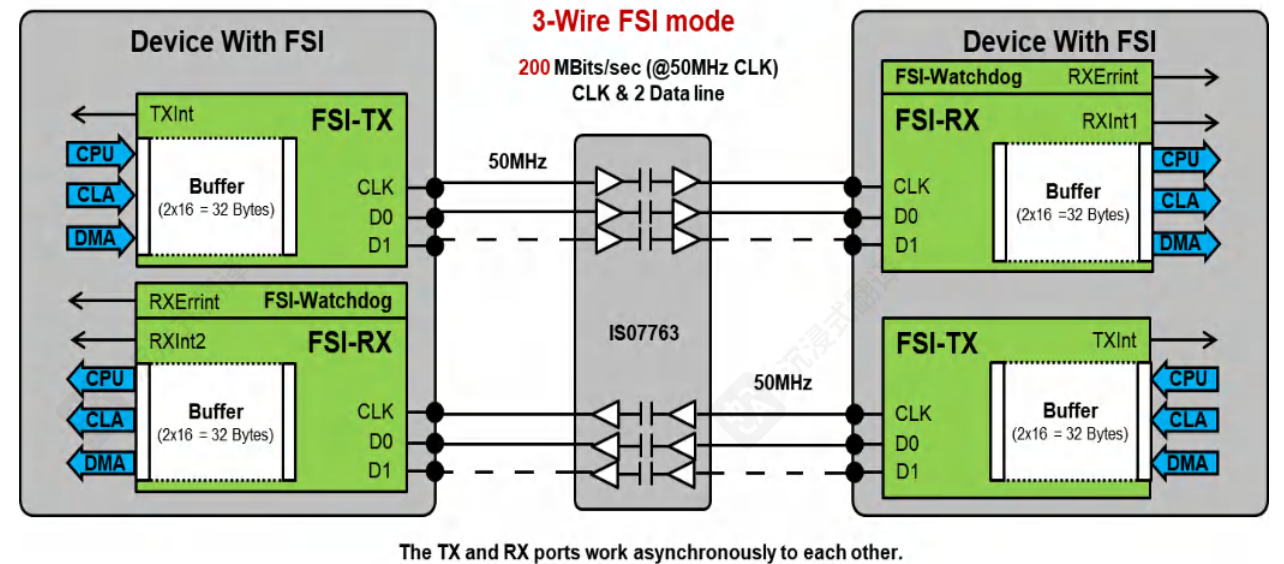


图5-4. 全双工3线FSI实现

Note

单数据 (D0) 和时钟 (CLK) 是 FSI 在 100Mbps 通信中信号数量最少且隔离成本最低的最小信号集。图 5-4 显示，信号 D1 是一条可选数据线，用于实现 FSI 的全部 200Mbps 数据速率。

The FSI physical interface consists of three wires, a clock and two data signals, where one of the data signals is optional (see [Figure 5-4](#)). Data is transferred on both the rising and falling edge which permits a 50 MHz maximum FSI clock frequency to transfer data at 100 Mbps with two wires (CLK and D0) and 200 Mbps with three wires (CLK, D0, and D1). The high through-put along with defined data packets (frames) that contain limited header and footer allows data to be transferred between devices with very little latency. The FSI module consists of independent transmitter and receiver cores which allow for simultaneous full speed communications in both directions with no concept of a master or slave. A real-time system using FSI for distributed control is showcased in [Distributed Multi-axis Servo Drive Over Fast Serial Interface \(FSI\) Reference Design](#).

Features that FSI offers over other commonly used communications peripherals include:

- Hardware implemented CRC at both the transmitter and receiver side eliminating the CPU overhead of a SW implementation
- Delay line control at the receiver module to compensate for channel-to-channel skew
- Line break detection using ping and data frame watchdogs
- FSI protocol has no concept of masters and slaves enabling devices to send feedback at any time without the master device having to make a request
- High bit rate with low signal count reduces the amount of isolators needed in the system

There are a number of system topologies which have components operating on both the “hot” (high voltage) and “cold” (low voltage) sides of the system that must communicate with each other. In this case digital isolators are used to bring data across an isolation barrier and the potential skew between signals that cross the isolation boundary can prove difficult to predict across many units of production. Even in systems without isolation, skew could be introduced by unequal signal trace lengths.

The delay line control feature at the receiver makes FSI well suited for this application as it can actively compensate for this skew ([Figure 5-5](#)) by adding delay to the individual FSI signals. See the [Fast Serial Interface \(FSI\) Skew Compensation](#) for more information on this differentiated feature. Also see the [TMDSFSIADAEVM](#) for hardware evaluation of FSI with digital isolators.

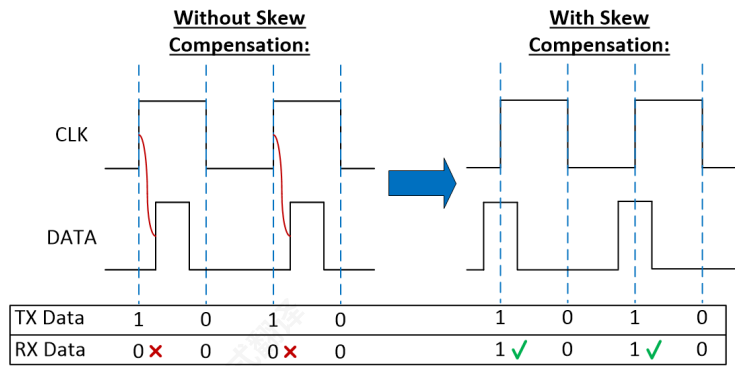


Figure 5-5. FSI Skew Compensation

Note

Data is always transmitted and received on both the rising and falling edges of the FSI clock.

While only one data line is show in [Figure 5-5](#) there is option of second data line in all FSI implementations

5.3.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

FSI 物理接口由三根线组成，包括一个时钟和两个数据信号，其中一个可选的（见图 5-4）。数据在上升沿和下降沿上传输，允许 FSI 时钟频率最高为 50 MHz，以两根线 (CLK 和 D0) 实现 100 Mbps 传输，或以三根线 (CLK、D0 和 D1) 实现 200 Mbps 传输。高吞吐量以及定义的数据包（帧），这些数据包包有限的头部和尾部，使得设备间数据传输延迟极小。FSI 模块由独立的发射器和接收器核心组成，允许双向同时全速通信，没有主从概念。一个使用 FSI 进行分布式控制的实时系统在《分布式多轴伺服驱动快速串行接口 (FSI) 参考设计》中进行了展示。

FSI提供的其他常用通信外设相比的功能包括：

- 硬件实现的CRC，在发射器和接收器两端消除SW实现的CPU开销
- 接收模块中的延迟线控制，以补偿通道间偏移
- 使用ping和数据帧看门狗进行行中断检测
- FSI协议没有主从概念，允许设备随时发送反馈，而无需主设备发起请求
- 高比特率与低信号计数减少了系统中所需的隔离器数量

系统拓扑结构中存在一些组件，这些组件在系统的“热”侧（高压）和“冷”侧（低压）运行，并且必须相互通信。在这种情况下，使用数字隔离器来跨越隔离屏障传输数据，而跨越隔离边界时信号之间的潜在时序偏差可能难以在多个生产单元中预测。即使在没有隔离的系统中，时序偏差也可能由于信号迹线长度不等而引入。

接收器端的延迟线控制功能使 FSI 非常适合该应用，因为它可以通过向单个 FSI 信号添加延迟来主动补偿这种时序偏差（图 5-5）。有关此差异化功能的更多信息，请参阅 Fast Serial Interface (FSI) 时序偏差补偿。另请参阅 TMDSFSIADAEVM，用于使用数字隔离器评估 FSI 的硬件。

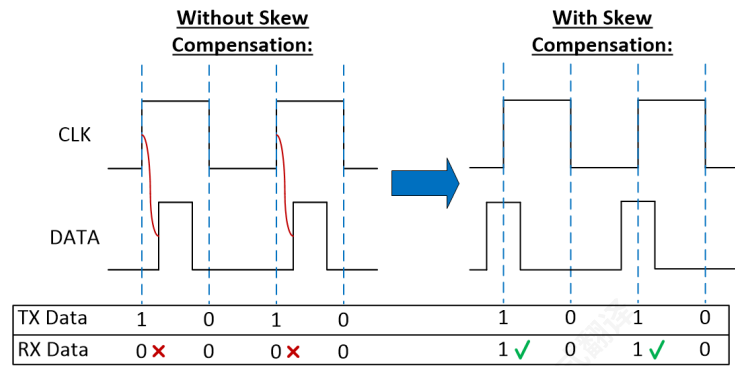


图 5-5。FSI 时序偏差补偿

Note

数据始终在 FSI 时钟的上升沿和下降沿上进行传输和接收。

虽然图5-5中只显示了一条数据线，但在所有FSI实现中都有第二条数据线的选项

5.3.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

5.3.4 Hardware Platforms and Software Examples

- [TMDSF5IADAEVM](#)
- [LAUNCHXL-F280049C LaunchPad](#)
- [LAUNCHXL-F280039C LaunchPad](#)
- [LAUNCHXL-F280025C LaunchPad](#)
- [TIDM-02006](#)

5.3.5 Documentation

- [Fast Serial Interface \(FSI\) Skew Compensation](#)
- [Using the Fast Serial Interface \(FSI\) With Multiple Devices in an Application](#)

5.4 Custom Tests and Data Pattern Generation Using the Embedded Pattern Generator (EPG)

5.4.1 Value Proposition

Unique pattern generation can be a useful addition to a real-time system, either for self testing purposes or as keep alive signals to other ICs in the system. This is typically accomplished with SW routines that toggle GPIOs manually, if possible. The EPG module removes the CPU overhead involved in these applications in addition to providing greater flexibility in patterns that can be generated.

5.4.2 In Depth

The Embedded Pattern Generator (EPG) module is a customizable pattern and clock generator that could servemany test and application scenarios that require a simple pattern generator or a periodic clock generator. TheEPG module can also be used to capture an incoming serial stream of data. The EPG module allows users todesign new clock generators,pulse width modulators (PWMs), serial communication modules, and so forth.

The EPG module can be used as an independent peripheral or it can be used along side other peripherals suchas CAN. When used alongside other peripherals, such as CAN, EPG can provide signal patterns for testing andrunning diagnostic inside the C2000™ device.

When used for clock generation, the EPG can generate:

- Independent clock generation and clock division
- Synchronous clock generation with programmable offsets([Figure 5-6](#))

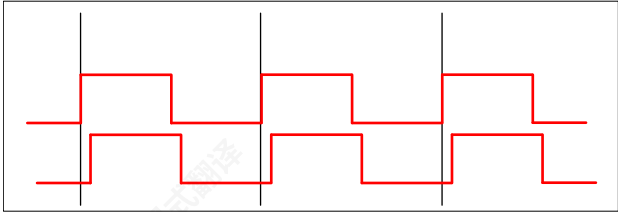


Figure 5-6. Clocks with Offset

When used for pattern generation, the EPG can generate:

- Independent serial data stream generation
- Serial data stream and the associated clock generation([Figure 5-7](#))
- Skew clock with respect to serial data
- Synchronous data stream with programmable offset with respect to one another

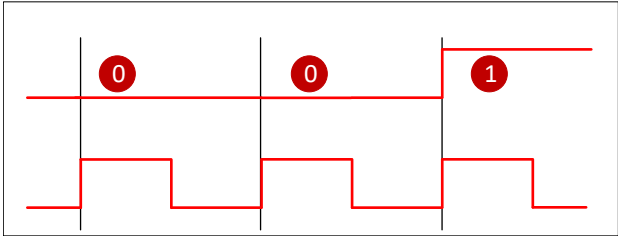


Figure 5-7. Serial Stream and Clock

5.3.4 硬件平台和软件示例

- [TMDSF5IADAEVM](#)
- [LAUNCHXL-F280049C 启动板](#)
- [LAUNCHXL-F280039C 启动板](#)
- [LAUNCHXL-F280025C 启动板](#)
- [TIDM-02006](#)

5.3.5 文档

- [快速串行接口 \(FSI\) 偏移补偿](#)
- [在应用程序中使用快速串行接口 \(FSI\) 与多个设备](#)

5.4 使用嵌入式模式发生器 (EPG) 进行自定义测试和数据模式生成

5.4.1 价值主张

独特的模式生成可以是对实时系统的一个有用补充，无论是用于自检目的，还是作为系统内其他 IC 的保持活动信号。这通常通过 SW 例程手动切换 GPIO 来实现（如果可能）。EPG 模块除了提供更灵活的可生成模式外，还消除了这些应用程序中 CPU 的开销。

5.4.2 深入了解

嵌入式模式生成器（EPG）模块是一个可定制的模式与时钟生成器，可用于多种测试和应用场景，这些场景需要简单的模式生成器或周期性时钟生成器。EPG模块还可以用于捕获传入的串行数据流。EPG模块允许用户设计新的时钟生成器、脉冲宽度调制器（PWM）、串行通信模块等。

EPG模块可以作为独立的外设使用，也可以与其他外设（如CAN）一起使用。当与其他外设（如CAN）一起使用时，EPG可以提供用于C2000™设备内部测试和运行诊断的信号模式。

用于时钟生成时，EPG可以生成：

- 独立时钟生成与时钟分频
- 具有可编程偏移量的同步时钟生成（图5-6）

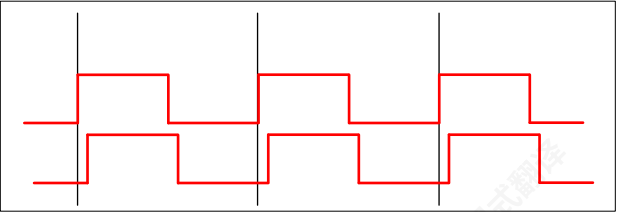


图5-6。带偏移量的时钟

当用于模式生成时，EPG可以生成：

- 独立串行数据流生成
- 串行数据流及相关时钟生成(图5-7)
- 相对于串行数据的偏斜时钟
- 具有可编程偏移的同步数据流，相对于彼此

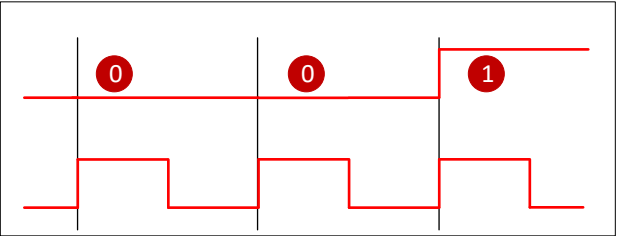


图5-7。串行流与时钟

5.4.3 Device List

- [TMS320F28003x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

5.4.4 Hardware Platforms and Software Examples

- [C2000Ware EPG example for clock generation](#)
- [C2000Ware EPG example for serial data](#)
- [TMS320F280039C LaunchPad](#)
- [TMS320F2800137 LaunchPad](#)
- [TMDSCNCDF2800157](#)

5.4.5 Documentation

- [Designing With the C2000™ Embedded Pattern Generator\(EPG\)](#)
- [TMS320F28003x Real-Time Microcontrollers TRM](#)
- [TMS320F280013x Real-Time Microcontroller TRM](#)
- [TMS320F280015x Real-Time Microcontroller TRM](#)

6 Safety Key Technologies

6.1 Non-Intrusive Run Time Monitoring and Diagnostics as Part of the Control Loop

6.1.1 Value Proposition

CPU resource management is often a key concern when evaluating the ability to control the system. Understanding the bandwidth required by the different functions is important not only during the development of the system but also while the system is in its final use. The ERAD (Embedded Real Time and Diagnostics) module provides this capability on the C2000 MCU, decreasing time spent during development, but also ensuring stable and safe performance in the end application.

6.1.2 In Depth

ERAD is a hardware module with enhanced bus comparators and system event counters that sits within the MCU bus architecture as shown in [Figure 6-1](#). ERAD on its own can generate system level interrupts and flags and can also feed into other peripherals such as the CLB for further enhancing the capabilities. For complete understanding of the relationship between these two modules, see the CLB documentation.

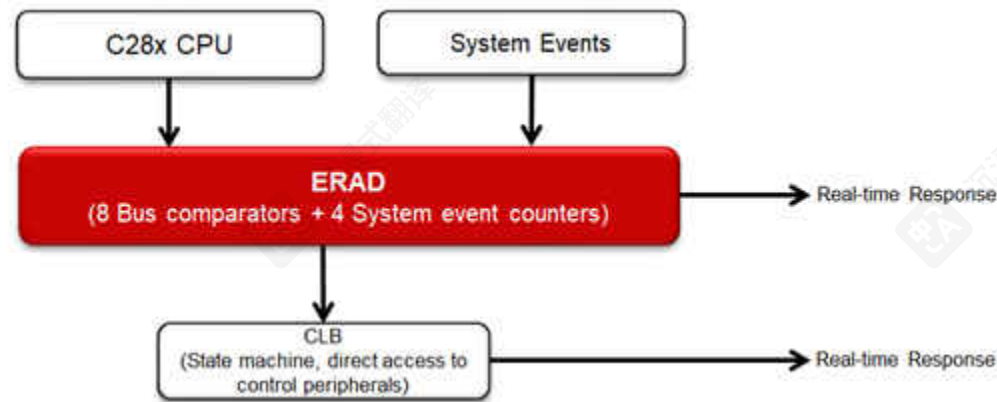


Figure 6-1. ERAD Block Diagram

Enhanced bus comparator monitors some of the critical CPU internal buses and signals that convey information about the CPU code execution and pipeline. Most of the critical CPU interfaces are monitored by EBC units, which can be configured to match any address/data/instruction patterns that appear on the CPU interface and generate events based on this. A simple example would be to generate an event when an address of interest is accessed for a read or write. Additionally these events can be exported to CLB to define state machines to keep track of sequences of these events.

5.4.3 设备列表

- [TMS320F28003x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

5.4.4 硬件平台和软件示例

- [C2000Ware EPG 时钟生成示例](#)
- [C2000Ware EPG 示例用于串行数据](#)
- [TMS320F280039C LaunchPad](#)
- [TMS320F2800137 LaunchPad](#)
- [TMDSCNCDF2800157](#)

5.4.5 文档

- [使用 C2000™ 嵌入式模式生成器\(EPG\)进行设计](#)
- [TMS320F28003x 实时微控制器 TRM](#)
- [TMS320F280013x 实时微控制器 TRM](#)
- [TMS320F280015x 实时微控制器 TRM](#)

6 安全关键技术

6.1 非侵入式运行时监控和诊断作为控制回路的一部分

6.1.1 价值主张

CPU资源管理在评估系统控制能力时通常是关键问题。了解不同功能所需的带宽不仅在进行系统开发期间重要，而且在系统最终使用时也同样重要。ERAD（嵌入式实时和诊断）模块在C2000微控制器上提供此功能，减少了开发期间的的时间，但也确保了最终应用中的稳定和安全性能。

6.1.2 深入了解

ERAD 是一个带有增强总线比较器和系统事件计数器的硬件模块，它位于 MCU 总线架构中，如图 6-1 所示。ERAD 本身可以生成系统级别的中断和标志，并且还可以输入到其他外设（如 CLB）以进一步增强其功能。要完全理解这两个模块之间的关系，请参阅 CLB 文档。

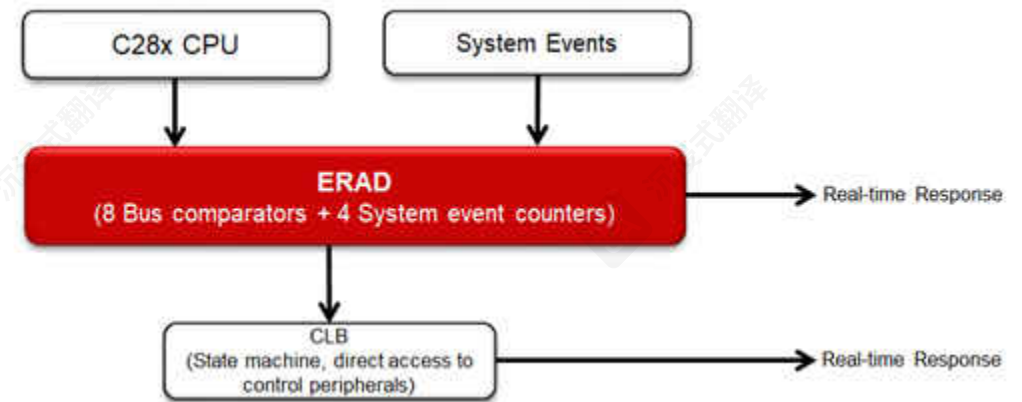


图 6-1. ERAD 框图

增强型总线比较器监控部分关键的CPU内部总线和信号，这些总线和信号传递有关CPU代码执行和流水线的信息。大多数关键的CPU接口由EBC单元监控，这些单元可以配置为匹配CPU接口上出现的任何地址/数据/指令模式，并根据此模式生成事件。一个简单的例子是在访问感兴趣的地址进行读或写时生成事件。此外，这些事件可以导出到CLB，以定义状态机来跟踪这些事件序列。

On the other hand, system event counters monitor various events in the system like interrupts and peripheral activity which is another important requirement for any real-time system to monitor and profile. All the key system events like interrupts, DMA triggers and important peripheral events (via interrupts, CLA events, and so forth) are available for profiling inside of the ERAD. Hence, these can be time-stamped, counted and measured with relation to the CPU activity. Enhanced counter block allows the system events as well as CPU activity to be measured and analyzed with reference to other events too.

Example: Stack Overflow Detection

A frequently encountered issue with embedded code development is detecting an overflowing stack. ERAD can simply use a single enhanced bus-comparator unit and then map the stack end address with a small margin to generate an interrupt when there is a write that is attempted. With the below example (Figure 6-2), just use a comparator with address 0x99FF0 and mask the last 4-bits for comparison, so any write access between addresses 0x99FF0 to 0x99FFF would generate an event/interrupt, which can then be handled appropriately based on kind of the application.

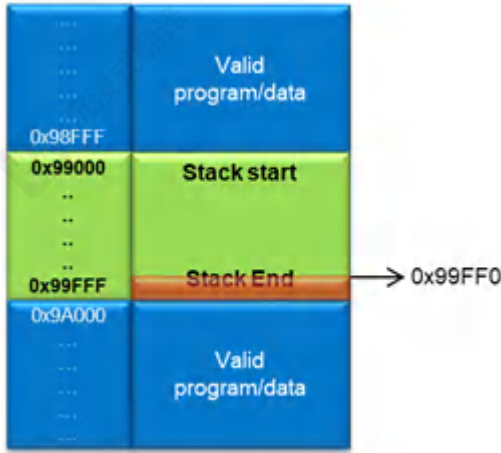


Figure 6-2. Stack Overflow Protection

6.1.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

6.1.4 Hardware Platforms and Software Examples

- [F28388D controlCARD evaluation module](#)
- [C2000 MCU F280049C LaunchPad™ development kit](#)
- [F280039C controlCARD evaluation module](#)
- [F280025 controlCARD evaluation module](#)

6.1.5 Documentation

- [ERAD Introduction video](#)
- [Embedded Real-Time Analysis and Response for Control Applications](#)
- [TMS320F2838x Real-Time Microcontrollers TRM](#)
- [TMS320F28004x Real-Time Microcontrollers TRM](#)
- [TMS320F28003x Real-Time Microcontrollers TRM](#)
- [TMS320F28002x Real-Time Microcontrollers TRM](#)

另一方面，系统事件计数器监控系统中的各种事件，如中断和外设活动，这是任何实时系统监控和分析的重要要求。所有关键系统事件，如中断、DMA触发器和重要外设事件（通过中断、CLA事件等）都可以在ERAD内部进行性能分析。因此，这些事件可以被时间戳标记、计数和测量，以与CPU活动相关。增强型计数器模块允许系统事件以及CPU活动与其它事件一起被测量和分析。

示例：Stack Overflow 检测

嵌入式代码开发中经常遇到的问题是检测溢出栈。ERAD 可以简单地使用一个增强型总线比较器单元，然后将栈末地址映射一个小余量，在尝试写入时生成中断。通过下面的示例（图6-2），只需使用地址为0x99FF0的比较器，并将最后4位进行屏蔽比较，因此任何在地址0x99FF0到0x99FFF之间的写入访问都会生成事件/中断，然后可以根据应用程序的类型进行适当处理。

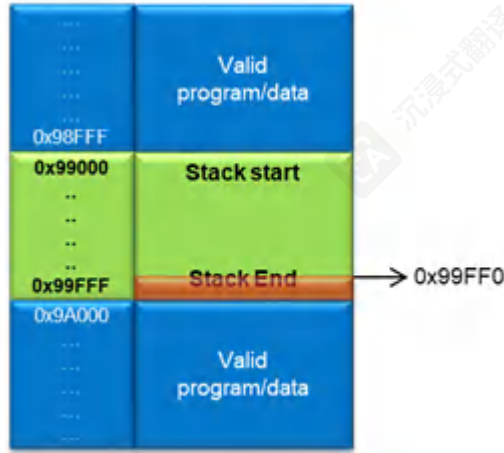


图 6-2。Stack Overflow 保护

6.1.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F28004x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

6.1.4 硬件平台和软件示例

- [F28388D controlCARD 评估模块](#)
- [C2000 MCU F280049C LaunchPad™ 开发套件](#)
- [F280039C controlCARD 评估模块](#)
- [F280025 controlCARD 评估模块](#)

6.1.5 文档

- [ERAD 简介 视频](#)
- [嵌入式实时分析与响应控制应用](#)
- [TMS320F2838x 实时微控制器 TRM](#)
- [TMS320F28004x 实时微控制器 TRM](#)
- [TMS320F28003x 实时微控制器 TRM](#)
- [TMS320F28002x 实时微控制器 TRM](#)

6.2 Hardware Built-In Self-Test of the C28x CPU

6.2.1 Value Proposition

Safety applications require both start up and run time diagnostics of critical device components. The most critical is the main CPU; in the case of a C2000 device this is the C28x real-time core. Furthermore, without deep understanding of the CPU this is not feasible for a user to create such diagnostics in software. On supported C2000 MCUs, a block of logic is provided to automate and perform these tests to high level of diagnostic coverage to satisfy safety standards like ISO 26262 or IEC 61508.

6.2.2 In Depth

The Hardware Built-In Self-Test (HWBIST) provides a very high diagnostic coverage on the CPU, including the FPU, TMU, and VCU, at a transistor level during start-up and application time. This logic utilizes Design for Test (DfT) structures inserted into the device for rapid execution of high quality manufacturing tests, but with an internal test engine rather than external automated test equipment (ATE). This technique has proven to be effective in providing high coverage in less time.

HWBIST tests are triggered by the software. The user may select to run all tests or only a subset of the tests based on the execution time allocated for the diagnostic. This time sliced test feature enables the HWBIST to be used effectively as a runtime diagnostic with execution of test in parallel with the application. HWBIST execution failure such as detection of a logic failure or time out without the self-test completing triggers an NMI. The maximum level of diagnostic coverage provided by HWBIST depends on the device as shown in [HWBIST Supported Diagnostic Coverage by Device](#).

Table 6-1. HWBIST Supported Diagnostic Coverage by Device

F2838xD/S	F2837xD/S	F2807x	F28002x
> 99% DC	> 99% DC	> 99% DC	> 99% DC

To configure and run HWBIST it is required to use the HWBIST APIs provided by the C2000 Software Diagnostic library. The functions provided enable running a single micro-run or running the full HWBIST test as well as configuring a self-test mode within HWBIST to allow a few different types of faults to be injected to check that it's functioning correctly.

6.2.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

6.2.4 Hardware Platforms and Software Examples

- [C2000 Software Diagnostic Library in C2000Ware \(F28002x, F2838x\)](#)
- [C2000 SafeTI™ Diagnostic Software Library \(F2837xD/S, F2807x\)](#)

6.2.5 Documentation

- [Safety Manual for TMS320F28002x](#)
- [Safety Manual for TMS320F2837xD, TMS320F2837xS, and TMS320F2807x](#)
- [C2000™ Hardware Built-In Self-Test](#)

6.2 C28x CPU 硬件内置自检

6.2.1 价值主张

安全应用需要关键设备组件的启动和运行时诊断。最关键的是主CPU；对于C2000设备来说，这是C28x实时核心。此外，没有对CPU的深入理解，用户无法在软件中创建此类诊断。在支持的C2000微控制器上，提供了一个逻辑块来自动化和执行这些测试，以实现高级别的诊断覆盖率，以满足ISO 26262或IEC 61508等安全标准。

6.2.2 深入解析

硬件内置自检 (HWBIST) 在启动和应用时间期间，以晶体管级别对 CPU 提供非常高的诊断覆盖率，包括 FPU、TMU 和 VCU。该逻辑利用插入到设备中的可测试性设计 (DfT) 结构，以快速执行高质量的制造测试，但使用内部测试引擎而非外部自动化测试设备 (ATE)。这种技术已被证明在更短的时间内提供高覆盖率。

HWBIST 测试由软件触发。用户可以选择运行所有测试或仅测试的一个子集，基于分配给诊断的执行时间。这种时间分片测试功能使 HWBIST 能够作为运行时诊断有效使用，测试与应用程序并行执行。HWBIST 执行失败，如检测到逻辑故障或自检未完成时超时，会触发 NMI。HWBIST 提供的最大诊断覆盖率取决于设备，如 HWBIST 按设备支持的诊断覆盖率所示。

表6-1。按设备划分的HWBIST支持诊断覆盖范围

F2838xD/S	F2837xD/S	F2807x	F28002x
> 99% DC	> 99% DC	> 99% DC	> 99% DC

要配置和运行HWBIST，需要使用C2000软件诊断库提供的HWBIST API。提供的函数可以运行单个微运行，也可以运行完整的HWBIST测试，以及在HWBIST中配置自检模式，以允许注入几种不同类型的故障来检查其是否正常工作。

6.2.3 设备列表

- [TMS320F2838xD/S](#)
- [TMS320F2837xD/S](#)
- [TMS320F2807x](#)
- [TMS320F28003x](#)
- [TMS320F28002x](#)

6.2.4 硬件平台和软件示例

- [C2000 软件诊断库（C2000Ware 中，F28002x，F2838x）](#)
- [C2000 SafeTITM 诊断软件库（F2837xD/S，F2807x）](#)

6.2.5 文档

- [TMS320F28002x 安全手册](#)
- [TMS320F2837xD、TMS320F2837xS 和 TMS320F2807x 安全手册](#)
- [C2000TM 硬件内置自检](#)

6.3 Zero CPU Overhead Cyclic Redundancy Check for Embedded On-Chip Memories

6.3.1 Value Proposition

During development of any control system great care is taken to understand the interactions and timings of the code that will be executed by the device. However, no matter the previous effort, if that code is unintentionally modified or corrupted the behavior of the system is no longer guaranteed. Many systems, especially those concerned with safety, will periodically check the goodness of the memories, but this comes at the expense of CPU and system bandwidth. The Background CRC(BGCRC) module on C2000 MCUs solves this problem, implementing a CRC engine that can check memory integrity with no CPU overhead or system performance impact.

6.3.2 In Depth

The memory blocks present in C2000 MCUs are parity or ECC protected. This helps in detecting memory corruptions if any. But parity/ECC feature can detect errors only if application does a read access to the memory. Hence it is very important to do a periodic readback of the memories to detect such issues earlier. Adding a CRC check for the memories confirms that the memory contents are unaltered.

The BGCRC module in C2000 MCUs can be used to perform readback of the memories in the background, hence ensuring uninterrupted execution of application without consuming any CPU cycles. Once triggered, it reads the memory contents in the background, computes CRC and compare it with the programmed golden value and flag error if there is mismatch. It also supports Scrub mode, where it does not perform a CRC comparison, but check for Parity/ECC errors. The reads happen during the idle time (when none of the other memory owners such as CPU, CLA, DMA are accessing the memory block) and hence the functional accesses are not impacted.

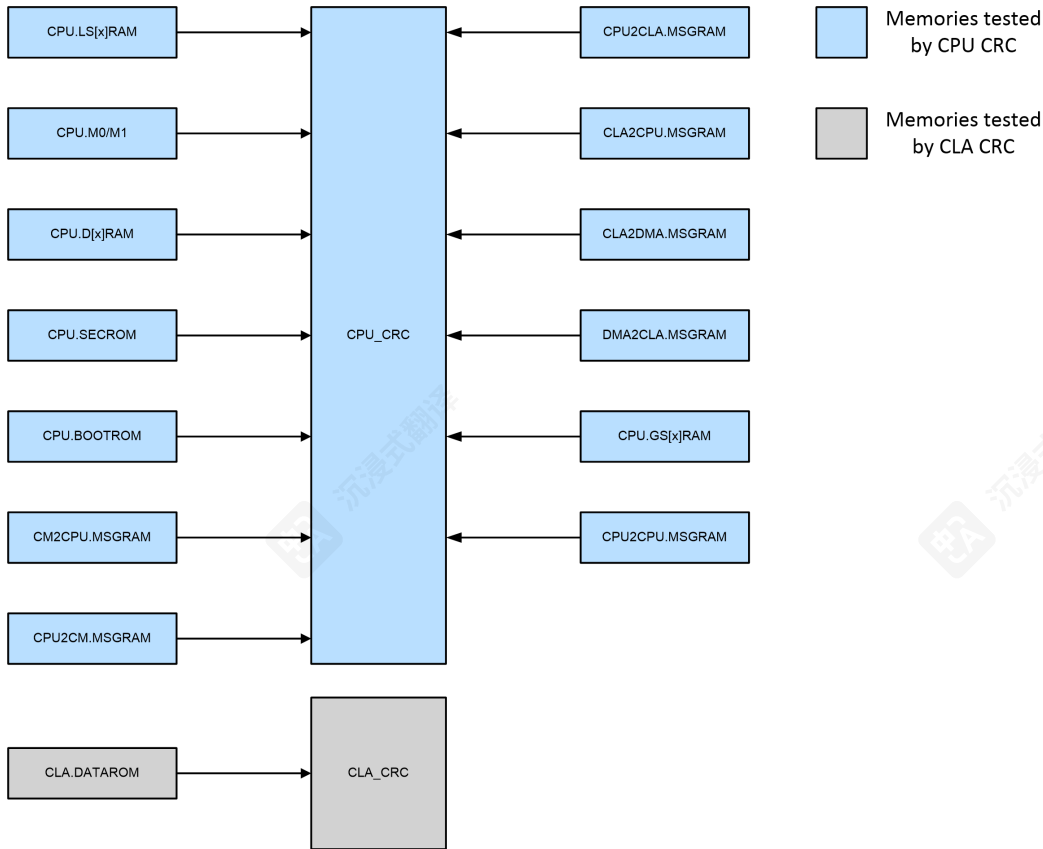


Figure 6-3. BGCRC Implementation on the TMS320F2838xD Device

6.3.3 Device List

- TMS320F2838xD/S

6.3 嵌入式片上存储器的零CPU开销循环冗余校验

6.3.1 价值主张

在开发任何控制系统时，都会非常小心地理解设备将要执行的代码的交互和时序。然而，无论之前的努力如何，如果该代码被无意修改或损坏，系统的行为将不再得到保证。许多系统，尤其是那些涉及安全的系统，会定期检查内存的良好性，但这会占用CPU和系统带宽。C2000微控制器的背景CRC(BGCRC)模块解决了这个问题，实现了一个CRC引擎，可以在没有任何CPU开销或系统性能影响的情况下检查内存完整性。

6.3.2 深入探讨

C2000微控制器中的内存块具有奇偶校验或ECC保护。这有助于检测任何内存损坏。但是，奇偶校验/ECC功能只有在应用程序对内存进行读取访问时才能检测到错误。因此，定期对内存进行读回以尽早发现此类问题非常重要。为内存添加CRC校验可以确认内存内容未被更改。

C2000微控制器中的BGCRC模块可用于在后台对内存进行读回，从而确保应用程序在不消耗任何CPU周期的情况下不间断执行。一旦触发，它会在后台读取内存内容，计算CRC并与编程的黄金值进行比较，如果存在不匹配则标记错误。它还支持Scrub模式，在该模式下它不会执行CRC比较，但会检查奇偶校验/ECC错误。读取操作发生在空闲时间（当没有其他内存所有者，如CPU、CLA、DMA访问内存块时），因此功能访问不会受到影响。

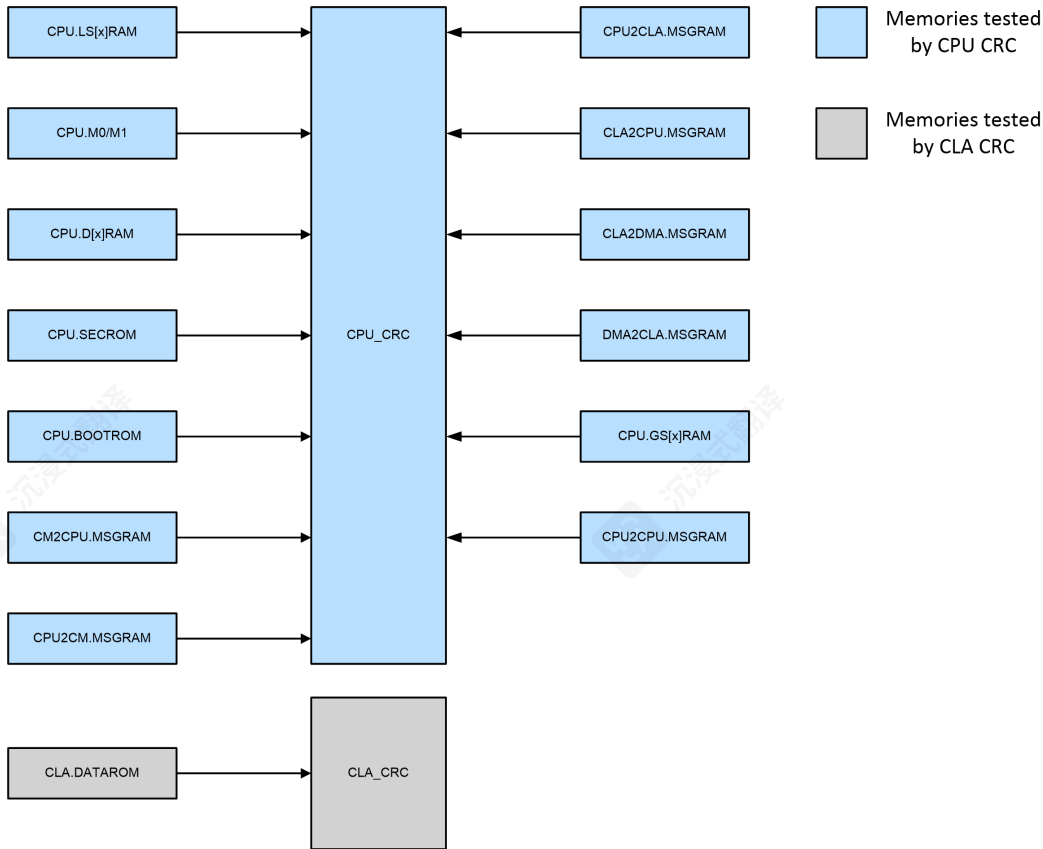


图6-3。在TMS320F2838xD设备上的BGCRC实现

6.3.3 设备列表

- TMS320F2838xD/S

- [TMS320F28003x](#)
- [TMS320F28002x](#)

6.3.4 Hardware Platforms and Software Examples

- [F2838x BGCRC Example](#)
- [F28003x BGCRC Example](#)
- [F28002x BGCRC Example](#)

6.3.5 Documentation

- [CRC Engines in C2000™ Devices](#)

6.4 Boot Code Authentication Prior To Code Execution

6.4.1 Value Proposition

A real-time control system typically has multiple FirmWare(FW) updates over its lifetime. While these can be done in person, it is much more commonplace and convient to update the FW remotely using a bootloader on the device. With a remote update there is increased concern to ensure the contents of the new FW are not only correct but have not been manipulated by an outside source. The secure boot feature on C2000 MCUs provides a method to verify the contents of the new FW prior to code execution.

6.4.2 In Depth

One of the C2000 MCU features related to the application flash boot is the ability to authenticate the user application code in flash before execution. This ascertains the integrity of the application code by ensuring that it has not been tampered with, after getting programmed into the Flash memory.

The secure flash boot is realized using the 128-bit AES-CMAC Authentication algorithm that is run on the application code contents returning a pass/fail status and proceeds to execute the application code only if the authentication succeeds. Cipher-based Message Authentication Code (CMAC) is an AES-based authentication algorithm that constructs an authentication tag from a block of input data. The input data block is fed 128 bits at a time, into the cryptoengine/software (based on the CPU subsystem), along with a 128-bit CMAC key (Figure 6-4).

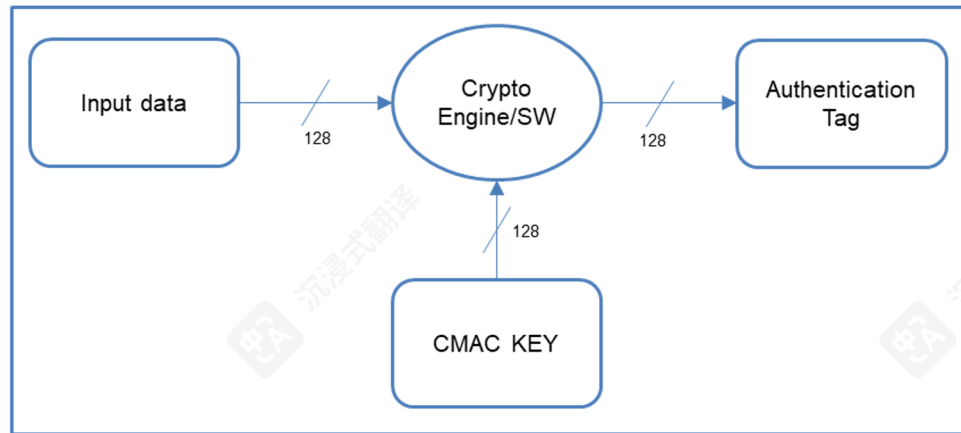


Figure 6-4. CMAC Operation

6.4.3 Device List

- [TMS320F2838xD/S](#)
- [TMS320F28003x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

- [TMS320F28003x](#)
- [TMS320F28002x](#)

6.3.4 硬件平台和软件示例

- [F2838x BGCRC 示例](#)
- [F28003x BGCRC 测试 例](#)
- [F28002x BGCRC 示例](#)

6.3.5 文档

- [C2000™ 设备中的 CRC 引擎](#)

6.4 代码执行前的引导代码认证

6.4.1 价值主张

实时控制系统在其生命周期中通常有多个 FirmWare(FW) 更新。虽然这些更新可以在现场进行，但使用设备上的引导加载程序远程更新 FW 要普遍得多且更方便。通过远程更新，需要更加关注确保新 FW 的内容不仅正确，而且没有被外部来源篡改。C2000 MCU 上的安全启动功能提供了一种在代码执行前验证新 FW 内容的方法。

6.4.2 深入探讨

C2000 MCU 与应用闪存启动相关的功能之一是能够在执行前认证闪存中的用户应用代码。这通过确保应用代码在编程到闪存内存后未被篡改，从而确认了应用代码的完整性。

安全闪存启动是通过运行在应用程序代码内容上的 128 位 AES-CMAC 认证算法实现的，该算法返回通过/失败状态，并且仅在认证成功时才执行应用程序代码。基于密码的消息认证码 (CMAC) 是一种基于 AES 的认证算法，它从一块输入数据中构造一个认证标签。输入数据块以 128 位为单位被输入到 cryptoengine/软件（基于 CPU 子系统），以及一个 128 位的 CMAC 密钥（图 6-4）。

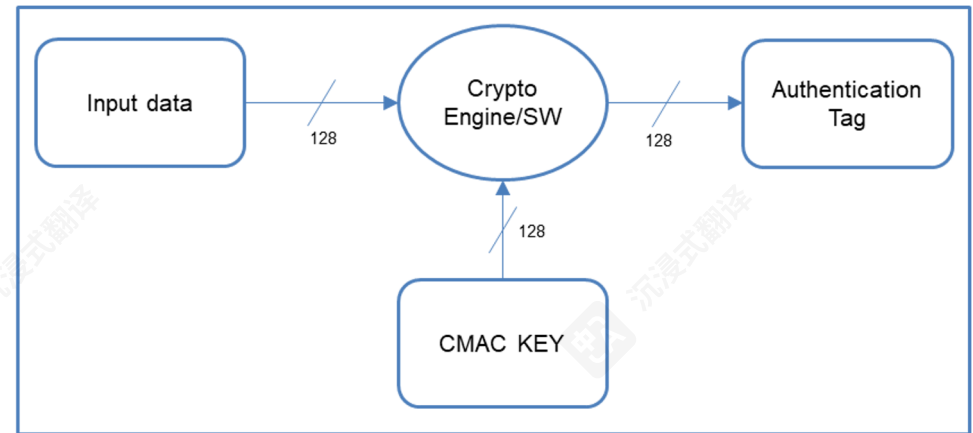


图6-4。CMAC操作

6.4.3设备列表

- [TMS320F2838xD/S](#)
- [TMS320F28003x](#)
- [TMS320F280013x](#)
- [TMS320F280015x](#)

6.4.4 Hardware Platforms and Software Examples

- C2000Ware Secure Boot Example for F2838xD
- F28388D controlCARD evaluation module
- TMS320F280039C LaunchPad
- TMS320F2800137 LaunchPad
- TMDSCNCD2800157 controlCARD

6.4.4.1 Documentation

- [Secure BOOT on C2000 Device](#)
- [TMS320F2838x Real-Time Microcontrollers With Connectivity Manager TRM](#)
- [TMS320F28003x Real-Time Microcontrollers TRM](#)
- [TMS320F280013x Real-Time Microcontrollers TRM](#)
- [TMS320F280015x Real-Time Microcontrollers TRM](#)

7 References

7.1 Device List

- TMS320F2838xD/S
- TMS320F2837xD/S
- TMS320F2807x
- TMS320F28004x
- TMS320F28003x
- TMS320F28002x
- TMS320F280013x
- TMS320F280015x

7.2 Hardware/Software Resources

- [TIDM-1007 Interleaved CCM Totem Pole Bridgeless Power Factor Correction \(PFC\) Reference Design](#)
- [TIDM-HV-1PH-DCAC Single-Phase Inverter Reference Design With Voltage Source and Grid Connected Modes](#)
- [TMDXIDDK379D C2000 DesignDRIVE Development Kit for Industrial Motor Control](#)
- [TMDSHVMTRINSPIN High Voltage Motor Control Kit with InstaSPIN-FOC and InstaSPIN-MOTION enabled for F280049C device lab7 and lab8](#)
- [Vienna Rectifier-Based Three Phase Power Factor Correction Reference Design Using C2000 MCU](#)
- [TIDM-02002 Bidirectional CLLLC resonant dual active bridge for HEV/EV onboard charger](#)
- [TIDA-00961 Highly Efficient 1.6kW High Density GaN Based 1MHz CrM Totem-pole PFC Converter](#)
- [F28388D controlCARD evaluation module](#)
- [C2000 MCU F28379D LaunchPad™ development kit](#)
- [C2000 MCU F280049C LaunchPad™ development kit](#)
- [C2000 MCU F280039C LaunchPad™ development kit](#)
- [C2000 MCU F280025 LaunchPad™ development kit](#)
- [C2000 MCU F2800137 LaunchPad™ development kit](#)
- [C2000 DesignDRIVE Development Kit for Industrial Motor Control](#)
- [Valley switching boost power factor correction \(PFC\) reference design](#)
- [TMDSIDDK379D](#)
- [TMDSCNCD28379D](#)
- [TMDSCNCD280049C](#)
- [TMDSCNCD280039C](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)
- [TIDM-1022 Valley Switching Boost Power Factor Corrector](#)

7.3 Documentation

- Texas Instruments: [C2000 Real-Time Control Peripherals Reference Guide](#)
- Texas Instruments: [TMS320F2837xD Dual-Core Real-Time Microcontrollers Technical Reference Manual](#)
- Texas Instruments: [TMS320F28004x Real-Time Microcontrollers Technical Reference Manual](#)

6.4.4 硬件平台和软件示例

- C2000Ware F2838xD 安全启动示例
- F28388D controlCARD 评估模块
- TMS320F280039C LaunchPad
- TMS320F2800137 LaunchPad
- TMDSCNCD2800157 控制卡

6.4.4.1 文档

- [C2000 设备的 Secure BOOT](#)
- [TMS320F2838x 实时微控制器带连接管理器 TRM](#)
- [TMS320F28003x 实时微控制器 TRM](#)
- [TMS320F280013x 实时微控制器 TRM](#)
- [TMS320F280015x 实时微控制器 TRM](#)

7 参考文献

7.1 设备列表

- TMS320F2838xD/S
- TMS320F2837xD/S
- TMS320F2807x
- TMS320F28004x
- TMS320F28003x
- TMS320F28002x
- TMS320F280013x
- TMS320F280015x

7.2 硬件/软件资源

- [TIDM-1007 交错式 CCM 无桥式功率因数校正 \(PFC\) 参考设计](#)
- [TIDM-HV-1PH-DCAC 单相逆变器参考设计，支持电压源和并网模式](#)
- [TMDXIDDK379D C2000 DesignDRIVE 工业电机控制开发套件](#)
- [TMDSHVMTRINSPIN 高压电机控制套件，支持 InstaSPIN-FOC 和 InstaSPIN-MOTION，适用于 F280049C 设备的 lab7 和 lab8](#)
- [基于维也纳整流器的三相功率因数校正参考设计，使用 C2000 MCU](#)
- [TIDM-02002 双向 CLLLC 谐振双有源桥，用于 HEV/EV 车载充电器](#)
- [TIDA-00961 高效 1.6kW 高密度氮化镓基 1MHz CrM 总线型 PFC 转换器](#)
- [F28388D 控制CARD 评估模块](#)
- [C2000 MCU F28379D LaunchPad™ 开发套件](#)
- [C2000 MCU F280049C LaunchPad™ 开发套件](#)
- [C2000 MCU F280039C LaunchPad™ 开发套件](#)
- [C2000 MCU F280025 LaunchPad™ 开发套件](#)
- [C2000 MCU F2800137 LaunchPad™ 开发套件](#)
- [C2000 设计驱动工业电机控制开发套件](#)
- [山谷切换式升压功率因数校正 \(PFC\) 参考设计](#)
- [TMDSIDDK379D](#)
- [TMDSCNCD28379D](#)
- [TMDSCNCD280049C](#)
- [TMDSCNCD280039C](#)
- [TMDSCNCD2800137](#)
- [TMDSCNCD2800157](#)
- [TIDM-1022 Valley 开关升压功率因数校正器](#)

7.3 文档

- 德州仪器：C2000实时控制外设参考指南 • 德州仪器：TMS320F2837xD双核实时微控制器技术参考手册
- 德州仪器：TMS320F28004x 实时微控制器技术参考手册

- Texas Instruments: [Enhancing the Performance Capabilities of the C2000™ MCU Family](#)
- Texas Instruments: [TMS320C28x Extended Instruction Sets Technical Reference Manual](#)
- Texas Instruments: [TMS320C28x CPU and Instruction Set Reference Guide](#)
- Texas Instruments: [TMS320F28002x Real-Time Microcontrollers Data Sheet](#)
- Texas Instruments: [TMS320F2838x Real-Time Microcontrollers Technical Reference Manual](#)
- Texas Instruments: [TMS320F28002x Real-Time Microcontrollers Technical Reference Manual](#)
- Texas Instruments: [Designing with the C2000™ Configurable Logic Block](#)
- Texas Instruments: [How to Migrate Custom Logic From an FPGA/CPLD to C2000™ Microcontrollers](#)
- Texas Instruments: [CLB Tool User's Guide](#)
- Texas Instruments: [Quick Response Control of PMSM Using Fast Current Loop](#)
- Texas Instruments: [Fast Current Loop Library](#)
- Texas Instruments: [C2000 Position Manager PTO API Reference Guide](#)
- Texas Instruments: [Distributed Multi-axis Servo Drive Over Fast Serial Interface \(FSI\) Reference Design](#)
- Texas Instruments: [Fast Serial Interface \(FSI\) Skew Compensation](#)
- Texas Instruments: [Using the Fast Serial Interface \(FSI\) With Multiple Devices in an Application](#)
- Training: [How the C2000 Configurable Logic Block \(CLB\) tool integrates custom logic in my design](#)
- [ADC Specifications for TMS320F2838xD/S](#)
- [ADC Specifications for TMS320F2837xD/S](#)
- [ADC Specifcations for TMS320F2807x](#)
- [ADC Specifications for TMS320F28004x](#)
- [C2000™ F2837xD Microcontroller Workshop](#)
- [CLA Hands On Workshop](#)
- [CLA Usage in Valley Switching Boost Power Factor Correction \(PFC\) Reference Design](#)
- [CLA FAQ on E2E](#)
- [C2000 F2837xD Microcontroller 1-Day Workshop Section 1.6: Control Peripherals](#)

8 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision E (June 2022) to Revision F (March 2023)	Page
• Updated the numbering format for tables, figures and cross-references throughout the document.....	4
• Added F280013x and F280015x device to applicable devices as well as any support hardware tools.....	4
• Updated Section 3.7.2	36
• Updated Section 3.9	40
• Updated Section 3.9.2	40

- 德州仪器：提升C2000TM微控制器系列性能能力
- 德州仪器：TMS320C28x扩展指令集技术参考手册
- 德州仪器：TMS320C28x CPU和指令集参考指南
- 德州仪器：TMS320F28002x实时微控制器数据手册
- 德州仪器：TMS320F2838x实时微控制器技术参考手册
- 德州仪器：TMS320F28002x实时微控制器技术参考手册
- 德州仪器：使用C2000TM可配置逻辑块进行设计
- 德州仪器：如何将自定义逻辑从FPGA/CPLD迁移到C2000TM微控制器
- 德州仪器：CLB工具用户指南
- 德州仪器：使用快速电流环对PMSM进行快速响应控制
- 德州仪器：快速电流环库
- 德州仪器：C2000位置管理器PTO API参考指南
- 德州仪器：基于快速串行接口 (FSI) 的分布式多轴伺服驱动参考设计
- 德州仪器：快速串行接口 (FSI) 偏移补偿
- 德州仪器：在应用中使用快速串行接口 (FSI) 与多个设备
- 培训：C2000 可配置逻辑块 (CLB) 工具如何将自定义逻辑集成到我的设计中
- [TMS320F2838xD/S 的 ADC 规格](#)
- [TMS320F2837xD/S 的 ADC 规格](#)
- [TMS320F2807x 的 ADC 规格](#)
- [TMS320F28004x 的 ADC 规格](#)
- [C2000TM F2837xD 微控制器研讨会](#)
- [CLA 实践研讨会](#)
- [CLA 在山谷切换式升压功率因数校正 \(PFC\) 参考设计中的应用](#)
- [CLA 在 E2E 上的常见问题解答](#)
- [C2000 F2837xD 微控制器 1 天工作坊 第 1.6 节：控制外设](#)

8 版本历史

注意：先前版本的页码可能与当前版本的页码不同。

从版本 E（2022 年 6 月）到版本 F（2023 年 3 月）的变更	Page
• Updated the numberin g format for tables, fi gures and cross-references throu ghout the document.....	4
• 已将 F280013x 和 F280015x 设备添加到适用设备以及任何支持硬件工具.....	4
• Updated Section 3.7.2	36
• 更新 d Section 3.9 ..	40
• Upda ted Section 3.9.2 ..	40

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